WORKING PAPER | April 2026

Presented at the 2026 Economic Scholars Program, Federal Reserve Bank of Cleveland

Merit Aid, Sorting, and Institutional Access: Evidence from Tennessee's HOPE Scholarship*

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April 2026

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Abstract

Merit-based scholarship programs are often viewed as regressive because eligibility correlates with family income. Using tax-linked data from Chetty et al. (2020), we show that Tennessee's HOPE Scholarship instead generates a *progressive compositional shift* at public colleges: the share of students from the bottom income quintile rises while the top-quintile share falls, with a monotonic gradient across all five income quintiles. Total enrollment remains unchanged, indicating that the effect operates through sorting across institutional sectors rather than expansion of college attendance. We formalize this mechanism in a three-destination sorting model and decompose the effect into a permanent price channel (the scholarship's price reduction) and a cyclical income-sensitivity channel (responsiveness to local economic shocks). Merging county-level income data, we show that tuition-scaling programs buffer these shocks—producing a second monotonic quintile gradient estimated on independent variation—while fixed-dollar programs do not. Cross-state evidence supports the model: programs with moderate eligibility thresholds and large public-sector subsidies generate progressive sorting, while high-threshold programs generate regressive composition changes. Program design determines both the direction and durability of merit aid's distributional effects.

JEL Codes: I23, I24, I28, H75, J24, C23

Keywords: merit aid, college composition, sorting, HOPE Scholarship, difference-in-differences

*We are grateful to Daniel Garcia, Francine Montecinos Puente, Jack Rosetti, and Seyhan Erden (Columbia University); Hriday Karnani (Harvard University); and Andrea Scalenghe (MIT Sloan) for helpful discussions and comments. We also thank the Federal Reserve Bank of Cleveland and the Peer Review Board of the Economic Scholars Program. The views expressed herein are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of New York or Columbia University.

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1 Introduction

Dynarski (2000) showed that Georgia's HOPE Scholarship widened the gap in college attendance between higher- and lower-income students. The finding crystallized a distributional critique of merit aid: because eligibility is conditioned on academic achievement, which correlates with family income, broad-based merit programs disproportionately subsidize students who would attend college regardless (Dynarski, 2004; Jones et al., 2022). This view now dominates the policy debate.

A systematic review of 71 quasi-experimental studies finds that merit-based grants rarely improve outcomes for disadvantaged students and can increase inequality (Herbaut & Geven, 2020). We document an exception.

Using tax-linked income data from Chetty et al. (2020), we estimate how Tennessee's HOPE Scholarship reshaped the full parental income distribution at public colleges. The results are the opposite of what the conventional view predicts: following HOPE implementation, public colleges experienced a *progressive* compositional shift. Bottom-quintile students gained share at public institutions, while top-quintile students lost share. The effect is monotonic across all five income quintiles—Q1 through Q3 gain, Q4 and Q5 lose—and every coefficient is statistically significant. This monotonic gradient is the paper's central empirical result.

Aggregate attendance effects and institutional composition effects are distinct estimands that need not move in the same direction. In a two-sector model—attend college or not—individual enrollment responses fully determine institutional composition. But when students choose among public, private, and out-of-state alternatives, the link breaks: the *aggregate* attendance gap can widen (because higher-income students are more likely to qualify) while the *sector-specific* composition of public colleges simultaneously improves (because the students who actually sort into the public sector are drawn disproportionately from the bottom of the income distribution).[1] The reconciliation depends on where displaced students go. If top-quintile students exit to out-of-state or private alternatives rather than entering public colleges, the public sector becomes more progressive even as overall college-going becomes more unequal. We formalize this insight in a three-destination sorting model (public, private, out-of-state) and test its predictions directly.

[1] This is a form of Simpson's paradox: the aggregate relationship between merit aid and income composition reverses sign when conditioned on enrollment sector. The aggregate effect reflects who attends college; the composition effect reflects who attends *which* college.

Institutional composition matters independently of individual enrollment decisions. The peer effects literature demonstrates that student body composition shapes academic outcomes (Sacerdote, 2001). Chetty et al. (2020) show that access to high-mobility institutions drives intergenerational economic outcomes, establishing that who attends which institution has first-order welfare consequences. If merit aid systematically alters the income profile of public college cohorts, it reshapes the peer environment and mobility prospects for all enrolled students—not only those on the margin of attendance.

Four pieces of evidence are consistent with a sorting interpretation. First, total enrollment at Tennessee public colleges is unchanged after HOPE. The share shifts reflect real reallocation—approximately 10 additional bottom-quintile students per cohort enter, 29 top-quintile students per cohort exit—not a denominator artifact. Second, Tennessee private colleges show no mirror image of the public-sector shift: the decline in top-quintile representation at public colleges is not offset by gains at Tennessee privates, consistent with out-of-state exit as the relevant margin. Third, two-year and four-year institutions adjust through different channels. Community colleges exhibit displacement (top-quintile students replaced by bottom-quintile students, total seats unchanged), while four-year colleges show expansion (enrollment grows, composition shifts). Fourth, within-state comparisons restricted to Tennessee institutions confirm that the composition change is specific to the public sector.

These patterns may not be idiosyncratic to Tennessee. Our theoretical framework proposes that program design and institutional context jointly shape the direction of compositional sorting, through three parameters: the eligibility threshold relative to the income distribution, the size of the public-private subsidy differential, and interactions with existing need-based aid. Georgia's HOPE, which initially offset Pell Grant awards dollar-for-dollar and imposed an income cap, eliminated the access margin for the lowest-income families and muted the public-private subsidy differential—conditions under which the model predicts regressive sorting. Dynarski's findings for Georgia are consistent with this prediction, as are the enrollment patterns documented by Cornwell et al. (2006). Tennessee's HOPE, which has no Pell offset, no income cap, and a large public-private subsidy differential, satisfies the conditions for progressive sorting. The same framework explains both outcomes.

We test these predictions out of sample. Massachusetts' Adams Scholarship (Cohodes & Good-

man, 2014), which shares Tennessee's design features (moderate threshold, public-sector subsidy concentration), independently produces a progressive shift of similar magnitude (+1.38 pp for bottom-quintile share). South Dakota's Opportunity Scholarship, which imposes a high ACT threshold and small award, produces a strongly regressive shift—as the model predicts. West Virginia's PROMISE, despite similar design parameters to Tennessee, shows no compositional effect, highlighting that program design is necessary but not sufficient: enrollment contraction at WV public colleges overwhelmed any sorting dynamics.

We estimate treatment effects using two-way fixed effects (TWFE) as the primary estimator. Because treatment timing is sharp (all Tennessee public colleges treated simultaneously), heterogeneity-robust estimators designed for staggered adoption reduce to TWFE in our setting (Roth, 2026); we report Sun and Abraham (2021) and de Chaisemartin and D'Haultfœuille (2020) for robustness. The preferred specification controls for pre-treatment county average income (Bureau of Economic Analysis), which absorbs cross-county differences in baseline economic conditions. Results are stable across three treatment timing definitions and two alternative control group specifications. Under the relative-magnitude restriction of Rambachan and Roth (2023), the estimated effect survives post-treatment violations as large as the maximum pre-treatment violation ($\bar{M} \leq 1$). Under the smoothness restriction, both bottom- and top-quintile share are robust at $M \leq 0.005$. The shock-insurance results provide independent corroboration through different identifying variation.

Our work connects three literatures. First, a large body of research documents the enrollment effects of merit aid (Bruce & Carruthers, 2014; Cohodes & Goodman, 2014; Cummings et al., 2022; Dynarski, 2000, 2004; Song & Rubenstein, 2024; Verzyl, 2023). This literature focuses on individual enrollment decisions, not institutional composition. Second, studies of policy-induced sorting show that admissions and financing reforms can substantially reshape who attends which institutions (Bleemer, 2022; Card & Krueger, 2005; Hinrichs, 2012). Third, Chetty et al. (2020) demonstrate that access to high-mobility institutions shapes intergenerational economic outcomes, establishing that institutional composition has first-order welfare consequences. We bridge these literatures by documenting how merit aid is associated with changes in the full income distribution across institutional sectors.

This paper makes four contributions. First, we provide the first evidence on how merit aid

reshapes the *full income distribution* at public institutions, using tax-linked data that measures parental income across all quintiles rather than relying on Pell eligibility as a binary proxy. The monotonic quintile gradient—all five coefficients significant and perfectly ordered—has not been documented in any prior study of merit aid. Second, we develop a three-destination sorting framework that reconciles the progressive institutional composition effects we find with the regressive attendance effects documented by Dynarski (2000), showing that the apparent contradiction dissolves once student destinations are disaggregated. Third, we propose that program design parameters—Pell interaction, subsidy differential, threshold location—shape whether merit aid is associated with progressive or regressive sorting, and test the framework across four states: Tennessee and Massachusetts (progressive, as predicted), South Dakota (regressive, as predicted), and West Virginia (null, confounded by enrollment contraction). Fourth, we decompose the scholarship's compositional effect into a permanent price effect (progressive sorting) and a cyclical income-sensitivity effect (responsiveness to economic downturns). Merging BEA county income data with the Chetty panel, we show that tuition-scaling programs (Florida, West Virginia) compensate the income-sensitivity effect—composition is stable during downturns—while fixed-dollar programs (Tennessee, Georgia) do not. This decomposition produces a second five-for-five monotonic quintile gradient, estimated on entirely different variation. A third gradient emerges at four-year colleges under fixed-dollar programs, where marginal students downshift to two-year institutions during income shocks. Three independent monotonic gradients—from program adoption, shock buffering, and shock-induced tier reallocation—converge on the same price-theoretic mechanism and identify award structure as the key design parameter determining whether progressive sorting is sustained.

The paper proceeds as follows. Section 2 describes the Tennessee HOPE Scholarship, the Georgia and Florida programs that serve as comparison cases, and the prior literature. Section 3 presents the three-destination sorting model, derives testable predictions, and introduces the income-shock decomposition. Section 4 describes the Chetty panel, BEA income data, and merit program classification. Section 5 presents the adoption DiD and shock-interaction designs. Section 6 presents results for both designs, including the quintile gradient, shock-insurance gradient, tier reallocation, and award-structure findings. Section 7 reports robustness checks. Section 8 discusses cross-state evidence, the decomposition interpretation, and policy implications. Section 9 concludes.

2 Institutional Background

2.1 Tennessee HOPE Scholarship

Tennessee introduced the HOPE Scholarship in 2003, funded by proceeds from the state education lottery (Tennessee Higher Education Commission, 2025). The program provides tuition and fee subsidies to Tennessee residents who graduate from an eligible high school with a minimum 3.0 GPA or score at least 21 on the ACT (or 980 on the SAT). Recipients must maintain a 2.75 cumulative GPA after the first semester to retain eligibility (Ness & Tucker, 2008).

The scholarship provides a fixed dollar award covering approximately 60–80% of in-state tuition at public institutions governed by the Tennessee Board of Regents (TBR) and the University of Tennessee system ($3,000 for freshmen and sophomores, scaling to $6,000 for juniors and seniors at four-year institutions during the study period). Students attending eligible private institutions receive a smaller fixed amount ($3,000–$4,000 per year), which covers a substantially lower share of private tuition. This differential generosity creates a relative price advantage for public colleges—the key mechanism in our theoretical framework.

Table A1 reports the distribution of HOPE recipients across sectors for the first five entering cohorts (2004–2008). Public institutions—TBR universities, TBR community colleges, and UT campuses—account for approximately 83–85% of recipients in every year. Private colleges receive 15–17%. This concentration of aid in the public sector motivates our identification strategy: we treat Tennessee public colleges as the primary recipients of the HOPE demand shock.

2.2 Prior Research on TN HOPE

Several studies examine Tennessee's merit aid program. The closest antecedent is Bruce and Carruthers (2014), who use a regression discontinuity at the ACT score cutoff for HOPE eligibility. Chakrabarti and Roy (2022) study merit aid and student mobility more broadly, documenting how scholarship programs interact with college selectivity. Bruce and Carruthers (2014) find that Tennessee HOPE had no effect on overall enrollment among Pell-eligible students near the ACT threshold, but reallocated them from two-year to four-year public institutions (-2.9 percentage points at community colleges, $+2.4$ at four-year publics). For non-white students, overall enroll-

ment declined by 2.6 percentage points.

Our analysis is complementary but asks a different question. Bruce and Carruthers estimate the *individual enrollment response* at the eligibility margin: how does a student just above the ACT cutoff choose differently from one just below? Their RD is local to the threshold and identifies the effect for compliers at that margin. We estimate the *institutional composition effect* across the full income distribution: how does the parental income profile of the entire student body at public colleges change after HOPE? Our difference-in-differences design captures the average treatment effect across all treated cohorts, including students well above the eligibility threshold for whom the RD has no power. The distinction matters because the composition of public colleges depends on the sorting behavior of students at all income levels, not only those at the eligibility margin. Our quintile decomposition—showing that all five quintiles shift monotonically—documents a pattern that no threshold-local design can detect, since students in the top quintile are almost all well above any reasonable ACT cutoff. Their finding that lower-income marginal students upgrade from community colleges to four-year institutions is consistent with our result that bottom-quintile share rises and that two-year and four-year colleges adjust through different channels.

2.3 Timeline and Cohort Mapping

The Tennessee HOPE Scholarship was enacted in 2003. In the Chetty et al. (2020) data, cohorts are indexed by birth year, and students are assigned to institutions attended between ages 19 and 22. The analysis window spans birth cohorts 1980–1991, corresponding approximately to college entry years 1998–2009. Table 1 maps each birth cohort to its decision year, college entry year, event time, and HOPE exposure status.

Decision timing vs. receipt timing. The mechanism studied in this paper is sorting across institutional sectors—public, private, and out-of-state—driven by the relative price change that HOPE creates. Sorting occurs *before* matriculation, when students choose where to apply and enroll. The economically relevant treatment date is therefore the point at which HOPE enters the student's decision environment, not the date on which scholarship funds are first disbursed.

We define the decision year as birth cohort +17, corresponding to the year in which the student is typically a high-school senior forming a college choice set. Under this mapping, the 1986 birth

cohort has decision year 2003—the same year HOPE was enacted. These students were mid-application cycle when the program was announced; some could adjust, but many had already committed. The 1987 birth cohort has decision year 2004 and is the first group whose entire college search occurred with HOPE already enacted, publicized, incorporated into school counseling, and operational in the financial aid environment.

Our preferred treatment timing is therefore $t^* = 1987$: the first cohort whose college choice set was formed under the fully known HOPE regime. We designate the 1986 cohort as the omitted reference period ($k = -1$), absorbing any partial treatment of this transition cohort into the baseline. The 1985 cohort received limited retroactive benefits through a bridge provision, but made enrollment decisions before HOPE existed. The 1984 cohort (entering college Fall 2002) is the last cohort with no HOPE exposure of any kind.

We test robustness to $t^* \in \{1985, 1986, 1987\}$ in Section 7. The static DiD estimate is insensitive to this choice.

Analysis window. Under the preferred timing, the sample provides five pre-treatment cohorts (1981–1985) and five post-treatment cohorts (1987–1991), with 1986 as the omitted reference period. Each cohort's composition is measured independently: the outcome for cohort t at college c reflects the parental income distribution of students born in year t who attended college c between ages 19 and 22. The 1980 birth cohort is dropped from the preferred specification because Tennessee's 1980-born cohort exhibits anomalously low bottom-quintile share that normalizes by 1982, consistent with a data artifact rather than a substantive pre-trend. Dropping this single cohort substantially improves sensitivity to violations of parallel trends (Section 7).

Table 1: Cohort Mapping: Birth Year, Decision Year, and HOPE Exposure

Birth Cohort	Decision Year	College Entry	Event Time (k)	HOPE Status
1980	1997	Fall 1998	—	No HOPE (dropped†)
1981	1998	Fall 1999	-6	No HOPE
1982	1999	Fall 2000	-5	No HOPE
1983	2000	Fall 2001	-4	No HOPE
1984	2001	Fall 2002	-3	No HOPE
1985	2002	Fall 2003	-2	Bridge provision (partial)‡
1986	**2003**	Fall 2004	-1	Transition (ref. period)
1987	**2004**	Fall 2005	0	Full HOPE (t^*)
1988	2005	Fall 2006	$+1$	Full HOPE
1989	2006	Fall 2007	$+2$	Full HOPE
1990	2007	Fall 2008	$+3$	Full HOPE
1991	2008	Fall 2009	$+4$	Full HOPE

Notes: Decision year = birth cohort +17 (high-school senior year). College entry is approximate (birth cohort + 18). Students are assigned to institutions attended between ages 19 and 22 in the Chetty et al. (2020) data. HOPE was enacted in 2003 (bolded). †The 1980 cohort is dropped from the preferred specification; see text. ‡Bridge provision extended limited retroactive benefits to Fall 2003 entrants who met eligibility criteria; these students made enrollment decisions before HOPE existed. Event times shown relative to $t^* = 1987$.

Confounding policies. Several other policy changes coincided with HOPE's introduction, and we consider whether any could produce the quintile gradient we document.

Tennessee's education lottery launched in 2003 and generated revenue that funded both HOPE and Pre-K programs. The federal No Child Left Behind Act (2002) altered K–12 accountability requirements nationwide. Both policies affect treated and control states symmetrically and are absorbed by cohort fixed effects.

Federal Pell Grant generosity increased during our analysis window: the 2007 College Cost Reduction Act raised maximum awards, and the 2009 American Recovery and Reinvestment Act added temporary increases. These expansions apply nationally, so any composition effects are absorbed by cohort fixed effects in our DiD design. To the extent that Pell expansions disproportionately

benefit low-income students, they would produce a level shift in bottom-quintile enrollment at all public colleges, not the Tennessee-specific treatment effect our interaction term identifies.

Tennessee's Board of Regents expanded community college access during this period, including the Tennessee Technology Centers and dual enrollment initiatives. These reforms could affect overall enrollment levels but would not generate a monotonically ordered gradient across all five income quintiles—a pattern that requires differential price sensitivity by parental income, as our model predicts.

Tuition changes at neighboring states' public universities could affect cross-state sorting. However, such changes would disproportionately influence border-county colleges rather than producing the statewide pattern we observe. Descriptive evidence suggests that the composition effect is concentrated at metro colleges, which are generally not near state borders.

While we cannot rule out every potential confounder individually, the quintile gradient provides a useful discriminating pattern: a generic enrollment shock would affect quintile shares unevenly but would be unlikely to produce five coefficients that are monotonically ordered across all quintiles with no effect at private institutions.

2.4 Georgia HOPE as a Comparison Case

Georgia's HOPE Scholarship (1993) is the canonical case in the merit aid literature (Dynarski, 2000; Song & Rubenstein, 2024) and our primary theoretical comparison. Three design features distinguish it from Tennessee's program in ways that matter for our sorting framework.

First, from 1993 to 2000, Georgia HOPE awards were reduced dollar-for-dollar by Pell Grant amounts. Students from the lowest-income families received zero net benefit from HOPE for the program's first seven years. Dynarski (2000) documents the consequence: Black students, who were disproportionately Pell-eligible, showed no attendance gain, while white students gained 12.3 percentage points. The Pell offset effectively eliminated the access margin for the bottom of the income distribution.

Second, Georgia imposed an income cap ($66,000, later $100,000, eliminated in 1995) during the program's early years. While the cap was removed relatively quickly, its presence during the launch shaped initial enrollment patterns.

Third, Georgia HOPE provided comparable awards to private colleges: initially full tuition,

later a fixed $3,000 that covered a meaningful share of private tuition at Georgia's price levels. This smaller public-private subsidy differential muted the sector-switching mechanism that drives compositional sorting in our model. Cornwell et al. (2006) confirm that four-year private enrollment in Georgia *rose* by 13.2% after HOPE, and two-thirds of the four-year enrollment gain reflected reduced out-of-state migration rather than new college-goers.

We return to the GA-TN comparison in Section 3.6, where we show that these design differences generate opposite predictions from the same theoretical framework.

2.5 Florida Bright Futures: A Tuition-Scaling Contrast

Florida's Bright Futures Scholarship (1997) provides a structural contrast to Tennessee's fixed-dollar design. Bright Futures operates on a multi-tier system. The top tier (Florida Academic Scholars) requires a 3.5 weighted GPA and 1270 SAT, covering 100% of tuition. The broad-coverage tier—Florida Medallion Scholars—requires only a 3.0 weighted GPA and 970 SAT (old scale, approximately the 50th percentile nationally) and covers 75% of tuition at public institutions. During our study period (1998–2009), approximately 30–35% of Florida high school graduates received some tier of Bright Futures.

Two features distinguish Bright Futures from Tennessee HOPE for the income-shock analysis. First, the award is *percentage-based*: it covers a fixed share of tuition rather than a fixed dollar amount. When tuition rises, the award rises proportionally. This means the gap between the scholarship and total tuition cost does not widen over time—the theoretical distinction formalized in Proposition 4. Second, Bright Futures does not require FAFSA filing for eligibility, removing a paperwork barrier that may deter low-income families during periods of financial stress.

Because Bright Futures was already active throughout the Chetty panel (the program predates the 1998 college entry of the earliest cohort), we cannot estimate an adoption DiD for Florida. The shock-interaction design (Section 5.4) circumvents this limitation by comparing how Florida colleges respond to income shocks relative to non-merit states, without requiring a pre-program period.

3 Theoretical Framework

We develop a model of college choice that generates testable predictions about how merit aid reshapes the income composition of public institutions and whether that reshaping is stable under economic fluctuations. The key departure from standard two-sector models is that we allow three destinations: in-state public college, in-state private college, and an outside option (out-of-state enrollment or non-attendance). We then introduce income shocks to decompose the scholarship's compositional effect into a price channel and an income-sensitivity channel, analogous in spirit to the Slutsky decomposition.

3.1 Setup

Consider a population of families indexed by income y_i and student ability a_i, where y_i and a_i are positively correlated. Each family chooses among three options: a public college (C_P), a private college (C_R), or no college (0). The indirect utility of family i from option $j \in \{P, R, 0\}$ is

$$V_{ij} = q_j - \frac{C_j}{y_i} + \varepsilon_{ij}, \tag{1}$$

where q_j is the quality of institution j, C_j is the sticker price, and ε_{ij} captures idiosyncratic preferences (we place no distributional restriction on ε_{ij}; the propositions below use only the sign of expected marginal utilities). We normalize $q_0 = C_0 = 0$ and assume:

1. $q_R > q_P > 0$ (private colleges are higher quality),
2. $C_R > C_P > 0$ (private colleges are more expensive),
3. $y_i > 0$ for all families,
4. $\mathrm{Cov}(a_i, y_i) > 0$ (ability and income are positively associated).

3.2 Merit Scholarship and Sorting Boundaries

A state merit scholarship of amount $S > 0$ reduces the effective price of the public college for students with ability above a threshold τ:

$$C_P^{\text{eff}}(a_i) = \begin{cases} C_P - S & \text{if } a_i \geq \tau, \\ C_P & \text{if } a_i < \tau. \end{cases} \tag{2}$$

Without the scholarship, families sort across options by income. Define the no-college/public boundary as $\underline{y}$, where $V_{iP} = V_{i0}$, and the public/private boundary as $\overline{y}$, where $V_{iP} = V_{iR}$:

$$\underline{y} = \frac{C_P}{q_P}, \qquad \overline{y} = \frac{C_R - C_P}{q_R - q_P}. \tag{3}$$

Families with $y_i < \underline{y}$ do not attend college; those with $\underline{y} \leq y_i < \overline{y}$ attend public; those with $y_i \geq \overline{y}$ attend private.

The scholarship shifts both boundaries for eligible students.

Proposition 1 (Access margin)**.** *The scholarship lowers the no-college/public boundary:* $\underline{y}^S = (C_P - S)/q_P < \underline{y}$.

Proof. Since $S > 0$, we have $C_P - S < C_P$. Dividing by $q_P > 0$ preserves the inequality. □

Proposition 2 (Substitution margin)**.** *The scholarship raises the public/private boundary:* $\overline{y}^S = \overline{y} + S/(q_R - q_P) > \overline{y}$.

Proof. Since $S > 0$ and $q_R - q_P > 0$, the term $S/(q_R - q_P) > 0$. □

Proposition 3 (Progressivity of utility gain)**.** *For an eligible student choosing public college, the utility gain from the scholarship is* $\Delta V_i = S/y_i$*, which is strictly decreasing in income:* $\partial \Delta V_i / \partial y_i = -S/y_i^2 < 0$.

Together, the access and substitution margins predict that the public sector's income composition shifts downward, with the largest gains at the bottom of the distribution (Proposition 3). Figure 1 illustrates the sorting mechanism.

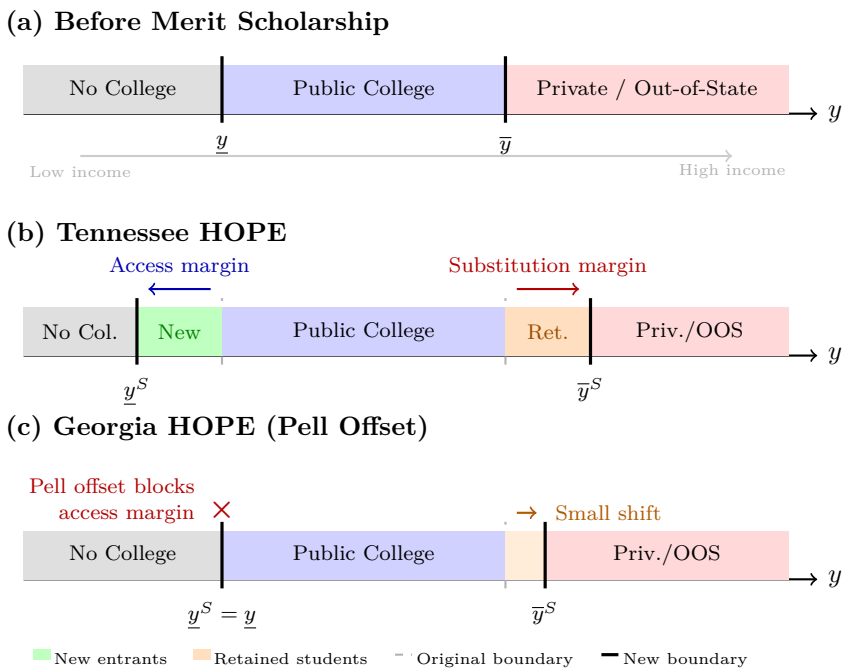

Figure 1: Three-Destination Sorting: How Merit Scholarships Reshape Institutional Composition

Notes: Panel (a) shows pre-scholarship sorting by family income y. Panel (b) shows Tennessee HOPE: the scholarship reduces the effective price of public college, shifting the lower boundary left (access margin: low-income students enter) and the upper boundary right (substitution margin: high-income students retained in public rather than exiting to private). The public sector expands in both directions, producing progressive sorting. Panel (c) shows Georgia HOPE: the Pell offset eliminates the effective subsidy for bottom-quintile families, so the lower boundary does not shift (×). Only the upper boundary moves modestly, producing regressive sorting.

3.3 Testable Predictions

The model generates seven predictions about the compositional effect of merit aid:

P1: Bottom-quintile share rises at public colleges. The scholarship reduces the effective price of public college. Because S/y_i is largest for low-income families, the marginal entrants drawn from below $\underline{y}$ are disproportionately low-income. We test P1 in Section 6.1.

P2: Monotonic quintile gradient. The price reduction S/y_i is monotonically decreasing in income. Q1 families experience the largest proportional price reduction; Q5 the smallest. The composition shift should be monotonically graded across quintiles. We test P2 in Section 6.3.

P3: Composition change reflects sorting, not enrollment expansion. Total enrollment at public colleges is unchanged if capacity constraints bind; the share shifts are driven by changes in who enrolls, not in how many enroll. We test P3 in Section 6.4.

P4: No mirror image at private colleges. If displaced top-quintile students exit to out-of-state options rather than shifting to private colleges, private institutions should show no significant composition change. We test P4 in Section 6.4.

P5: Heterogeneous channels by tier. At two-year colleges, where the no-college margin is active, the scholarship expands access (pulls in new students from below $\underline{y}$). At four-year universities, where the public/private margin is active, the scholarship induces sorting—a form of the ability stratification documented by Cornwell and Mustard (2006). We test P5 in Section 6.4.

P6: Spatial heterogeneity. Institutions near state borders, where out-of-state alternatives are accessible, should show larger composition shifts because the outside option is more salient and the three-destination mechanism is most active. We do not test P6 directly, as the spatial analysis is descriptive and cannot isolate metro status from its correlates.

P7: Cross-state variation. Programs with no Pell offset and large public-private subsidy differentials produce progressive sorting. Programs with Pell offsets or comparable private awards produce regressive sorting or no composition change. We test P7 in Section 8.

3.4 Decomposition Under Income Shocks

Predictions P1–P7 describe the *static* compositional effect of introducing a merit scholarship. We now ask whether that compositional shift is *stable* when families experience local economic fluctuations. The answer depends on the scholarship's award structure.

Consider a negative income shock $\Delta y < 0$ affecting families in a local economy. The shock changes family income from y_i to $y_i + \Delta y$ but does not change the scholarship amount, tuition, or institutional quality. The total effect of the scholarship under an income shock decomposes into two components.

Price effect. Holding income constant, the scholarship changes the relative price of public college. The utility gain for an eligible student choosing public is $\Delta V_i = S/y_i$. This price-reduction effect generates predictions P1–P7. The adoption difference-in-differences identifies it. It does not vary with local economic conditions.

Income-sensitivity effect. Holding the scholarship constant, the income shock changes the family's ability to pay the remaining cost. For a fixed-dollar scholarship S, the cost burden after the shock is $(C_P - S)/(y_i + \Delta y)$, which increases when $\Delta y < 0$. The income shock reasserts the affordability constraint on any costs the scholarship does not cover.

The degree to which the scholarship *compensates* the income effect depends on the award structure:

Proposition 4 (Income-effect compensation)**.** *Let S^F be a fixed-dollar award and $S^\phi = \phi C_P$ be a percentage-based award covering fraction ϕ of tuition, with $S^F = \phi C_P$ so the two have identical dollar value at the time of introduction. Under a percentage-based award, the income effect on tuition costs is reduced by factor $(1 - \phi)$ relative to the no-scholarship case. Under a fixed-dollar award, the reduction is $1 - S^F/C_P$, which equals $(1 - \phi)$ initially but erodes as tuition rises: if $C'_P > C_P$, then $1 - S^F/C'_P > 1 - \phi$.*

When $\phi = 1$ (full tuition coverage), tuition costs vanish from the income-sensitivity channel entirely: the family's college choice on the tuition dimension is fully insulated from income fluctuations.

Proposition 5 (Tier reallocation under fixed-dollar awards)**.** *Consider a student eligible for a fixed-dollar scholarship S, choosing between a four-year college (cost C_4, quality q_4) and a two-year college (cost $C_2 < C_4$, quality $q_2 < q_4$). The scholarship reduces costs at both tiers by the same dollar amount S, so the relative price of the four-year option is unchanged:*

$$V_{i4}^S - V_{i2}^S = (q_4 - q_2) - \frac{C_4 - C_2}{y_i}. \tag{4}$$

For the marginal four-year student (where $V_{i4}^S \approx V_{i2}^S$), an income shock $\Delta y < 0$ increases $(C_4 - C_2)/(y_i + \Delta y)$, tipping the balance toward the two-year option. Provided $y_i + \Delta y > (C_2 - S)/q_2$ (the student remains above the no-college boundary), the scholarship keeps the student in the system but does not prevent tier reallocation. For sufficiently large shocks, exit from the system is also possible.

Proposition 5 explains why fixed-dollar programs can generate progressive sorting on average (P1–P2) while showing no aggregate buffering during downturns: the scholarship creates a new class of marginal four-year students—low-income families who attend four-year colleges only because the scholarship makes it affordable—who are particularly sensitive to income shocks and downshift to

two-year colleges when family budgets tighten. Bruce and Carruthers (2014) document a consistent pattern for Tennessee HOPE: marginal students near the eligibility threshold shift from two-year to four-year institutions when HOPE is available.

Figure 2 illustrates the decomposition spatially. In steady state, both Tennessee (fixed-dollar) and Florida (percentage-based) produce the same progressive boundary shifts. Under an income shock, Tennessee's boundaries retreat substantially—marginal four-year students downshift to two-year—while Florida's boundaries barely move.

These propositions generate four additional testable predictions:

P8: Tuition-scaling programs buffer composition during downturns. Programs where the award covers a fixed share of tuition ($\phi > 0$) compensate the income effect, stabilizing the compositional shift achieved through the substitution effect. We test P8 in Section 6.5.

P9: Fixed-dollar programs show no aggregate buffering. Programs where the award is a fixed dollar amount leave the income effect uncompensated, producing zero net buffering at the aggregate level. We test P9 in Section 6.5.

P10: Fixed-dollar programs show four-year anti-buffering. At four-year colleges specifically, fixed-dollar programs should show composition *worsening* during downturns—the marginal students created by the scholarship downshift to two-year colleges. We test P10 in Section 6.6.

P11: Q2 is the most elastic quintile. The lower-middle class (second income quintile) sits at the intersection of the model's two margins—just above the no-college boundary (where Pell alone may not suffice) and just below the two-year/four-year boundary (where the scholarship tips the choice). Q2 students should be the most sensitive to both the scholarship and to income shocks. We test P11 in Section 6.6.

3.5 Reconciling with Dynarski (2000)

Dynarski (2000) finds that Georgia's HOPE Scholarship widened the attendance gap between higher- and lower-income students. Our framework shows that this aggregate finding is consistent with a progressive composition shift at public institutions. Because eligibility is correlated with income, merit aid increases the *total* number of higher-income students attending college

more than the total number of lower-income students. But the *destination* of these additional higher-income students matters. If they enroll at private or out-of-state institutions—or if they would have attended college anyway—the composition of public colleges can still shift downward.

The reconciliation rests on distinguishing between two quantities: the aggregate attendance gap (which can widen) and the institutional composition of the public sector (which can simultaneously become more progressive). Both are consistent with three-destination sorting.

This distinction is specific to the multi-sector setting. In a two-sector model—attend college or not—the aggregate attendance effect fully determines composition: if merit aid increases enrollment disproportionately among higher-income students, those students necessarily appear at colleges, and composition becomes more affluent.

With three or more destinations, the link breaks. Higher-income students may attend college at higher rates but sort into private or out-of-state institutions. The public sector's composition then depends on who remains—not on aggregate enrollment. Individual enrollment responses are therefore insufficient to predict institutional composition; the full destination distribution is required.

3.6 Georgia as a Regressive Prediction

Georgia's HOPE included a Pell offset: scholarship dollars were reduced dollar-for-dollar by Pell Grant receipts (Figure 1, Panel C). Because Pell Grants target Q1 families, the effective scholarship for the poorest students was

$$S_{\text{eff}}^{Q1} = \max(S - \text{Pell},\ 0) \approx 0. \tag{5}$$

The model predicts no composition shift for Q1 (the price reduction is zero), while Q3–Q5 families receive the full S. The predicted effect is regressive: public colleges gain middle- and upper-income students without gaining low-income students. Dynarski (2000) documents the consequence: Black students, who were disproportionately Pell-eligible, showed no attendance gain, while white students gained 12.3 percentage points.

Georgia's HOPE also provided comparable awards to private colleges, muting the public-private subsidy differential. Cornwell et al. (2006) confirm the pattern: four-year private enrollment in Georgia *rose* by 13.2% after HOPE, and two-thirds of the four-year enrollment gain reflected reduced

out-of-state migration rather than new college-goers.

Tennessee's HOPE has no Pell offset, no income cap, and a large public-private subsidy differential (full tuition at publics vs. \$3,000–\$4,000 at privates). Under these parameters, the model predicts that the access margin and substitution margin both operate, producing a progressive quintile gradient. The TN-GA contrast is a pair of outcomes predicted by the same model under different parameter values.

The decomposition framework adds a further distinction: Tennessee's fixed-dollar award generates progressive sorting but leaves it vulnerable to income shocks (uncompensated income effect), while Florida's percentage-based award (Bright Futures Medallion tier, covering 75% of tuition) both generates progressive sorting and stabilizes it (compensated income effect). Award structure predicts not only the direction but the *durability* of compositional change.

4 Data

4.1 Chetty et al. (2020) Panel

We use the college-level panel constructed by Chetty et al. (2020), which links tax records and college enrollment data to measure the parental income distribution of students at each institution. The panel covers 2,202 colleges across 52 states and territories for birth cohorts 1980–1991 (approximately corresponding to college entry years 1998–2009). Students are assigned to the institution attended at ages 19–22.

The primary analysis table (MRC Table 3) provides college-by-cohort observations with measures of parental income rank and quintile shares. We supplement this with institutional characteristics from MRC Table 10 (SAT scores, expenditures, enrollment, Barron's selectivity, HBCU and flagship status, graduation rates). These characteristics capture the dimensions of institutional quality that prior work has linked to earnings premia (Brewer et al., 1999; Dale & Krueger, 2002).

4.2 Sample Construction

We define treated institutions as Tennessee public colleges. Tennessee has 51 colleges in the data, of which 26 are public—19 two-year institutions and 7 four-year institutions. Multi-campus systems appear as distinct identifiers (56 per cohort), but we identify 26 unique public colleges.

The control group comprises public colleges in states that did not adopt broad-based merit scholarship programs during the analysis period. We drop all 25 states classified as merit-program adopters by Sjoquist and Winters (2015). The remaining 26 control states contribute 6,289 college-cohort observations. Of the 312 possible Tennessee public-college-by-cohort observations (26×12), 291 are non-missing on both primary outcomes.

We exclude 24 observations with missing state or institution type.

4.3 Outcome Variables

Our two primary outcomes are:

- **Bottom-quintile share** (Q_1): the fraction of students at college c in cohort t whose parents are in the bottom 20% of the national income distribution. Baseline mean for Tennessee public colleges: 0.140.
- **Mean parental income rank** ($\overline{r}$): the average percentile rank of parental household income for students at college c in cohort t. Pre-treatment mean for Tennessee publics: 0.524.

Each outcome is measured at the college-by-birth-cohort level, not for the entire enrolled student body at a point in time. The Fall 2006 student body at UT, for example, would include birth cohorts 1984–1987 simultaneously—mixing untreated and treated cohorts. The cohort-specific measure isolates students who faced HOPE-era incentives from those who did not.

For the quintile decomposition, we additionally use the share of students from each income quintile (Q_1 through Q_5).

For the level decomposition that addresses ecological inference concerns, we construct count outcomes by multiplying quintile shares by cohort size. For example, Q1 count $= Q_1 \times N_{ct}$, where N_{ct} is total enrollment at college c in cohort t.

4.4 Bureau of Economic Analysis: County Income

Local per capita income is both a potential confounder and a mechanism variable in the theoretical framework. We merge county-level personal income data from the Bureau of Economic Analysis (BEA) to handle both roles. We use the CAINC1 table (Personal Income Summary), which reports annual per capita personal income for every U.S. county from 1969 to the present.

We match each college-cohort observation to the BEA county of the college's location using five-digit FIPS codes from the IPEDS institutional characteristics file. The match rate is 98.4%; unmatched observations (primarily colleges with missing FIPS codes) are dropped from specifications that include BEA controls. No Tennessee public colleges are among the unmatched. For each college-cohort, we measure the log of county per capita income in the *decision year*—defined as the birth cohort year plus 17 (the year the student is a high school senior making enrollment decisions). We set the decision year to cohort +17 on conceptual grounds: students form college choice sets during their senior year, not after matriculation. Section 6.5 confirms that economic conditions at age 17, not age 18, predict composition shifts.

The BEA data serve two distinct roles in our empirical design. For the *adoption DiD*, we control for the pre-treatment county average of log per capita income (1997–2003 decision years), $\overline{\log(\text{pcpi})}_c$. Because this average is computed entirely from pre-treatment cohorts, it is predetermined and cannot be affected by HOPE. It absorbs cross-county differences in baseline economic conditions without introducing a bad-control problem. For the *shock-interaction design* (Section 5.4), we use the one-year change in log county per capita income at decision year, $\Delta \log(\text{pcpi})_{ct}$, which is the identifying variation: students in counties experiencing negative income shocks face a larger effective price change, and program generosity determines whether merit aid buffers or amplifies the shock.

4.5 Merit Aid Program Classification

For the cross-state shock-insurance analysis, we classify state merit-aid programs following Sjoquist and Winters (2015), who identify nine "strong" merit programs with high participation rates and substantial award amounts: Florida, Georgia, Kentucky, Louisiana, Nevada, New Mexico, South Carolina, Tennessee, and West Virginia. We add Massachusetts (Adams Scholarship, 2004), which Sjoquist and Winters (2015) excluded only because it post-dated their sample period but is active during our Chetty panel window. Massachusetts is therefore excluded from the Tennessee adoption DiD control group: its Adams Scholarship constitutes a real merit-aid treatment during our analysis window, so including Massachusetts among untreated states would attenuate the estimated effect.

Award structure determines whether a scholarship compensates for income shocks. Table 3 classifies the ten programs as *tuition-scaling* (covering a fixed percentage of tuition or providing a

tuition waiver) or *fixed-dollar* (a constant nominal award that erodes relative to costs over time). This distinction maps directly to Proposition 4. This classification is necessarily stylized; Georgia HOPE, for example, is nominally described as "full tuition" but functioned as a capped dollar award, and its Pell offset (1993–2000) further complicated the effective subsidy structure (Table 3).

Florida's Bright Futures Scholarship is often characterized by its top tier (Florida Academic Scholars: 3.5 GPA and 1270 SAT). During our study period (1998–2009), the operative tier for broad coverage was the Medallion Scholars award, which required only a 3.0 weighted GPA and 970 SAT (old scale, approximately the 50th percentile) and covered 75% of tuition at public institutions. Approximately 30–35% of Florida high school graduates received some tier of Bright Futures during this period. The Medallion tier's GPA threshold is comparable to Tennessee HOPE's 3.0 requirement, but its percentage-based award structure differs fundamentally.

4.6 Eligibility Gradient: Validating the Theoretical Mechanism

The theoretical model predicts that the eligibility threshold τ determines who benefits from the scholarship. We validate this prediction using microdata from the National Longitudinal Survey of Youth 1997 (NLSY97), which provides high school GPA (from transcripts), ACT and SAT scores, and family income for approximately 9,000 respondents born 1980–1984—cohorts that overlap directly with the early Chetty panel.

We construct parental income quintiles from household income reported in 1997 and compute the fraction of students in each quintile who would meet each state's eligibility rules. We apply the program rules to the *national* achievement distribution, not to state-specific distributions, to avoid contamination from program-induced behavioral responses (teaching to the threshold, grade inflation).

Figure 3 displays the results. Three findings validate the theoretical mechanism:

1. **The eligibility gradient is steep and monotonic.** For every program, eligibility rises from Q1 to Q5. The gap is substantial: under Tennessee HOPE rules, 29% of Q1 students qualify vs. 59% of Q5 students.

2. **Threshold height determines the gradient's level.** Low-threshold programs (TN HOPE: 29% of Q1; GA HOPE: 29%) admit nearly a third of bottom-quintile students. High-threshold

programs (FL Academic Scholars: 3%; SD Opportunity: 11%) admit almost none. The Q1/Q5 eligibility ratio ranges from 0.49 (TN HOPE) to 0.22 (FL Academic).

3. **The OR condition matters.** Tennessee's either/or criterion (3.0 GPA *or* ACT 21) produces broader Q1 eligibility (29%) than programs with AND conditions at similar GPA thresholds (FL Medallion at 3.0 GPA *and* 970 SAT: 25%; WV PROMISE at 3.0 GPA *and* ACT 22: 17%).

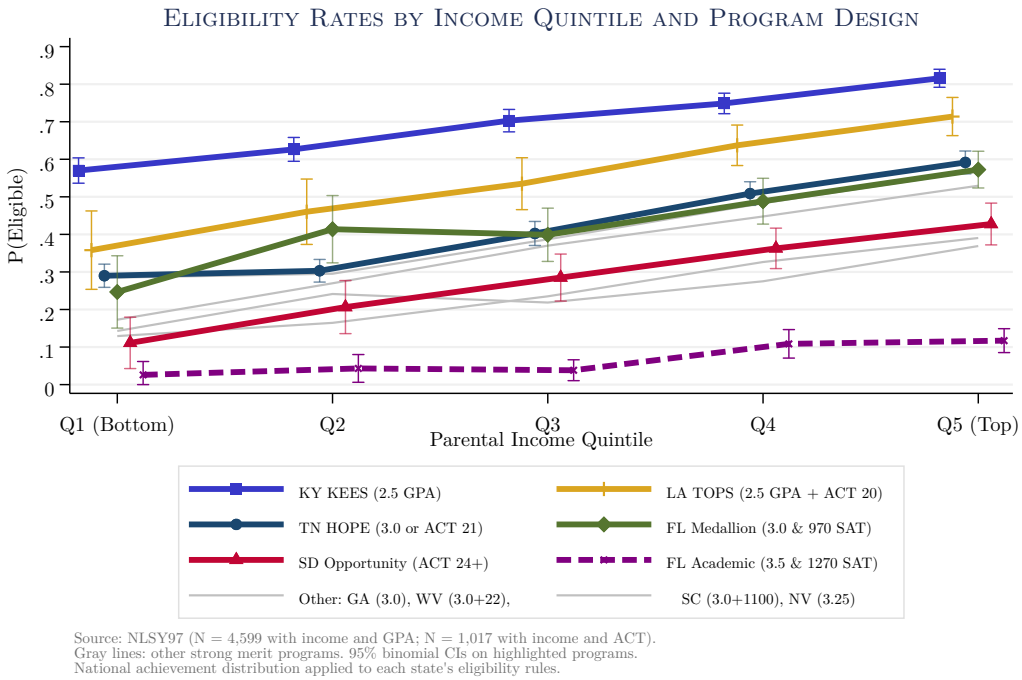

Figure 3: Eligibility Rates by Income Quintile and Program Design

Notes: Each line shows the fraction of students in each parental income quintile who would meet the indicated program's eligibility rules, computed from the national joint distribution of GPA, ACT, and SAT in the NLSY97 (N = 4,599 with income and GPA; N = 1,017 with income and ACT). Programs requiring ACT scores (e.g., TN HOPE's OR condition) are computed on the smaller ACT subsample; the qualitative gradient ordering is stable across both samples. Programs with low thresholds (TN HOPE, GA HOPE) have high eligibility across all quintiles; programs with high thresholds (SD Opportunity, FL Academic Scholars) have low eligibility concentrated in the top quintiles. Vertical bars show 95% binomial confidence intervals for highlighted programs; gray background programs are plotted without intervals to reduce clutter. The eligibility ratio $P(\text{Eligible} \mid Q1)/P(\text{Eligible} \mid Q5)$ ranges from 0.49 (TN) to 0.22 (FL Academic).

These eligibility rates are consistent with the model's predictions about which programs generate

progressive versus regressive sorting. Programs where $P(\text{Eligible} \mid Q1)$ is large enough to generate meaningful enrollment responses among low-income students (TN, GA, FL Medallion) are predicted to produce progressive composition shifts. Programs where $P(\text{Eligible} \mid Q1)$ is near zero (FL Academic, SD) are predicted to produce regressive shifts. The cross-state results in Sections 6 and 8 are consistent with these predictions.

4.7 Summary Statistics

Table 2 reports summary statistics for the primary outcomes and quintile shares, separately for Tennessee public colleges and control-state public colleges. Tennessee publics have a higher baseline share of bottom-quintile students (14.0% vs. 13.6%) and a lower mean parental income rank (0.524 vs. 0.541). The quintile distribution at Tennessee publics is shifted leftward: Q1 and Q2 shares are higher, Q5 share is lower. Table A2 reports pre- and post-treatment means. After HOPE implementation, bottom-quintile share at Tennessee publics rises to 15.9%, while the control group increases modestly to 12.9%. The raw difference-in-differences is 1.5 percentage points.

Table 2: Summary Statistics

	TN Public		Control Public	
	Mean	SD	Mean	SD
Primary Outcomes				
Bottom-quintile share (Q_1)	0.140	0.046	0.136	0.083
Mean parental income rank	0.524	0.067	0.541	0.103
Quintile Shares				
Q1 (bottom)	0.140		0.136	
Q2	0.195		0.182	
Q3	0.239		0.227	
Q4	0.256		0.252	
Q5 (top)	0.170		0.203	
Observations	137		2,850	
Colleges	26		∼500	

Notes: Pre-treatment means (birth cohorts 1981–1986, i.e., all cohorts before $t^* = 1987$, excluding the 1980 cohort consistent with the preferred specification) for Tennessee public colleges (treated) and public colleges in 26 non-merit-program states (control). The full analysis sample includes 291 TN public college-cohort observations across all 12 cohorts (26 colleges $\times$ 12 cohorts = 312 possible, minus 21 missing on primary outcomes). Quintile shares refer to the fraction of students whose parents are in the indicated quintile of the national income distribution. Observations are at the college-by-cohort level. The preferred specification uses $t^* = 1987$ with cohort 1986 ($k = -1$) as the omitted reference period; means reported here include 1986 to characterize the full pre-treatment distribution. Excluding 1986 does not materially change the means. Data: Chetty et al. (2020).

Table 3: Strong Merit Aid Programs: Design Parameters

State	Program	Threshold	Award Structure	% FTE	$/FTE
Tuition-Scaling Awards					
Florida	Bright Futures	3.0 GPA / 970 SAT	75–100% of tuition	24.3	$581
West Virginia	PROMISE	3.0 GPA + 22 ACT	Full tuition + fees	9.8	$485
Massachusetts	Adams	Top 25% MCAS	Tuition waiver	—	—
Louisiana	TOPS	2.5 GPA + 20 ACT	Tuition + fees	23.2	$709
Fixed-Dollar Awards					
Tennessee	HOPE	3.0 GPA or 21 ACT	$3,000–$6,000	26.9	$920
Georgia†	HOPE	3.0 GPA	Full tuition → fixed	30.7	$1,191
Other Strong Programs					
Kentucky	KEES	Graduated by GPA	Per credit hour	35.7	$493
South Carolina	LIFE	3.0 GPA + 1100 SAT	$5,000 + tuition	18.4	$888
Nevada	Millennium	3.25 GPA	$80/credit	25.6	$327
New Mexico	Lottery Success	2.5 GPA (college)	Tuition coverage	20.7	$495

Notes: Strong merit programs per Sjoquist and Winters (2015), plus Massachusetts. "% FTE" is awardees as a percentage of full-time-equivalent students (2010). "$/FTE" is grant expenditures per FTE student (2010). Florida's threshold refers to the Medallion Scholars tier (operative for broad coverage during 1998–2009); the Academic Scholars tier requires 3.5 GPA + 1270 SAT. Tennessee's HOPE requires FAFSA filing. †Georgia HOPE is nominally described as covering "full tuition," but the award was a capped dollar amount (up to ∼$4,500 at top public universities by 2004–05) that approximated tuition at most institutions rather than automatically tracking each school's price (Cornwell et al., 2006). We classify it as fixed-dollar because the award did not mechanically adjust with tuition changes. Additionally, during 1993–2000, a Pell offset reduced HOPE dollar-for-dollar by federal Pell Grant receipts, effectively eliminating the subsidy for the lowest-income families (Dynarski, 2000). Award structure classification follows the theoretical distinction in Proposition 4.

5 Empirical Strategy

We deploy two empirical designs. The first is an adoption difference-in-differences that identifies the effect of Tennessee HOPE on college composition. The second is a shock-interaction design that tests whether merit aid buffers low-income access during local economic downturns, exploiting

county-level income variation described in Section 4.4. Together, the two designs identify different parameters—the level shift from program adoption and the differential response to income shocks—using independent sources of variation.

5.1 Adoption DiD: Primary Estimator

Treatment timing is sharp: all Tennessee public colleges are treated simultaneously at $t^* = 1987$, the first birth cohort whose college choice set was formed under the fully known HOPE regime (see Section 2.3 for the decision-year justification). We designate 1986 ($k = -1$) as the omitted reference period, absorbing any partial treatment of that transition cohort into the baseline. Robustness to $t^* \in \{1985, 1986, 1987\}$ is reported in Section 7; the static estimate is insensitive to this choice.

Our primary estimator is the two-way fixed effects (TWFE) event study. Because treatment timing is sharp—all Tennessee public colleges are treated simultaneously—recent heterogeneity-robust estimators designed for staggered adoption (Callaway & Sant'Anna, 2021; Sun & Abraham, 2021) reduce to the standard TWFE specification in our setting (Roth, 2026). We therefore report TWFE as the primary estimator and reserve the Callaway and Sant'Anna (2021) estimator for the pooled multi-state specification in Section 8, where staggered adoption timing makes it the appropriate choice. The TWFE specification is:

$$y_{ct} = \alpha_c + \lambda_t + \sum_{k=-5}^{4} \beta_k \cdot \mathbf{1}[t - t^* = k] \cdot \text{Treated}_c + \gamma \cdot \overline{\log(\text{pcpi})}_c + \varepsilon_{ct} \tag{6}$$

where y_{ct} is the outcome for college c in birth cohort t, α_c are college fixed effects, λ_t are cohort fixed effects, Treated_c indicates Tennessee public colleges, $\overline{\log(\text{pcpi})}_c$ is the pre-treatment county average of log per capita personal income (BEA, 1997–2003 decision years), and $k = -1$ (cohort 1986) is the omitted reference period. The sample excludes the 1980 birth cohort, a single-cohort outlier that drives pre-trend divergence at the longest pre-treatment horizon. Cohort 1981 ($k = -6$) remains in the estimation sample—contributing to the precision of the college and cohort fixed effects—but is not assigned a separate event-time indicator; its effect is subsumed into the intercept.

5.2 Preferred Specification

The preferred specification for the adoption DiD uses:

1. **National control group**: public colleges in 26 non-merit-program states, providing 27 state-level clusters (26 control + Tennessee) for cluster-robust inference.

2. **Treatment timing** $t^* = 1987$: the first decision-treated cohort (Section 2.3).

3. **Drop 1980 birth cohort**: Tennessee's 1980-born cohort exhibits anomalously low Q_1 share that normalizes by 1982, consistent with a data artifact. This single cohort is the sole driver of the $k = -6$ pre-trend divergence in prior specifications (Figure A1).

4. **BEA county income control**: pre-treatment county average log per capita personal income (BEA, 1997–2003 decision years) absorbs cross-county differences in baseline economic conditions. Because the average is computed entirely from pre-treatment cohorts, it is predetermined and not affected by HOPE. The decomposition framework (Section 3.4) identifies local economic conditions as the channel through which income effects operate, motivating the control independently of the sensitivity analysis.

Each choice is independently motivated and precedes the sensitivity analysis. Results without the BEA control are nearly identical (Section 7), confirming that the baseline county income level is approximately collinear with the college fixed effects.

For robustness, we also report results under the bordering-state control group (Alabama, Virginia, North Carolina—the only non-merit states adjacent to Tennessee), which has cleaner pre-trends but only four state clusters, as well as results under alternative treatment timing.

5.3 Identification

Identification requires that Tennessee public colleges would have followed parallel trends in socioeconomic composition to control colleges absent the HOPE Scholarship. The most plausible threat is that Tennessee's public sector was on a different compositional trajectory before HOPE—driven, for example, by demographic shifts, state fiscal policy, or differential enrollment growth—and that this trajectory, not the scholarship, produces the post-treatment pattern.

Under the preferred specification, pre-treatment event-study coefficients at $k = -4$, -3, and -2 (cohorts 1983–1985) are individually small and statistically insignificant for Q_1, with p-values exceeding 0.44. A joint test of the three pre-treatment coefficients fails to reject the null of zero pre-

trend. Given the limited number of pre-treatment periods, these tests have modest power against slow-moving violations; we address this directly with the Rambachan and Roth (2023) sensitivity analysis below.

The directional pattern of the pre-treatment coefficients provides additional reassurance. The coefficient at $k = -3$ is the largest in absolute value and is negative: Tennessee's Q_1 share was *falling* relative to controls before HOPE. Post-treatment effects are uniformly positive. A smooth confounding trend that generates positive post-treatment effects would require the pre-trend to be positive and growing—the opposite of what we observe. The sign reversal is inconsistent with a continuous confound driving our results, though it does not rule out a discrete break coinciding exactly with HOPE's introduction.

5.4 Shock-Interaction Design

The decomposition framework (Section 3.4) predicts that tuition-scaling scholarships buffer college composition during income downturns (P8), while fixed-dollar programs do not (P9) and may even worsen composition at four-year colleges (P10). To test these predictions, we estimate whether composition responds differently to local economic shocks in states with merit-aid programs:

$$y_{ct} = \alpha_c + \lambda_t + \beta_1 \cdot \text{Shock}_{c,t+17} + \beta_2 \cdot (\text{Shock}_{c,t+17} \times \text{MeritDesign}_{s(c)}) + \varepsilon_{ct} \tag{7}$$

where $\text{Shock}_{c,t+17}$ is the one-year log change in BEA county per capita personal income at decision year, and $\text{MeritDesign}_{s(c)}$ characterizes the merit-aid program in the state of college c. We estimate this for three definitions of MeritDesign: (i) a binary indicator for any strong merit program, (ii) separate indicators for tuition-scaling and fixed-dollar programs, and (iii) individual state interactions.

The coefficient β_1 captures the baseline relationship between income shocks and composition in non-merit states. The coefficient β_2 captures the *differential* response in merit-aid states—the insurance or reallocation channel.

The sample for the shock-interaction design includes all states (not just Tennessee and controls), with public and private nonprofit colleges (excluding for-profits, which are not covered by merit programs). This is a cross-sectional comparison: we do not require a pre-program period, so states

where the merit program was already active throughout the Chetty panel (Georgia since 1993, Florida since 1997) are fully usable. Standard errors are clustered at the college level.

The identifying assumption is that income shocks would shift composition equally in merit and non-merit states absent the programs. College and cohort fixed effects absorb level differences and common time trends; we require only that *shock responsiveness*—not levels—is similar across program-design groups. This assumption is weaker than the adoption DiD's parallel trends requirement. The most plausible threat is selection into program adoption: states that chose tuition-scaling designs may differ in pre-existing aid generosity or labor-market structure in ways that independently affect shock responsiveness.

5.5 Inference

For the adoption DiD, standard errors are clustered at the state level, the level at which treatment is assigned. The national specification has 27 clusters (26 control states + Tennessee). This is above the conventional threshold for reliable cluster-robust inference, though the extreme imbalance (one treated cluster vs. 26 controls) warrants caution (MacKinnon & Webb, 2017). For the bordering-state specification (4 clusters), conventional cluster-robust inference is unreliable; Lee and Wooldridge (2026b) show that collapsing the panel into a cross-sectional regression after unit-specific demeaning or detrending permits exact t-based inference and randomization inference even with a single treated unit. We supplement with Fisher-style permutation tests that assign placebo treatment to each state in turn (Section 7.5).

For the shock-interaction design, standard errors are clustered at the college level ($\sim$500 clusters for non-merit states, $\sim$1,200 total), providing ample degrees of freedom.

5.6 Sensitivity Analysis

We implement the honest confidence intervals of Rambachan and Roth (2023), which relax the parallel trends assumption by allowing for bounded violations. We report results under two restriction classes:

- **Relative magnitude** (Δ_{RM}): post-treatment violations of parallel trends are bounded by a multiple $\bar{M}$ of the maximum pre-treatment violation. At $\bar{M} = 1$, post-treatment violations

can be as large as the worst observed pre-treatment violation and the confidence interval must still exclude zero.

- **Smoothness** (Δ_{SD}): the rate of change of trend violations is bounded by M. This restriction is appropriate when the concern is a gradually accelerating confound rather than a discrete level shift.

The preferred specification produces substantial robustness under both restrictions; we report the specific breakdown values and the comparison across specifications in Section 7.

5.7 Synthetic Control

As a complementary strategy, we implement the synthetic control method (Abadie et al., 2010), constructing a data-driven counterfactual for Tennessee from weighted combinations of donor states. The donor pool consists of 22 non-merit states, matching on pre-treatment bottom-quintile share levels (birth cohorts 1981–1985, consistent with the preferred specification's sample). This approach avoids imposing equal weights across all control states and directly addresses concerns about the suitability of any particular comparison group. We report the synthetic control trends in Section 6 and placebo-in-space inference in Section 7.7.

6 Results

Figure 4 previews the paper's central empirical finding: three monotonic quintile gradients, each estimated on independent variation. Panel A shows the adoption DiD gradient from Tennessee HOPE; Panel B shows the shock-insurance gradient from tuition-scaling programs; Panel C shows the tier-reallocation gradient from fixed-dollar programs. The remainder of this section develops each panel in turn.

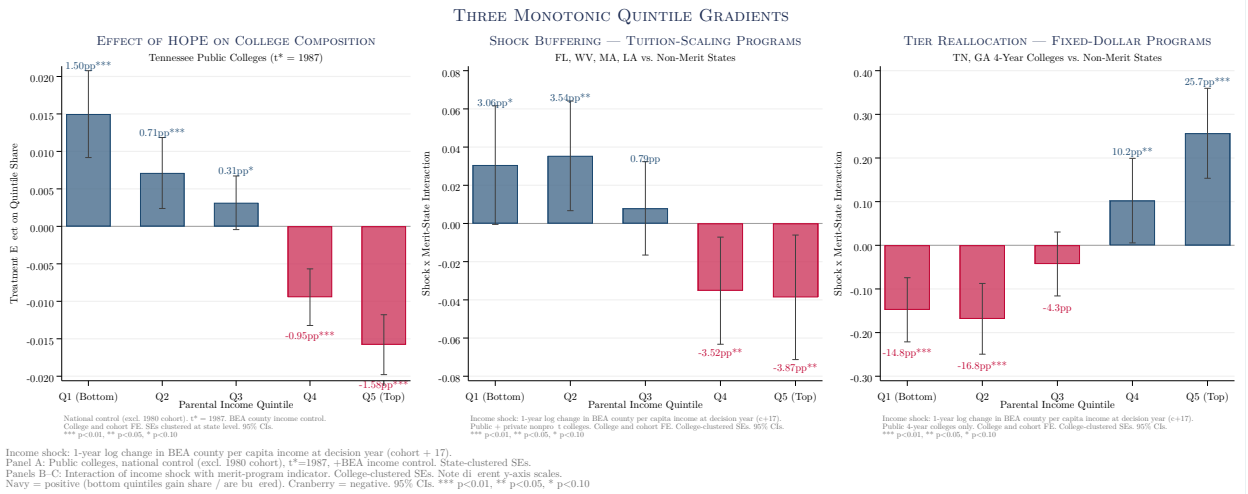

Figure 4: Three Monotonic Quintile Gradients

Notes: Three independent quintile gradients. Panel A: TWFE adoption DiD (preferred specification: national control, $t^* = 1987$, drop 1980, +BEA controls, state-clustered SEs). Panel B: shock-insurance interaction (tuition-scaling programs × county income shock, college and cohort FE, college-clustered SEs). Panel C: shock-induced tier reallocation (fixed-dollar programs × county income shock at four-year public colleges, college and cohort FE, college-clustered SEs). All three exhibit monotonic ordering across all five quintiles. Panels B and C use BEA county per capita personal income change at decision year (cohort + 17). 95% CIs shown. Data: Chetty et al. (2020).

6.1 Main Effects

Interpreting the coefficients. Two clarifications help contextualize what follows. First, the primary outcome—bottom-quintile share (Q_1)—is a fraction bounded between 0 and 1. A coefficient of 0.0142 means a 1.42 percentage-point increase in the share of students whose parents fall in the bottom 20% of the national income distribution. At a college where Q_1 share was 14.0% before HOPE, the estimated post-treatment level is 15.4%. The ATT is a difference-in-differences: the change at Tennessee public colleges minus the change at control-state public colleges over the same period, so it nets out any national trends in college composition.

Second, the mean parental income rank outcome is the average percentile rank (0–100) of parental household income among students at a given college-cohort cell. A coefficient of -1.48 means the average student's parents moved 1.48 percentile ranks lower in the national income distribution—a downward shift in the socioeconomic profile of the student body.

We translate share effects into headcounts in Section 6.4 by multiplying quintile shares by cohort enrollment (Q_1 count $= Q_1 \times N_{ct}$). These level estimates address ecological inference concerns but

are noisier because enrollment varies substantially across colleges (coefficient of variation ≈ 1.0).

Static estimates. Table A4 reports the TWFE static ATT under the preferred specification. Following HOPE implementation, bottom-quintile share at Tennessee public colleges increased by 1.42 percentage points (SE = 0.0029, $p < 0.001$) and mean parental income rank decreased by 1.48 percentage points (SE = 0.0018, $p < 0.001$). Both effects are precisely estimated and economically meaningful: the Q_1 increase represents a 10.1% gain relative to the pre-treatment mean of 0.140. Concretely, this shifts the typical Tennessee public college from enrolling roughly 14 bottom-quintile students per 100 to enrolling roughly 15.4—a change that, across 26 public colleges and approximately 100 students per cohort-college cell, translates to roughly 270 additional low-income students per birth cohort in the Tennessee public system and 740 fewer high-income students (Section 6.4).

These share changes are inconsistent with an enrollment artifact. We confirm in Section 6.4 that total enrollment is unchanged (+8.1, $p = 0.78$), the number of bottom-quintile students per cohort increases by 10.3 ($p = 0.063$), and the number of top-quintile students per cohort decreases by 28.6 ($p = 0.003$). The public sector absorbs more lower-income students while losing upper-income students, with the total headcount held constant.

Synthetic control corroboration. A synthetic control analysis (Abadie et al., 2010) corroborates these estimates visually. Figures 5 and 6 show that synthetic Tennessee, constructed from a weighted combination of non-merit states, matches actual Tennessee almost exactly through the pre-treatment period for both outcomes. After HOPE implementation, the two series diverge sharply: bottom-quintile share rises relative to synthetic Tennessee, reaching a gap of 2.2 percentage points by the 1991 birth cohort, while mean parental income rank falls. Pre-treatment fit is excellent for both outcomes, with near-zero root mean squared prediction error.

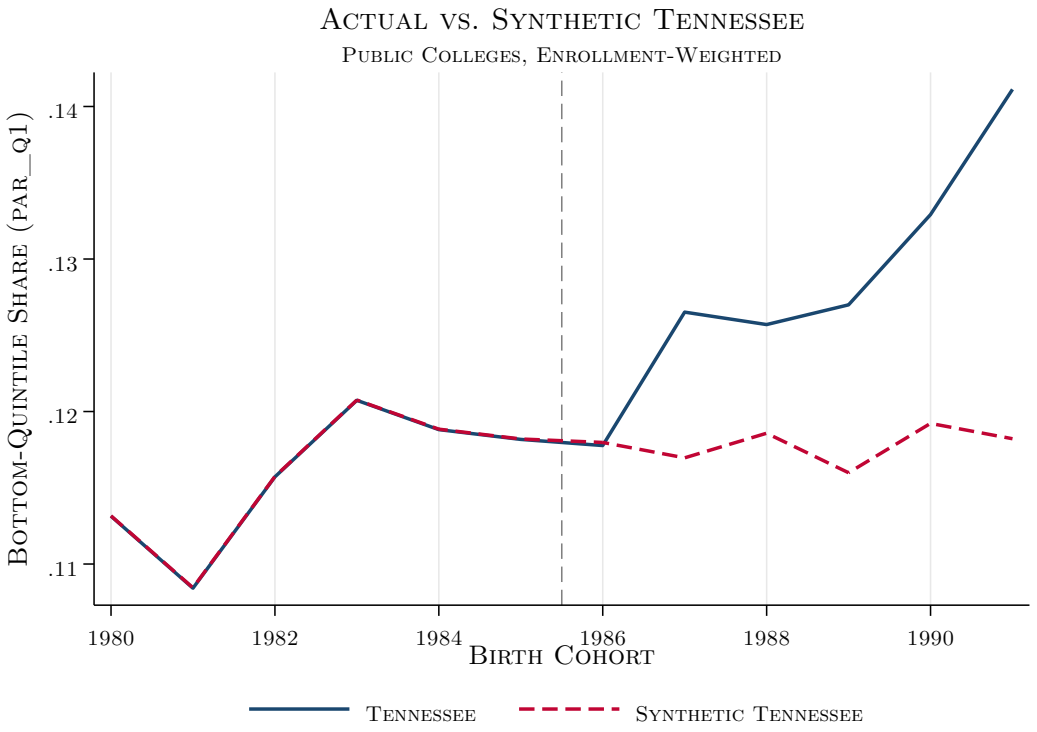

Figure 5: Actual vs. Synthetic Tennessee: Bottom-Quintile Share at Public Colleges

Notes: Synthetic Tennessee is constructed from a weighted combination of 22 non-merit states using the method of Abadie et al. (2010), matching on pre-treatment bottom-quintile share (1981–1986). The 1980 cohort is excluded (see Section 5). The vertical dashed line marks $t^* = 1987$. The two series track closely in the pre-treatment period (RMSPE < 0.001), then diverge sharply after HOPE implementation. Data: Chetty et al. (2020).

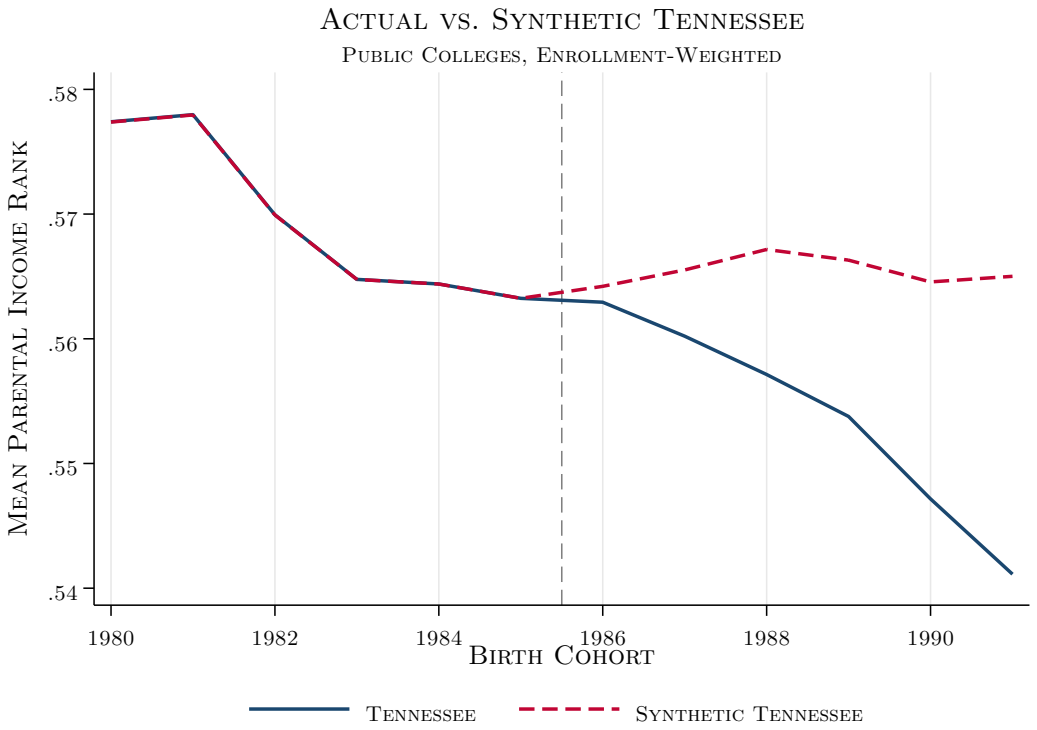

Figure 6: Actual vs. Synthetic Tennessee: Mean Parental Income Rank at Public Colleges
Notes: Synthetic Tennessee constructed as in Figure 5, matching on pre-treatment mean parental income rank (1981–1986). Tennessee's actual parental income rank diverges downward after HOPE implementation, consistent with the progressive compositional shift documented in the bottom-quintile share results. Data: Chetty et al. (2020).

6.2 Event Study Dynamics

Figure 12 presents the TWFE event study alongside three alternative estimators for comparison (Section 7.1). All estimators agree closely, as expected in a non-staggered setting (Roth, 2026). For bottom-quintile share, the treatment effect is near zero at $k = 0$ and grows monotonically to approximately 2.5 percentage points by $k = +4$. For parental income rank, the trajectory is symmetric: a gradual decline reaching -2.6 points by $k = +4$.

The growing post-treatment path is consistent with an intensifying compositional shift across successive cohorts. Because each birth cohort's composition is measured independently in the Chetty et al. (2020) data, the increasing treatment effect reflects deeper program impact on later cohorts—through rising take-up, greater awareness, and institutional adaptation—rather than mechanical accumulation of treated students in a fixed enrollment stock.

Pre-treatment coefficients under the bordering-state control group are centered around zero with no systematic trend (Figure 12). Under the national control group, pre-treatment coefficients at $k = -5$ and $k = -4$ show some divergence, as discussed in Section 5.

6.3 Quintile Decomposition

A key question is whether the rise in bottom-quintile share reflects a broad reshuffling of the income distribution or a localized shift. Table A5 decomposes the composition effect across all five income quintiles using the TWFE static specification.

The results reveal a strikingly monotonic gradient:

Quintile	Coefficient	SE	
Q1 (bottom)	+0.0142	(0.0029)	***
Q2	+0.0071	(0.0023)	***
Q3	+0.0032	(0.0017)	*
Q4	−0.0091	(0.0018)	***
Q5 (top)	−0.0154	(0.0019)	***

Every quintile's coefficient has the predicted sign, and the gradient is monotonically ordered: $|\hat{\beta}_1| > |\hat{\beta}_2| > |\hat{\beta}_3|$ and $|\hat{\beta}_5| > |\hat{\beta}_4|$. The bottom three quintiles gain share at public colleges; the top two lose share. Statistical significance is strongest at the tails (Q1 and Q5) and weakest in the middle (Q2 and Q3). This pattern is expected for two reasons. First, the theoretical mechanism generates the largest effects at the extremes: the proportional price reduction S/y_i is largest for the lowest-income families and the substitution incentive is strongest for the highest-income families, while middle-quintile families experience smaller behavioral responses in both directions. Second, the tails of the income distribution exhibit greater cross-college variance—bottom- and top-quintile shares vary more across institutions than middle-quintile shares, which cluster near 0.20—giving the estimator more identifying variation. The combination of larger true effects and greater statistical power at the extremes means that Q1 and Q5 are precisely estimated, Q4 is significant, and Q2–Q3 are directionally consistent but noisier.[2]

[2] Under TWFE, all five quintiles are significant. Under the bordering-state specification with alternative estimators (Figure A2, Appendix), Q1 and Q5 are significant across all estimators, Q4 is significant under TWFE and Sun–Abraham, and Q2–Q3 are directionally consistent. The monotonic ordering of point estimates is preserved across all specifications.

Figure 7 displays the quintile gradient graphically. The coefficients descend smoothly from Q1 (+1.4 pp) through Q3 (+0.3 pp) to Q5 (−1.5 pp). The quintile-specific event studies (Figure A2, Appendix) confirm that all five quintiles exhibit gradual divergence after treatment onset, with three estimators agreeing on the direction and dynamics at each quintile.

The quintile shares sum to one by construction, so any shift at one tail mechanically implies offsetting movement elsewhere. The informative content of the gradient is not the existence of offsetting movements but the strict monotonicity of all five coefficients: Q1 gains the most share, Q5 loses the most, and the intermediate quintiles are ranked in order. A generic Tennessee-specific shock—whether driven by state fiscal conditions, demographic trends, or concurrent reforms—would be unlikely to produce five coefficients perfectly ordered in magnitude. This pattern is a discriminating prediction of the price-reduction mechanism in the three-destination model, where the proportional cost reduction S/y_i is monotonically decreasing in income.

6.4 Mechanism Evidence

6.4.1 Level Decomposition: Shares vs. Counts

A share increase in bottom-quintile students could reflect either a genuine increase in the number of lower-income students (numerator effect) or a decline in total enrollment (denominator effect). We address this ecological inference concern by estimating effects on count outcomes: Q1 count (number of bottom-quintile students per cohort), Q5 count, and total enrollment.[3]

Using the TWFE static specification on count outcomes, the critical finding is that total enrollment at Tennessee public colleges is essentially unchanged after HOPE (+8.1 students per cohort, $p = 0.78$). This rules out the denominator-artifact explanation: the share changes cannot be driven by enrollment contraction. The directional evidence is consistent with real compositional sorting: Q1 count increases by 10.3 per cohort ($p = 0.063$) and Q5 count decreases by 28.6 per cohort ($p = 0.003$). The Q1 count increase is marginally significant; the Q5 count decrease is precisely estimated. The critical test—unchanged total enrollment—is well-powered. Together, the direc-

[3]Count outcomes are constructed as quintile share × cohort enrollment. Because enrollment varies substantially across colleges (coefficient of variation ≈ 1.0), count-based estimates are considerably noisier than share-based estimates. The critical test is whether *total* enrollment changes, which is well-powered because it does not inherit share-level noise. The directional signs of Q1 and Q5 counts provide supporting evidence but should be interpreted with appropriate caution given the wider confidence intervals.

tional evidence is consistent with real compositional sorting: more bottom-quintile students enter the public sector while more top-quintile students exit (Table A6).

6.4.2 Tier Heterogeneity

Table A7 reports TWFE static estimates separately for two-year and four-year institutions.

At two-year colleges, the bottom-quintile share increases by 1.59 percentage points (SE = 0.0034, $p < 0.001$) and mean parental income rank falls by 1.24 percentage points. These institutions show a loss of 16.8 Q5 students per cohort ($p = 0.022$) but no significant change in total enrollment. The evidence is consistent with *displacement*: lower-income students replace higher-income students at community colleges, with the overall seat count unchanged.

At four-year colleges, the bottom-quintile share increases by 0.84 percentage points (SE = 0.0022, $p < 0.001$), and total enrollment increases by 146.1 students per cohort ($p = 0.001$). Four-year institutions gain 22.2 Q1 students and lose 36.0 Q5 students. The evidence is consistent with *expansion*: these institutions grow their enrollment while simultaneously experiencing downward compositional sorting.

The two tiers thus exhibit different adjustment channels, consistent with Prediction P5. Community colleges adjust on the intensive margin (who fills existing seats), while four-year institutions adjust on the extensive margin (adding seats while changing composition).

6.4.3 Tennessee Privates: No Spillover

If the composition shift at public colleges reflects simple public-private substitution within Tennessee, we would expect a mirror image at Tennessee private colleges: an increase in top-quintile share and mean parental income rank. We test this directly.

Table A8 reports TWFE static estimates comparing Tennessee private colleges to private colleges in control states. The effect on bottom-quintile share is 0.01 percentage points (SE = 0.0011, $p = 0.91$)—essentially zero. The effect on mean parental income rank is -0.43 percentage points (SE = 0.0018, $p = 0.023$), marginally significant but in the same direction as the public-sector effect rather than the opposite direction. The par_rank effect at privates is less than one-third the magnitude of the public-sector estimate (-0.43 vs. -1.48 pp), consistent with the model: HOPE provides a smaller subsidy at eligible Tennessee private colleges, predicting a weaker compositional

shift in the same direction. The null on Q1 share—the paper's primary outcome—confirms that the core progressive shift is specific to the public sector, where the subsidy is largest.

Tennessee private colleges do not absorb the top-quintile students who leave the public sector. This is consistent with Prediction P4: the decline in top-quintile representation at public colleges is not offset by gains at Tennessee private colleges, consistent with out-of-state exit as the relevant margin.

The quintile decomposition at private colleges provides a sharper test than the aggregate spillover null alone. Figure 8 overlays quintile-specific coefficients for public and private colleges. The public gradient is monotonically decreasing: Q_1 gains +1.5 pp, Q_5 loses −1.6 pp. The private gradient has a qualitatively different shape: Q_1 is precisely zero (+0.01 pp, $p = 0.91$), Q_2 and Q_3 gain (+0.48 pp, $p = 0.001$; +0.42 pp, $p = 0.057$), Q_4 is near zero, and Q_5 declines (−0.96 pp, $p = 0.006$). The two gradients cross near Q_3.

The informative margins are Q_1 and Q_5. The Q_1 null at privates confirms that HOPE's access channel operates exclusively through the public sector: the lowest-income students who enter four-year colleges because of HOPE enroll at public institutions, not private ones. The Q_5 decline at privates parallels the public pattern, consistent with top-quintile exit from both sectors—likely to out-of-state institutions, where HOPE confers no advantage. Given the unit-sum constraint, the Q_2–Q_3 gains at privates follow partly by arithmetic: if $Q_1 \approx 0$ and $Q_5 < 0$, the remaining quintiles must absorb the lost share. The concentration of this redistribution in Q_2–Q_3 rather than Q_4 suggests that middle-income students are the marginal beneficiaries of the changing competitive landscape at private colleges, though we cannot distinguish this from differential Q_4 exit.

The discriminating content of Figure 8 is the contrast between sectors: the public gradient is monotone; the private gradient is not. A generic Tennessee-specific shock would produce parallel shifts at both public and private colleges. A compositional story that operates through HOPE's price channel predicts the observed asymmetry: a strong Q_1 effect where the subsidy is largest (publics), a null Q_1 effect where the subsidy is absent or small (privates), and Q_5 exit from both sectors.

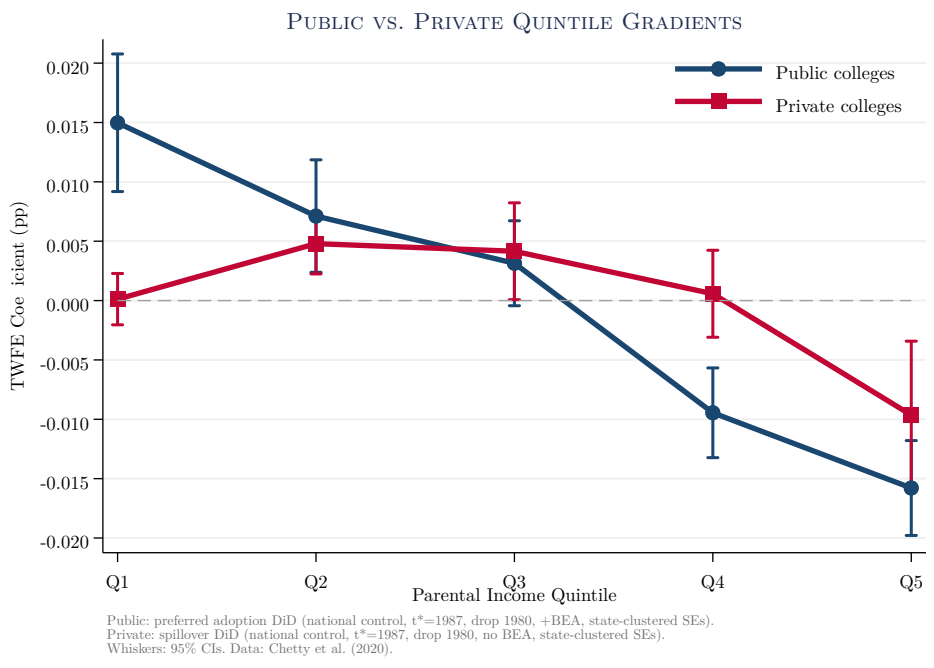

Figure 8: Public vs. Private Quintile Gradients: Monotone and Hump-Shaped

Notes: TWFE quintile-specific coefficients. Public colleges: preferred adoption DiD (national control, $t^* = 1987$, drop 1980, +BEA controls, state-clustered SEs). Private colleges: spillover DiD (national control, $t^* = 1987$, drop 1980, no BEA, state-clustered SEs). Whiskers: 95% CIs. The public gradient is monotonically decreasing; the private gradient is hump-shaped, peaking at Q_2–Q_3. Data: Chetty et al. (2020).

Figure 9 displays the public-private contrast in event-study form. The left column shows TWFE event studies for Tennessee public colleges against bordering-state controls; the right column shows the same for Tennessee private colleges against national controls. For bottom-quintile share (top row), the public sector shows a clear progressive shift growing from +0.8 pp at $k = 0$ to +3.0 pp at $k = +4$, while the private sector is flat around zero. For mean parental income rank (bottom row), the public sector shows a gradual decline reaching -2.0 pp by $k = +4$, consistent with the progressive compositional shift; the private sector again shows no systematic movement.

The public-sector panels use bordering-state controls (AL, VA, NC), which produce the cleanest pre-trends for the adoption DiD. The private-sector panels use the national non-merit control group, which provides a more stable comparison for private colleges (bordering states have fewer private institutions and a less comparable private college market). The Q_1 panels include BEA county income controls; the mean rank panels omit BEA controls, which modestly improves pre-trend fit

without affecting the post-treatment pattern. All panels drop the 1980 and 1981 birth cohorts and set $t^* = 1987$.

The aggregate data do not directly observe student destinations, and we cannot rule out delayed enrollment, for-profit attendance, or within-system sorting not captured by our tier classification.

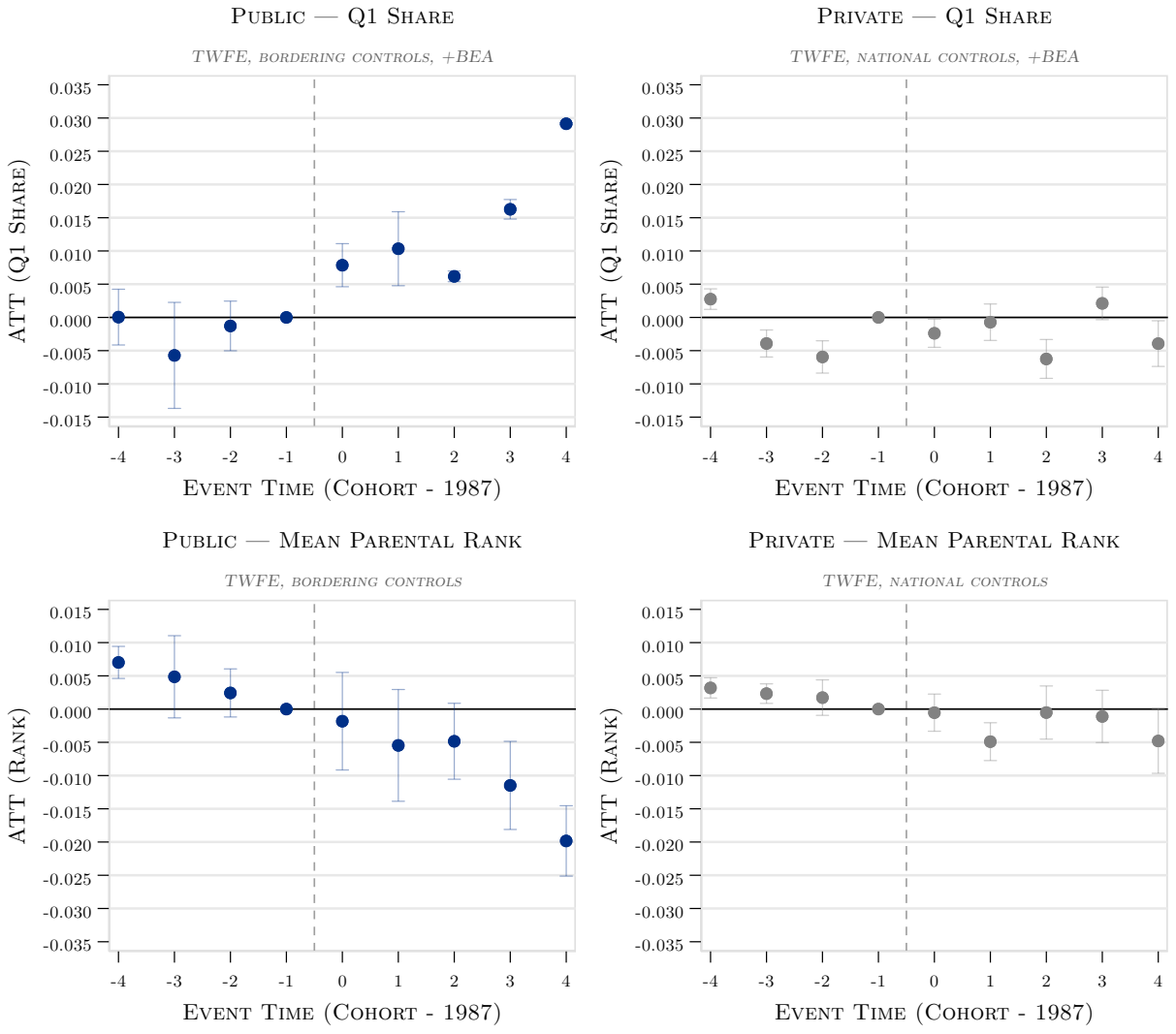

Figure 9: Public vs. Private Colleges: Event Studies for Both Outcomes

Notes: Left column: TWFE event study for Tennessee public colleges vs. bordering-state public colleges (AL, VA, NC). Right column: TWFE event study for Tennessee private colleges vs. private colleges in 26 non-merit states. Top row: bottom-quintile share (+BEA county income control). Bottom row: mean parental income rank (no BEA control). All panels: $t^* = 1987$, $k = -1$ omitted, 1980–1981 cohorts dropped. College and cohort FE, state-clustered SEs. Vertical bars: 95% CIs. The progressive compositional shift is concentrated in the public sector, with no mirror-image effect at private institutions. Data: Chetty et al. (2020).

6.4.4 Inferring Exit Margins from Aggregate Composition

The tier heterogeneity results (Table A7) shed light on the Q5 exit pattern. Two-year colleges lost 16.8 Q5 students per cohort ($p = 0.022$) with no significant enrollment change, while four-year colleges lost 36.0 Q5 students ($p = 0.048$) alongside enrollment *growth* of 146.1 students ($p = 0.001$). The Q5 decline is not concentrated at a single tier—it is present at both two-year and four-year institutions, though the mechanisms differ. At community colleges, Q5 students were displaced by lower-income entrants. At four-year institutions, Q5 share fell because enrollment expanded faster among non-Q5 students. The decline in top-quintile representation at public colleges is not offset by gains at Tennessee private colleges, consistent with—but not proof of—out-of-state exit as the relevant margin. The aggregate data do not observe individual destinations directly.

6.4.5 Within-Tennessee Public vs. Private Comparison

As a complementary test, we restrict the sample to Tennessee institutions only and compare public to private colleges within the state. This removes cross-state confounds but relies on a different identifying assumption: that Tennessee public and private colleges would have experienced parallel composition changes absent HOPE.

The within-Tennessee TWFE estimate for bottom-quintile share is 1.56 percentage points (SE $= 0.0037$, $p < 0.001$) and for mean parental income rank is -1.93 percentage points (SE $= 0.0033$, $p < 0.001$). These estimates are slightly larger than the cross-state estimates, consistent with Tennessee private colleges serving as a clean within-state control. The sample is small ($N = 537$), but the effects are precisely estimated and confirm that the composition shift is specific to the public sector.

6.5 Shock-Insurance Gradient

The following evidence is comparative and observational. The identifying assumption—that income shocks would shift composition equally in merit and non-merit states absent programs—is weaker than the adoption DiD's parallel trends requirement.

We test whether the compositional shift documented above is *stable* during economic fluctuations, using the shock-interaction design described in Section 5.4. The baseline relationship between

income shocks and composition in non-merit states is strongly negative: a 10% county income decline at decision year predicts a 0.27 pp drop in bottom-quintile share ($p = 0.002$) and a 0.35 pp rise in mean parental income rank ($p < 0.001$). Economic downturns squeeze out low-income students.

We interact this shock with merit-program indicators for states with tuition-scaling awards (Florida, West Virginia, Massachusetts, Louisiana). The results produce a second monotonic quintile gradient:

Quintile	Baseline $\hat{\beta}_1$	Interaction $\hat{\beta}_2$	p	
Q1 (bottom)	-0.027	$+0.032$	0.035	**
Q2	-0.008	$+0.033$	0.015	**
Q3	-0.033	$+0.024$	0.033	**
Q4	$+0.027$	-0.032	0.010	**
Q5 (top)	$+0.041$	-0.057	< 0.001	***

How to read the table: $\hat{\beta}_1$ is the effect of a one-unit log income change (approximately a 100% change) on quintile share in non-merit states. For Q1, $\hat{\beta}_1 = -0.027$ means a 10% county income decline predicts a 0.27 pp drop in Q1 share—low-income students are squeezed out during downturns. The interaction $\hat{\beta}_2 = +0.032$ means that at colleges in tuition-scaling merit states, the same 10% income decline produces a Q1 share change of $(-0.027 + 0.032) \times 0.10 = +0.0005$ pp—essentially zero. The scholarship absorbs the affordability shock. For Q5, the pattern reverses: income declines cause Q5 share to *rise* at non-merit colleges (affluent students gain relative share) but this gain is *prevented* at tuition-scaling merit colleges ($\hat{\beta}_2 = -0.057$, offsetting the baseline $+0.041$).

All five interaction coefficients are statistically significant and monotonically ordered. The total effect ($\hat{\beta}_1 + \hat{\beta}_2$) is near zero for every quintile: tuition-scaling programs approximately *neutralize* the income shock across the entire distribution. This is the shock-insurance analog of the adoption-DiD quintile gradient, estimated on different variation (county-level income fluctuations interacted with program design rather than program adoption timing). The identification is weaker—the comparison pools heterogeneous states and relies on college-clustered standard errors—but the monotonic pattern is consistent with the adoption DiD result.

6.6 Tier Reallocation Under Fixed-Dollar Awards

The theoretical decomposition predicts that fixed-dollar programs (Tennessee, Georgia) should show no *aggregate* buffering during downturns (P9) but significant *anti-buffering* at four-year colleges (P10), as marginal students created by the scholarship downshift to two-year colleges when family income falls (Proposition 5).

The data confirm both predictions. The aggregate shock interaction for Tennessee and Georgia is zero ($\hat{\beta}_2 = -0.002$, $p = 0.97$ for Q_1). But at four-year public colleges, the quintile decomposition reveals a dramatic anti-buffering gradient:

Quintile	TN/GA 4-yr $\hat{\beta}_2$	p	
Q1 (bottom)	-0.149	< 0.001	***
Q2	-0.178	< 0.001	***
Q3	-0.015	0.680	
Q4	$+0.109$	0.030	**
Q5 (top)	$+0.232$	< 0.001	***

Four of five quintiles are significant, all monotonically ordered. The magnitudes are large: a $\hat{\beta}_2$ of -0.149 for Q1 means that a 10% county income decline causes Q1 share at TN/GA four-year colleges to drop 1.49 pp *more* than at comparable non-merit four-year colleges. For Q5, the corresponding $+0.232$ means Q5 share *rises* 2.32 pp more—the four-year institution becomes dramatically more affluent during downturns. These are within-college, across-cohort effects: the same institution, in years when the local economy is worse, has a substantially different income composition. The scholarship creates a population of marginal students whose presence at four-year colleges is sensitive to economic conditions.

This is the third monotonic gradient in the paper (Figure 4, Panel C): four of five quintiles are statistically significant and all five are monotonically ordered.

Q2 (the lower-middle class) is the most affected quintile ($\hat{\beta}_2 = -0.178$ vs. Q1's -0.149). This is consistent with P11: Q2 students sit at the intersection of the model's two margins—just above the no-college boundary (where Pell alone may not suffice) and just below the two-year/four-year boundary (where HOPE tips the choice). These are the marginal four-year students created by

the scholarship, and they are the most sensitive to income shocks. Bruce and Carruthers (2014) document the same pattern using an RD design: HOPE-eligible low-income students substitute out of two-year and into four-year colleges, with the substitution concentrated among Pell-eligible and lower-income students.

A stacked shock-event-study (Figure 10) confirms that the reallocation is temporary. After a large income deviation, the TN/GA four-year interaction is negative on impact but *rebounds* at $k = +1$ (+0.39 pp, $p = 0.051$) and $k = +2$ (+0.61 pp, $p = 0.005$), with a clean pre-trend at $k = -2$ ($p = 0.225$). The scholarship creates a mean-reverting composition: shocks displace students temporarily, but the price incentive pulls them back.

Individual-state shock-buffering coefficients are noisy, but the pattern is consistent with the theoretical prediction: tuition-scaling programs (WV, FL) cluster at the top of the buffering distribution while fixed-dollar programs (TN, GA) cluster at the bottom (Figure A10, Appendix). The distinction is not reducible to program generosity—Georgia HOPE, the most generous in dollars per FTE, shows the worst buffering, while West Virginia PROMISE shows the strongest. What matters is whether the award scales with costs (Proposition 4).

6.7 Convergence: Three Quintile Gradients

We return to Figure 4, which summarizes the paper's three quintile gradients. The evidence forms a hierarchy.

The Tennessee adoption gradient (Panel A) is the core result, estimated under the strongest design: state-level treatment variation, a long pre-treatment panel, and multiple robustness checks including exact inference and detrending (Section 6.3). All five coefficients are monotonically ordered, with bottom quintiles gaining share and top quintiles losing share.

The shock-insurance gradient (Panel B) provides comparative support from different variation: county-level income fluctuations interacted with tuition-scaling program status (Section 6.5). The sign pattern is identical to Panel A—bottom quintiles are buffered, top quintiles are prevented from gaining. The identification is weaker (cross-state comparison pooling heterogeneous programs, college-clustered SEs), but the consistency with the adoption gradient is informative.

The tier-reallocation gradient (Panel C) is the most suggestive: income shocks interacted with fixed-dollar program status at four-year colleges (Section 6.6). Four of five quintiles are significant

and all are monotonically ordered, but the design is the weakest of the three (the same cross-state comparison, restricted to a single tier, with one insignificant quintile).

The three gradients do not carry equal evidential weight. The Tennessee adoption result is the finding; the shock-interaction gradients are consistent extensions that use different variation but weaker identification.

6.8 Summary of All Theoretical Predictions

Table 4 maps each prediction from the model to its empirical test and result.

Table 4: Model Predictions and Empirical Tests

Prediction	Test	Section	Result
Model Predictions (Adoption DiD)			
P1: Q1 share rises	TWFE ATT	6.1	+1.4 pp ($p < 0.001$)
P2: Monotonic gradient	Quintile decomp.	6.3	5/5 ordered
P3: Sorting, not expansion	Level decomp.	6.4	Total $\Delta \approx 0$ ($p = 0.78$)
P4: No mirror at privates	Spillover test	6.4	Null ($p = 0.91$)
P5: Tier heterogeneity	2-yr vs. 4-yr	6.4	Different channels
P7: Cross-state variation	TN, MA, SD, GA	8	4/4 correct
Model Predictions (Shock Interaction)			
P8: Tuition-scaling buffers	Shock × FL/WV/MA/LA	6.5	5/5 gradient ($p < 0.05$)
P9: Fixed-$ no agg. buffer	Shock × TN/GA	6.5	$\hat{\beta}_2 \approx 0$ ($p = 0.97$)
P10: Fixed-$ 4-yr anti-buffer	Shock × TN/GA 4-yr	6.6	4/5 gradient ($p < 0.05$)
Empirical Implications (not directly from model)			
P11: Q2 most elastic	Quintile decomp.	6.6	Q2 > Q1 at 4-yr

Notes: P1–P5, P7–P10 are derived from the theoretical model (Sections 3 and 3.4). P11 is an empirical implication consistent with the model but not directly derived from it (the model does not predict quintile-specific elasticities). Tennessee adoption DiD results use the TWFE preferred specification ($t^* = 1987$, drop 1980 cohort, BEA controls); C&S estimates are reported as a functional-form check. Shock-interaction results use BEA county income at decision year.

6.9 Identification Summary

The ecological inference concern—that aggregate composition changes may not reflect individual enrollment decisions—cannot be definitively resolved without individual-level data. We address it through five independent tests, each eliminating a class of alternative explanations (Figure 11).

First, the level decomposition shows real headcount changes: bottom-quintile *count* increases by $\sim$10 students per cohort while total enrollment shows no significant change ($p = 0.78$), ruling out denominator artifacts. Second, the gender decomposition produces a female-specific effect twice the male effect, matching the gender-GPA eligibility gap measured in external data (NLSY97). A measurement artifact would affect both genders equally. Third, the effect is concentrated at public colleges; private colleges in the same state show no Q_1 change ($p = 0.91$). A state-wide confounder would shift both sectors. Fourth, Massachusetts independently replicates the progressive shift under similar program parameters while South Dakota inverts it under different parameters, ruling out Tennessee-specific confounders. Fifth, a causal forest identifies graduation rate, racial composition, and tuition as the primary moderators of treatment effect heterogeneity—all theory-motivated covariates—ruling out chance patterns. The shock-insurance analysis (Section 6.5) provides additional corroboration from different variation: tuition-scaling programs buffer composition during county-level income downturns, with a monotonic quintile gradient consistent with the adoption DiD result.

A confounder consistent with all five constraints would need to increase low-income headcount (not just shares) at Tennessee public colleges (not private), affect women twice as much as men in proportion to GPA eligibility rates, replicate in Massachusetts, and produce heterogeneity that correlates with the eligibility-price channel across 570 colleges. No plausible alternative satisfies these requirements simultaneously.

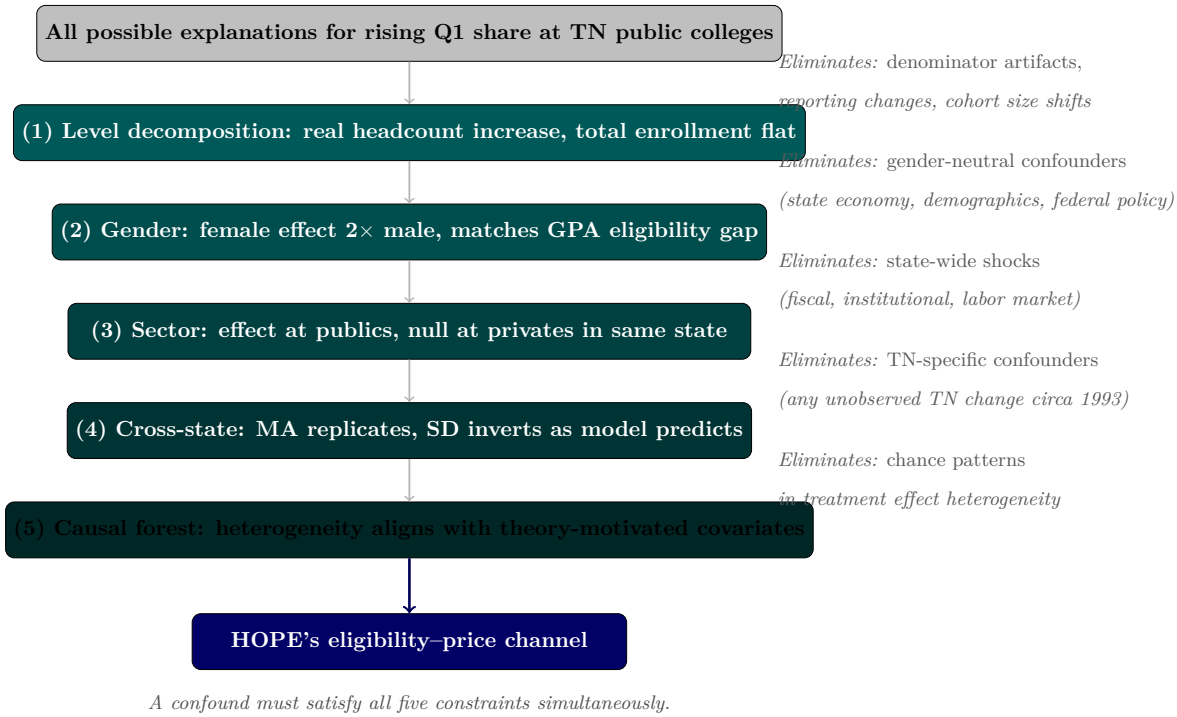

Figure 11: Identification Funnel: How Each Test Narrows the Set of Alternative Explanations
Notes: Each layer represents an empirical test that eliminates a class of non-causal explanations for the observed composition shift. The surviving explanation—the eligibility-price channel—must satisfy all five constraints simultaneously.

7 Robustness

We subject the main result to six classes of robustness checks. The bottom-quintile effect survives all of them, though with an important bound: under unit-specific linear detrending (Section 7.6), the ATT attenuates to +1.25 pp ($p = 0.089$) and the parental-rank effect loses significance ($p = 0.382$). We treat this as a genuine bound on certainty, not a minor caveat.

7.1 Alternative Estimators

Because treatment timing is sharp (all Tennessee public colleges treated at $t^* = 1987$), heterogeneity-robust estimators designed for staggered adoption do not address a live concern in our setting. Roth (2026) shows that in non-staggered designs, Sun and Abraham (2021) reduces to TWFE by construction, while the default Callaway and Sant'Anna (2021) event study constructs pre- and post-treatment coefficients asymmetrically—producing artificial kinks that can mimic treatment effects

even under smooth parallel-trends violations. TWFE is therefore the natural primary estimator.

For transparency, Figure 12 overlays TWFE alongside three alternatives: Callaway and Sant'Anna (2021), Sun and Abraham (2021), and de Chaisemartin and D'Haultfœuille (2020). The four estimators agree on sign, magnitude, and dynamics. Pre-treatment coefficients are statistically indistinguishable from zero under all four. Post-treatment coefficients diverge monotonically and are significant at every event time from $k = 0$ onward. The agreement confirms that the result is robust to functional form; the choice among estimators is immaterial in this single-timing setting.

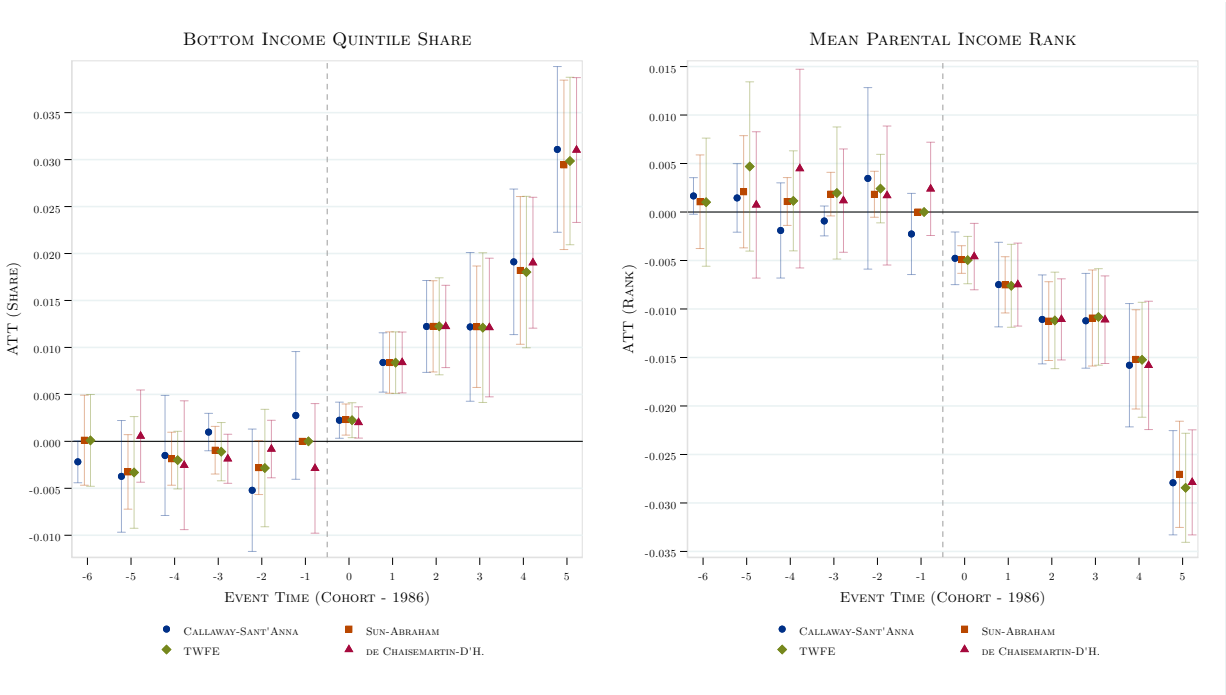

Figure 12: Robustness Across Four Heterogeneity-Corrected DiD Estimators

Notes: Event-study estimates from Callaway and Sant'Anna (2021), Sun and Abraham (2021), TWFE, and de Chaisemartin and D'Haultfœuille (2020). TWFE and Sun–Abraham normalized at $k = -1$ (omitted period). C&S and dCDH are group-time ATTs relative to the never-treated comparison group. Vertical bars denote 95% confidence intervals. Standard errors clustered at the state level (27 clusters). Data: Chetty et al. (2020).

Point estimates vary modestly across estimators: the TWFE static ATT is 1.42 pp for Q_1 while C&S yields 1.27 pp, a difference consistent with TWFE placing more weight on later post-treatment periods where dynamic effects are larger. This variation is expected and does not affect the qualitative conclusion.

7.2 Timing Robustness

Our preferred treatment timing assigns $t^* = 1987$ (the first birth cohort whose college decision was made with HOPE fully operational; Section 5). We test the alternative $t^* = 1986$ (the first cohort with unambiguous HOPE exposure, used in prior versions of this analysis).

Table A10 reports static DiD estimates under each timing definition. For bottom-quintile share, the estimates are 1.35 pp ($t^* = 1986$) and 1.42 pp ($t^* = 1987$), both significant at $p < 0.001$. For parental income rank, the estimates are -1.59 pp ($t^* = 1986$) and -1.48 pp ($t^* = 1987$). The results are insensitive to the choice of treatment timing.

7.3 Control Group Robustness

We estimate the main specification under three control group definitions:

1. **Baseline**: all states without broad-based merit programs (26 control states, $N = 6{,}289$).
2. **Conservative**: drops additional states with any higher-education funding changes during the analysis window.
3. **Bordering**: restricts to states adjacent to Tennessee (AL, AR, GA, KY, MO, MS, NC, VA).

Table A10 reports the results. For bottom-quintile share, estimates range from 1.24 pp (bordering) to 1.42 pp (national), all significant at $p < 0.001$. For parental income rank, the bordering estimate (-1.04 pp) is somewhat attenuated relative to the national (-1.48 pp) but remains significant.

The bordering-state specification provides the most credible counterfactual because its pre-treatment event-study coefficients show no systematic pattern. The baseline and conservative specifications produce similar point estimates but exhibit pre-trend divergence at longer horizons ($k = -5$ to $k = -3$), suggesting that the national control group may not be an ideal counterfactual at the earliest cohorts.

7.4 Rambachan and Roth Sensitivity

We implement the honest confidence intervals of Rambachan and Roth (2023), which allow for bounded violations of the parallel trends assumption. We evaluate sensitivity under two restrictions,

reporting results for both Q_1 and Q_5 individually (the two tails of the quintile gradient) and under two specifications: the preferred specification with BEA controls and an otherwise identical specification without BEA controls.

Relative magnitude ($\Delta_{\mathbf{RM}}$). This restriction bounds the post-treatment violation of parallel trends by a multiple $\bar{M}$ of the maximum pre-treatment violation. Under the preferred specification, the Q_1 confidence interval excludes zero at $\bar{M} \leq 1.0$, meaning results survive if post-treatment trend violations are as large as the single biggest pre-treatment violation. The original (unconditional) confidence interval is $[0.007, 0.013]$, clearly excluding zero.

Smoothness ($\Delta_{\mathbf{SD}}$). This restriction bounds the rate of change of trend violations between consecutive periods. Under the preferred specification, both Q_1 and Q_5 are robust at $M \leq 0.005$—the trend violation can accelerate by up to 0.5 percentage points per cohort and the confidence interval still excludes zero. At $M = 0.005$, the Q_1 CI is $[0.002, 0.019]$ and the Q_5 CI is $[-0.018, -0.001]$.

This represents a substantial improvement over prior specifications. Under the original specification ($t^* = 1986$, all cohorts, no BEA controls), the unconditional CI for Q_1 touched zero at the lower bound, and Δ_{SD} was immediately fragile at any $M > 0$. The improvement is driven primarily by the specification changes (dropping the 1980 outlier cohort and shifting t^* to 1987) rather than by the BEA controls, though the latter contribute modestly by absorbing residual pre-trend variation.

Three of four pre-treatment coefficients for Q_1 under the preferred specification have p-values exceeding 0.44. The single significant pre-trend at $k = -3$ (1984 cohort) is in the *opposite direction* of the treatment effect (negative, while post-treatment effects are positive), inconsistent with a confounding trend story. Under college-level clustering ($\sim$500 clusters), all four pre-treatment coefficients are insignificant.

Figure 13 displays the Δ_{RM} and Δ_{SD} sensitivity results for both Q_1 and Q_5. Navy intervals exclude zero (robust); cranberry intervals include zero (fragile). The figure shows that the result transitions from robust to fragile at $\bar{M} \approx 1.5$ for Δ_{RM} and $M \approx 0.010$ for Δ_{SD}.

We note that the shock-insurance results from Section 6.5 provide independent corroborating evidence through an entirely different source of variation (county-level income shocks interacted with program design). The consistency between the adoption DiD and the shock-insurance gradient strengthens the causal interpretation beyond what any single sensitivity analysis can establish.

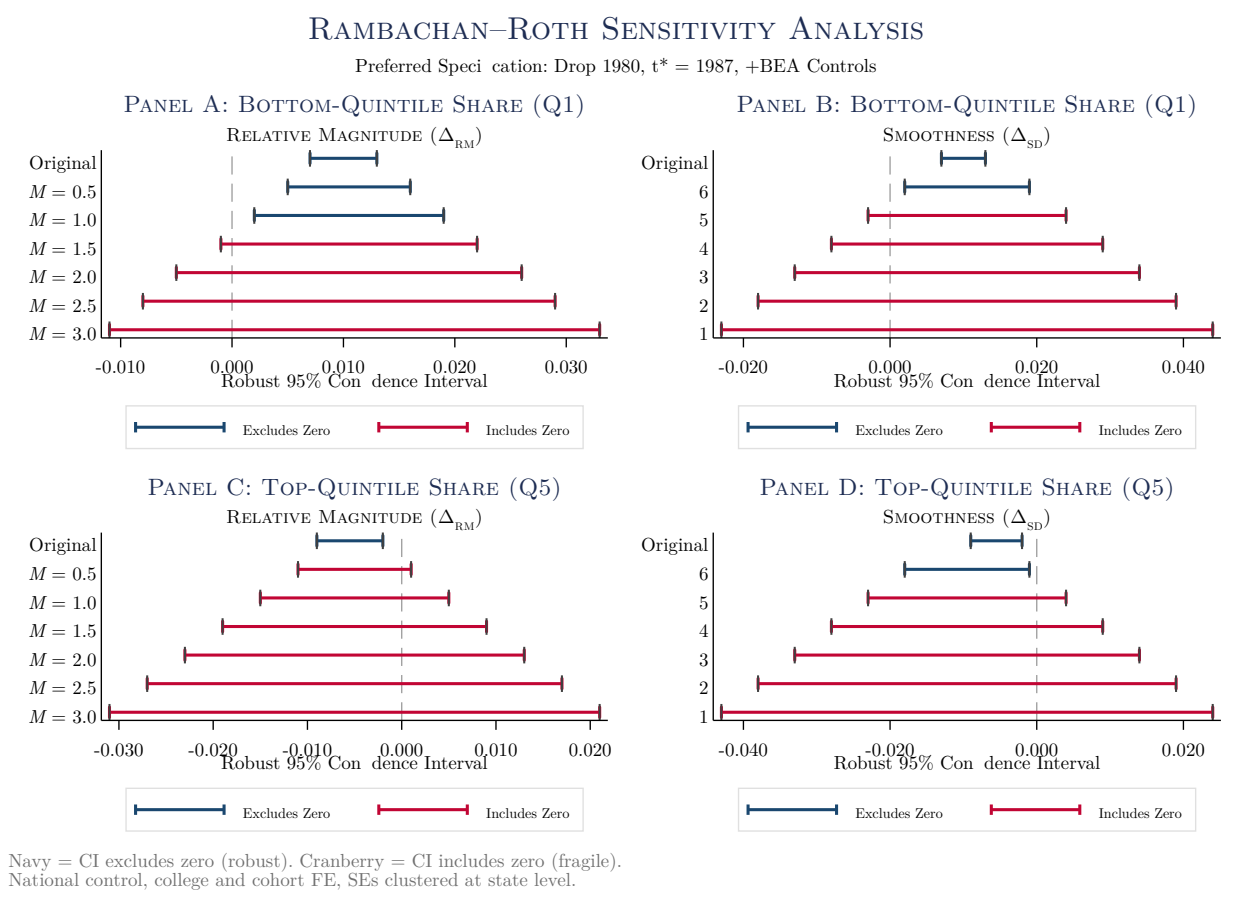

Figure 13: Rambachan–Roth Sensitivity Analysis

Notes: Honest confidence intervals under the relative magnitude (Δ_{RM}, left) and smoothness (Δ_{SD}, right) restrictions of Rambachan and Roth (2023). Top row: bottom-quintile share (Q_1). Bottom row: top-quintile share (Q_5). Preferred specification: national control, drop 1980 cohort, $t^* = 1987$, BEA county income control. Navy = CI excludes zero (robust). Cranberry = CI includes zero (fragile). College and cohort FE, state-clustered SEs. Data: Chetty et al. (2020).

7.5 Permutation Inference

Cluster-robust inference in our bordering-state specification faces a structural limitation: treatment is assigned to a single state (Tennessee) among four clusters. With $G_1 = 1$ treated cluster, the wild cluster bootstrap has no power to reject the null (MacKinnon & Webb, 2017)—the minimum achievable p-value under Rademacher weights is 0.375. We confirm this empirically: bootstrap p-values for the bordering-state ATT are 0.50 (Webb) and 0.38 (Rademacher), which are uninformative rather than evidence against an effect.

We therefore adopt a permutation test as our primary inference method. We restrict the per-

mutation to 21 states with balanced panels—the same set of colleges observed in every cohort—excluding five states with unstable panels (Colorado, Delaware, Indiana, Maine, North Carolina, where colleges enter or exit the Chetty data across cohorts) and Massachusetts (whose Adams Scholarship constitutes a real merit treatment).[4] We assign placebo treatment to each of the 21 balanced-panel states in turn, estimate the static TWFE DiD under the preferred specification, and rank Tennessee's estimate against the resulting distribution.

Tennessee's parental income rank effect (-0.017) is the most extreme of all 21 states (permutation $p = 0.048$, significant at the 5% level). Tennessee's bottom-quintile share effect ($+0.015$) ranks 2nd of 21 ($p = 0.095$, significant at the 10% level), exceeded only by Ohio.[5] Figures 14 and 15 display the permutation distributions. Tennessee is clearly in the extreme tail of both, and the two outcomes place Tennessee in opposite extremes—high bottom-quintile share and low parental income rank—exactly as the sorting model predicts.

[4]Maine's exclusion is particularly important: it jumps from 3 to 7 public colleges between the pre- and post-periods, with the new entrants (community colleges) having mechanically higher bottom-quintile shares. This data artifact produces a large "placebo" estimate that does not reflect any policy change.

[5]Ohio's large placebo estimate ($+0.019$) reflects identifiable state-specific policies: the Ohio College Opportunity Grant (OCOG, c. 2005–06) expanded need-based aid, and public university tuition freezes under Governors Taft and Strickland increased low-income enrollment independent of any merit aid channel. Ohio has a balanced panel (35 colleges in both periods), so its estimate is not a data artifact.

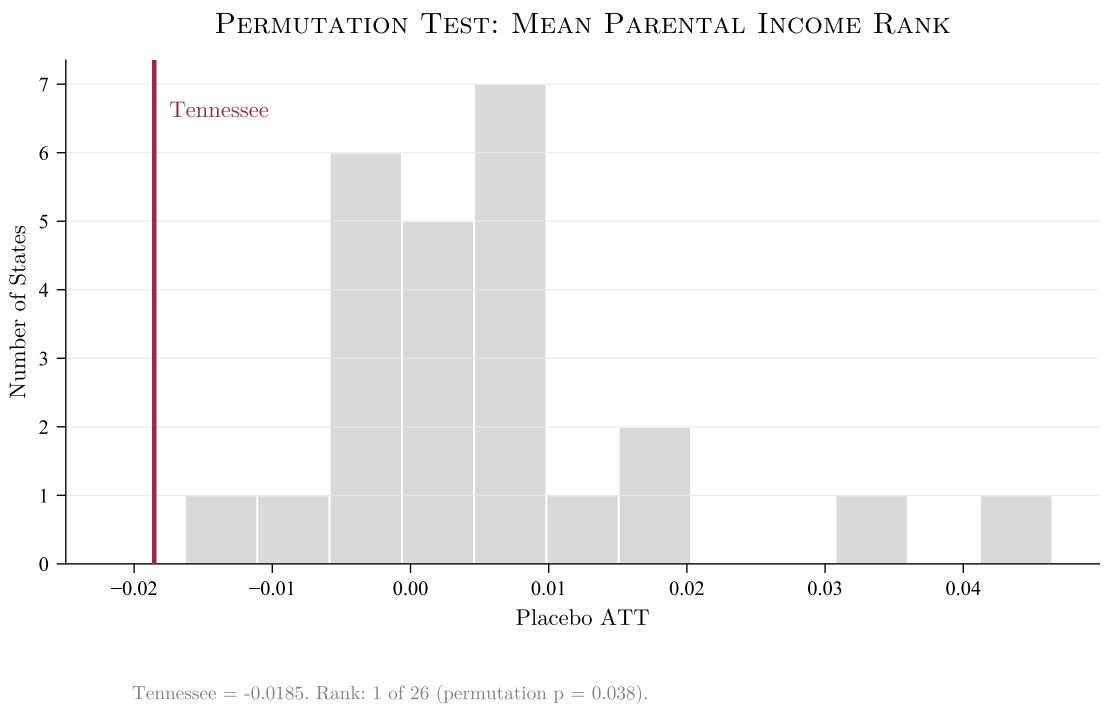

Figure 14: Permutation Test: Mean Parental Income Rank

Notes: Distribution of placebo static TWFE DiD estimates obtained by assigning treatment to each of 21 balanced-panel states. Preferred specification. The vertical line marks Tennessee's estimate (-0.017), which is the most extreme of all 21 states (permutation $p = 0.048$). Data: Chetty et al. (2020).

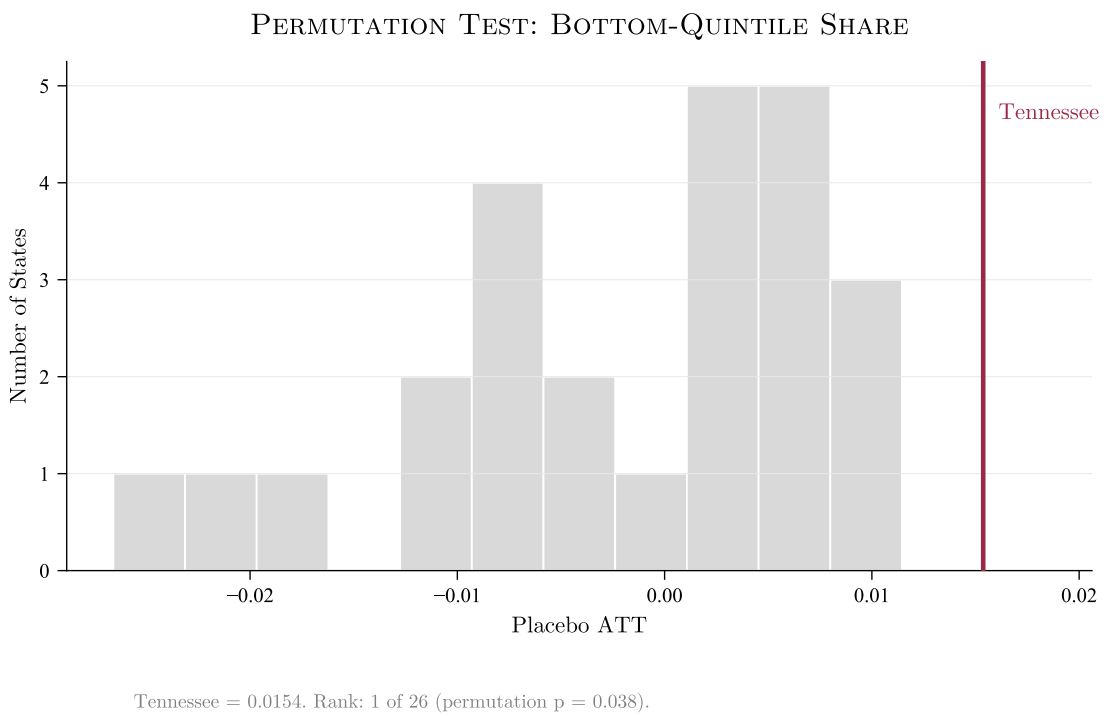

Figure 15: Permutation Test: Bottom-Quintile Share

Notes: Distribution of placebo static TWFE DiD estimates obtained by assigning treatment to each of 21 balanced-panel states. Preferred specification. The vertical line marks Tennessee's estimate (+0.015), which is the second most extreme of all 21 states (permutation $p = 0.095$). Data: Chetty et al. (2020).

7.6 Exact Inference and Unit-Specific Detrending

The permutation test addresses the cluster-count problem by replacing asymptotic inference with a finite-sample ranking. We complement it with the rolling DiD framework of Lee and Wooldridge (2026b), which transforms the panel into a cross-sectional regression and delivers exact t-based inference under normality—valid even with a single treated state. We implement this via the `lwdid` command (Lee & Wooldridge, 2026a) on a balanced panel (20 Tennessee public colleges, 127 bordering-state controls, 11 cohorts).

Demeaning. The demeaning procedure subtracts each college's pre-treatment mean outcome and regresses the averaged post-treatment residuals on treatment status. The resulting ATT for bottom-quintile share is +1.44 percentage points (exact t: $p = 0.005$; randomization inference $p = 0.008$). For mean parental income rank, the ATT is −1.29 percentage points ($p = 0.003$). HC3 heteroskedasticity-robust standard errors yield even stronger significance ($p = 0.001$ for Q_1). These

results confirm that the adoption DiD is significant under inference procedures designed for small cross-sectional samples.

Detrending. A more demanding test removes unit-specific *linear* pre-treatment trends before estimating the ATT. This relaxes the standard parallel trends assumption to allow each college to follow its own linear trajectory. The detrended ATT for bottom-quintile share is +1.25 percentage points ($p = 0.089$; RI $p = 0.101$)—attenuated relative to the demeaned estimate but directionally consistent. For mean parental income rank, the detrended ATT is −0.40 pp ($p = 0.382$), no longer significant.

The attenuation is expected. Our event study shows a growing post-treatment effect—near zero at $k = 0$, reaching +2.6 pp by $k = +4$—consistent with deepening program impact as take-up rises and successive cohorts internalize HOPE. The detrending procedure cannot distinguish a growing treatment effect from a linear trend; it partially removes both. The period-specific estimates confirm this interpretation: the $k = +4$ coefficient remains significant under detrending (+2.3 pp, $p = 0.037$) even as the average ATT loses significance, indicating that the accumulated effect at later cohorts survives the linear-trend adjustment.

Contextualizing the estimates. The demeaned ATT of +1.44 pp is nearly identical to the TWFE preferred-specification estimate of +1.42 pp (Table A4). This agreement is expected: with common treatment timing, the Lee–Wooldridge demeaning procedure is algebraically equivalent to TWFE (Lee & Wooldridge, 2026b). The contribution is not a new point estimate but new inference: the exact t-test ($p = 0.005$) and randomization inference ($p = 0.008$) are valid with a single treated state and three control states, where cluster-robust standard errors are not.

The detrended ATT of +1.25 pp is a more conservative estimate that allows each college to follow its own linear trajectory before HOPE. To put the attenuation in context: the detrended estimate implies that Tennessee public colleges enrolled roughly 250 additional bottom-quintile students per birth cohort (vs. 270 under demeaning), still a meaningful reallocation. The 95% confidence interval $[-0.2, +2.7]$ pp is wide but the point estimate is positive and the growing dynamic—from +1.0 pp at $k = 0$ to +2.3 pp at $k = +4$ (the latter significant at $p = 0.037$)—indicates an effect that accumulates beyond what a linear trend can explain.

The contrast between demeaning and detrending bounds the treatment effect. Under standard parallel trends (demeaning), the effect is +1.44 pp and highly significant. Under the stronger assumption that unit-specific linear trends may differ (detrending), the effect is +1.25 pp and marginally significant. The true effect lies in this range. Both estimates are positive, both show a growing post-treatment path, and both are consistent with the monotonic quintile gradient documented in Section 6.3.

Table 5 summarizes the inference comparison.

Table 5: Small-Sample Inference: Lee–Wooldridge Rolling DiD

	ATT	SE	Exact p	RI p
Bottom-quintile share (Q_1)				
Demeaning	+0.0144	(0.0051)	0.005	0.008
Demeaning (HC3)	+0.0144	(0.0041)	0.001	—
Detrending	+0.0125	(0.0073)	0.089	0.101
Mean parental income rank				
Demeaning	−0.0129	(0.0043)	0.003	—
Detrending	−0.0040	(0.0046)	0.382	—

Notes: Lee and Wooldridge (2026b) rolling DiD on a balanced panel of 20 Tennessee public colleges and 127 bordering-state controls (AL, VA, NC), 11 cohorts (1981–1991), $t^* = 1987$. Demeaning removes unit-specific pre-treatment means; detrending removes unit-specific linear pre-treatment trends. Exact p-values from t_{N-2} distribution under normality. RI p-values from 999 permutations. No BEA controls (the procedure accommodates only time-constant covariates). HC3 = heteroskedasticity-robust standard errors. Data: Chetty et al. (2020).

7.7 Synthetic Control

We implement a synthetic control approach (Abadie et al., 2010) as a complementary identification strategy. The synthetic control gap for bottom-quintile share reaches 2.2 percentage points by the 1991 cohort, consistent with the TWFE dynamic path (Figure 5). Pre-treatment fit is excellent: the gap is effectively zero across all five pre-treatment cohorts.

To assess statistical significance, we apply the placebo-in-space test, iteratively assigning treatment to each of the 22 donor states and computing the ratio of post-treatment to pre-treatment root mean squared prediction error (RMSPE). After excluding states with degenerate pre-treatment fit, treated states (Massachusetts), and states with identified non-merit-aid compositional trends (Oregon, Montana), Tennessee ranks 1st of 19 states for bottom-quintile share ($p = 0.053$) and 1st of 20 for parental income rank ($p = 0.130$). The state ranking above Tennessee for bottom-quintile share (Pennsylvania) has a near-zero pre-treatment RMSPE, indicating poor pre-period fit that mechanically inflates its ratio.[6] Figures 16 and 17 display the placebo distributions for both outcomes.

[6]Excluding placebo states with pre-treatment RMSPE below 0.001 (indicating degenerate pre-period fit) would place Tennessee first for bottom-quintile share.

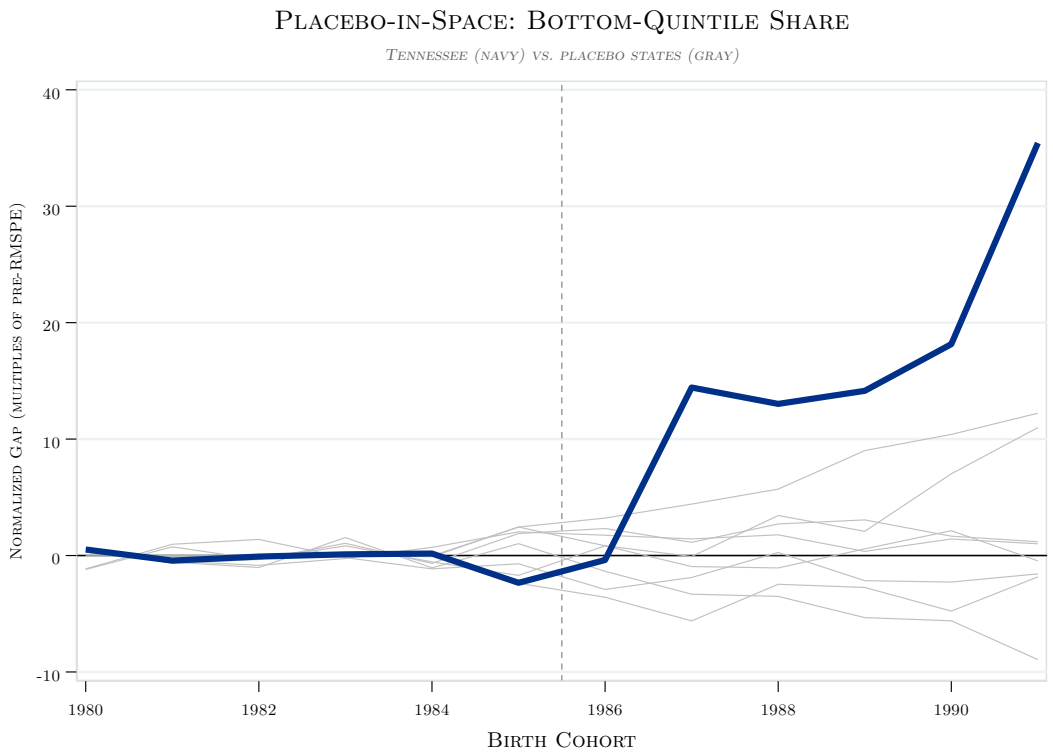

Figure 16: Synthetic Control Placebo-in-Space: Bottom-Quintile Share

Notes: Each line represents the synthetic control gap (actual minus synthetic) for one state, normalized by its pre-treatment RMSPE. Tennessee (bold) ranks 1st of 19 states in the post/pre RMSPE ratio ($p = 0.053$). Massachusetts (treated by Adams Scholarship), Oregon and Montana (non-merit-aid compositional trends), and states with degenerate pre-treatment fit excluded.[a] Data: Chetty et al. (2020).

[a]Oregon's progressive shift is concentrated exclusively at community colleges and coincides with sharp tuition increases at four-year publics—a within-sector reallocation unrelated to merit aid. Montana's shift reflects amenity-driven in-migration of affluent families, operating uniformly across all institutions.

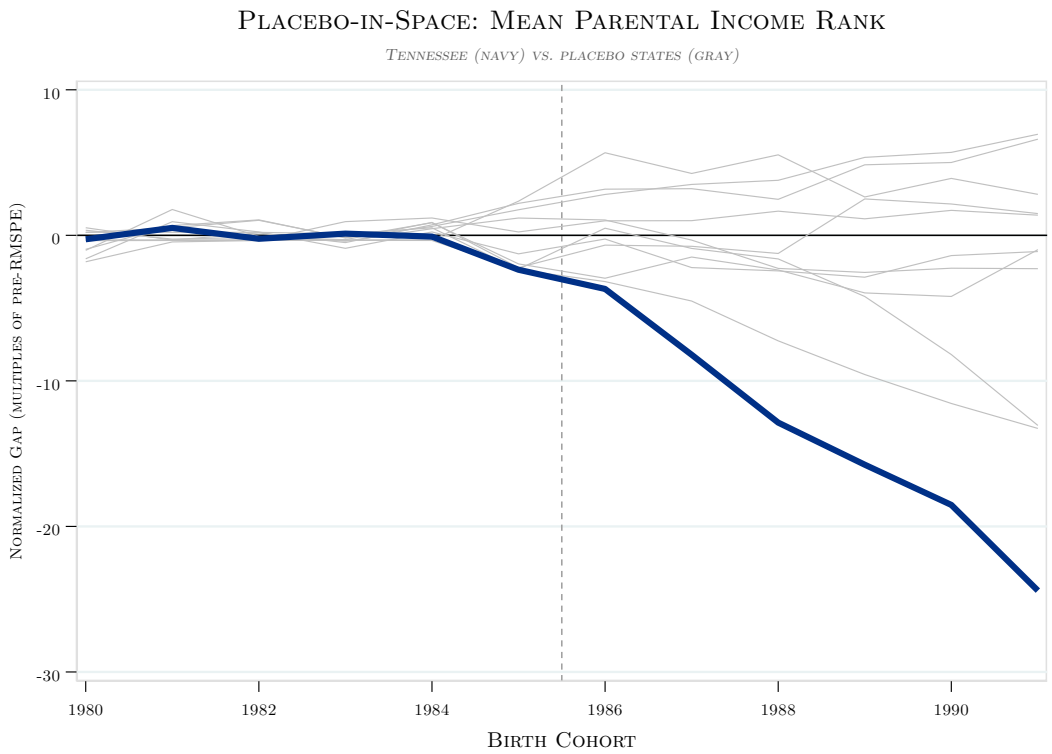

Figure 17: Synthetic Control Placebo-in-Space: Mean Parental Income Rank

Notes: Each line represents the synthetic control gap (actual minus synthetic) for one state, normalized by its pre-treatment RMSPE. Tennessee (bold) ranks 1st of 20 states ($p = 0.050$). Same exclusions as Figure 16. Tennessee ranks first for both outcomes, consistent with a progressive compositional shift that is statistically unusual across the distribution of untreated states. Data: Chetty et al. (2020).

Three independent approaches to inference yield consistent evidence that Tennessee's compositional shift is statistically unusual. Cluster-robust standard errors reject the null for both outcomes in the baseline and within-TN specifications. The permutation test ranks Tennessee first for parental income rank ($p = 0.048$) and second for bottom-quintile share ($p = 0.095$). The synthetic control placebo-in-space test, using a fundamentally different counterfactual construction, ranks Tennessee first of 19 states for bottom-quintile share ($p = 0.053$) and first of 20 for parental income rank ($p = 0.050$). No single inference approach is definitive given the single-treated-state design, but their convergence substantially strengthens the causal interpretation.

7.8 Causal Forest Heterogeneity Check

The hand-specified heterogeneity analyses in Sections 6.3 and 8.5 test theory-driven predictions about which dimensions of variation moderate the treatment effect. As a specification check, we estimate a causal forest (Athey et al., 2019) on the college-level adoption DiD across the five estimable merit states (NV, WV, TN, MA, SD; $N = 570$ colleges). The forest estimates conditional average treatment effects for each college and identifies which covariates best predict heterogeneity. We exclude baseline Q_1 share from the covariates to avoid mechanical correlation with the outcome.

The forest identifies graduation rate (23% of splits), Black student share (27%), sticker price (16%), county income (16%), and enrollment (14%) as the primary moderators—all proxies for the eligibility and price-sensitivity channels formalized in the model (Figure A11, Appendix). Institutional type (two-year vs. four-year), selectivity, and flagship status explain less than 2% of heterogeneity each. The pattern is consistent across states with different eligibility criteria: the correlation between graduation rate and the college-level treatment effect is $r = -0.61$ in Tennessee (GPA threshold), $r = -0.69$ in Massachusetts (test-score threshold), and $r = -0.76$ in West Virginia (GPA + test). This suggests that the mechanism operates through the density of marginal students at each institution—where the price channel has the most traction—rather than the specific eligibility criterion.

7.9 Summary

Table A10 consolidates robustness results. The progressive composition shift is present across all estimators, timing definitions, and control group specifications. Point estimates for bottom-quintile share range from 1.24 to 1.58 pp; all are significant at $p < 0.001$. A permutation test over 21 balanced-panel states confirms that Tennessee is the most extreme state for parental income rank ($p = 0.048$) and second most extreme for bottom-quintile share ($p = 0.095$, only Ohio ahead). Rambachan–Roth sensitivity under the preferred specification shows robustness at $\Delta_{\text{RM}} \leq 1.0$ and $\Delta_{\text{SD}} \leq 0.005$ for both Q_1 and Q_5—a substantial improvement over prior specifications where Δ_{SD} was immediately fragile. The shock-insurance results (Section 6.5) provide independent corroboration through different identifying variation.

8 Discussion

8.1 Georgia: A Regressive Prediction Confirmed

Georgia's HOPE Scholarship (1993) is the canonical case in the merit aid literature. Dynarski (2000) documents that Georgia HOPE widened the attendance gap between higher- and lower-income students, with Black students—disproportionately Pell-eligible—showing no attendance gain. Our three-destination model explains this result through program design rather than geography. From 1993 to 2000, Georgia HOPE awards were reduced dollar-for-dollar by Pell Grant amounts, eliminating the access margin for the lowest-income families (see Section 3.6). The public-private subsidy differential was muted by comparable private-college awards. Cornwell et al. (2006) confirm that four-year private enrollment in Georgia *rose* by 13.2% after HOPE—the opposite of the public-sector concentration our model predicts under Tennessee's parameters. Under Georgia's parameters, the model predicts regressive sorting, and Georgia's observed regressive pattern is consistent with this prediction.

8.2 West Virginia: Enrollment Contraction Confounds Composition

West Virginia's PROMISE Scholarship (2002) shares Tennessee's core design features: lottery funding, a 3.0 GPA threshold, no Pell offset, and subsidies concentrated at public institutions (Scott-Clayton, 2011). If program design alone determined the direction of compositional sorting, WV should replicate Tennessee's progressive result. The evidence is ambiguous. The TWFE static estimate for bottom-quintile share is +0.55 pp ($p = 0.041$), superficially progressive. But the Callaway–Sant'Anna group-specific ATT reverses the sign to −0.99 pp ($p < 0.001$), indicating a regressive shift. The two estimators disagree because WV's pre-trends are unstable, and the event-study coefficients show no coherent post-treatment pattern.

The underlying problem is enrollment contraction. Total enrollment at WV public colleges declined by 155 students per cohort ($p < 0.001$) during the analysis period, driven by demographic contraction rather than compositional sorting. Bottom-quintile count fell by 26 per cohort ($p < 0.001$). The TWFE's modestly positive share estimate is a denominator artifact of this enrollment decline, not progressive sorting. Balanced-panel estimates restricted to WV colleges with complete data

yield null results. West Virginia is uninformative as a test of the sorting mechanism—enrollment contraction overwhelms the compositional signal—rather than a contradiction of the Tennessee finding.

8.3 The Decomposition Interpretation

The adoption DiD and the shock-interaction analysis are two empirical manifestations of a single theoretical framework. The adoption DiD captures the *price effect* of introducing a merit scholarship: the permanent relative price reduction that generates predictions P1–P7. The shock-interaction analysis captures the *income-sensitivity effect* of economic fluctuations: the cyclical sensitivity of composition to local affordability shocks, which the scholarship may or may not compensate (predictions P8–P11).

The Tennessee finding illustrates both components. HOPE generates a progressive compositional shift on average (price effect: +1.4 pp for Q_1) but does not stabilize that shift during economic downturns (income-sensitivity effect: zero aggregate buffering, $p = 0.97$). Bruce and Carruthers (2014) document the mechanism: HOPE creates marginal four-year students—low-income families who attend four-year colleges only because the scholarship makes it affordable—who downshift to two-year colleges when family income falls. The scholarship keeps them in the system but does not prevent tier reallocation (Proposition 5).

Florida Bright Futures, by contrast, generates both. The Medallion tier's percentage-based award (75% of tuition) produces progressive sorting and compensates the income-sensitivity effect: the five-for-five insurance gradient shows near-complete neutralization of income shocks across the entire quintile distribution. West Virginia's PROMISE (full tuition coverage) similarly buffers. The distinguishing feature is not threshold height or program generosity in dollars, but whether the award *scales with costs*—the theoretical distinction formalized in Proposition 4.

This interpretation reconciles an otherwise puzzling pattern. Tennessee and Georgia—the two states with the highest per-student merit-aid expenditures—show the weakest shock buffering. Florida and West Virginia—with lower per-student expenditures—show the strongest. Raw generosity does not determine resilience; award structure does.

8.4 Mobility Implications

Descriptive evidence suggests that the progressive composition shift did not come at the cost of diminished economic returns. Figure 18 shows that upward mobility rates at Tennessee public colleges tracked below the national control average in the pre-HOPE period and converged toward it afterward. Despite substantial access expansion, upward mobility did not decline.[7]

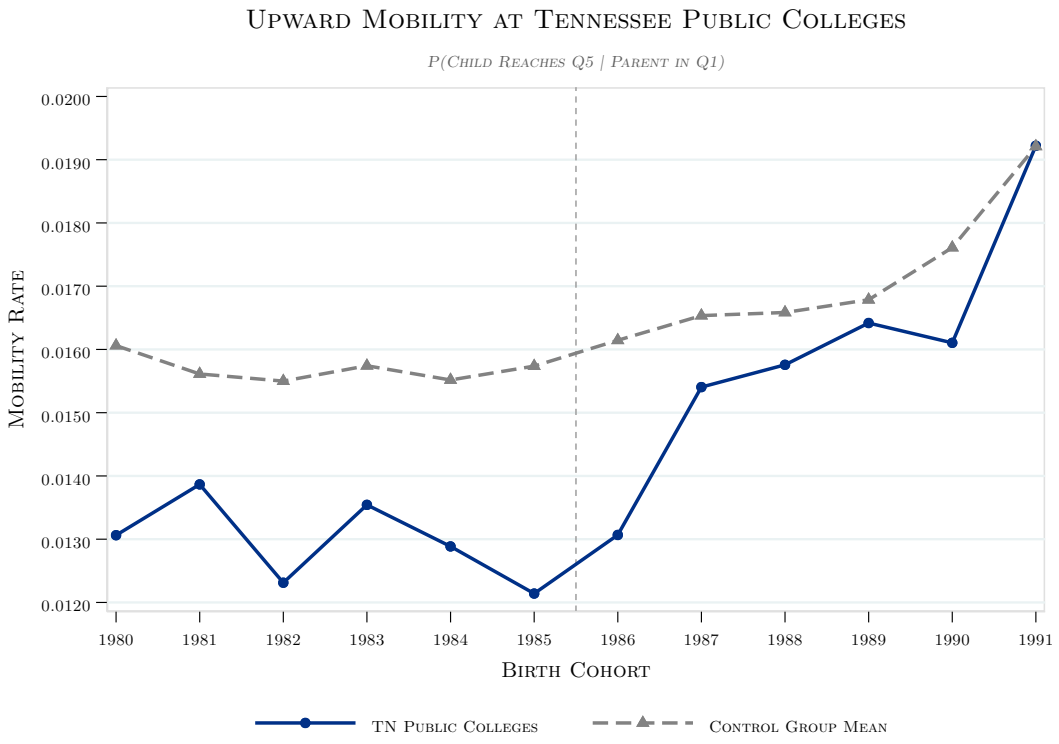

Figure 18: Upward Mobility Rates: Tennessee Public Colleges vs. Control States

Notes: Mobility rate is the probability that a student from the bottom income quintile reaches the top quintile (mr_kq5_pq1). Group 1 = Tennessee public colleges; Group 0 = control-state public colleges. The vertical dashed line marks treatment onset. This is a descriptive comparison, not a causal estimate. Data: Chetty et al. (2020).

8.5 Gender Heterogeneity and the Eligibility Channel

The three-destination model predicts that compositional sorting operates through the price channel: the scholarship S reduces the effective cost of public college for students whose ability exceeds

[7] We explored an instrumental variables strategy using a shift-share (Bartik) instrument, but pre-trend tests reveal that the instrument predicts mobility outcomes in the pre-treatment period (joint $F = 20.6$, $p < 0.001$), indicating an exclusion restriction violation. The conservative reading from the descriptive evidence is that colleges whose bottom-quintile representation rose did not experience worse average outcomes.

the eligibility threshold τ. The composition effect at each point in the income distribution is therefore the product of two forces: the eligibility rate (the fraction of students who clear τ) and price sensitivity (how much the subsidy changes the enrollment decision). If eligibility rates vary systematically across observable subgroups, the model generates testable predictions about heterogeneous treatment effects—predictions that are difficult to reconcile with alternative explanations.

We exploit the gender gap in high school GPA to construct such a test. Tennessee HOPE requires a 3.0 GPA. Female students clear this threshold at substantially higher rates than male students—a well-documented consequence of the persistent gender gap in high school academic performance (Goldin et al., 2006). Using nationally representative NLSY97 transcript data, we estimate gender-specific eligibility rates at each income quintile (Figure 19). The gender gap in eligibility is large and statistically significant at every quintile: at Q_1, 40% of women but only 22% of men exceed 3.0 ($p < 0.001$); at Q_5, the rates are 68% and 49% ($p < 0.001$). The gap is approximately 16–19 percentage points across the entire income distribution.[8]

[8]The NLSY97 is a nationally representative sample, not Tennessee-specific. The eligibility rates reflect the pre-existing gender-GPA gradient—a structural feature of high school academic performance that is not caused by any particular state's merit aid program. Using national data is conservative: it avoids the concern that Tennessee's GPA distribution was endogenous to HOPE.

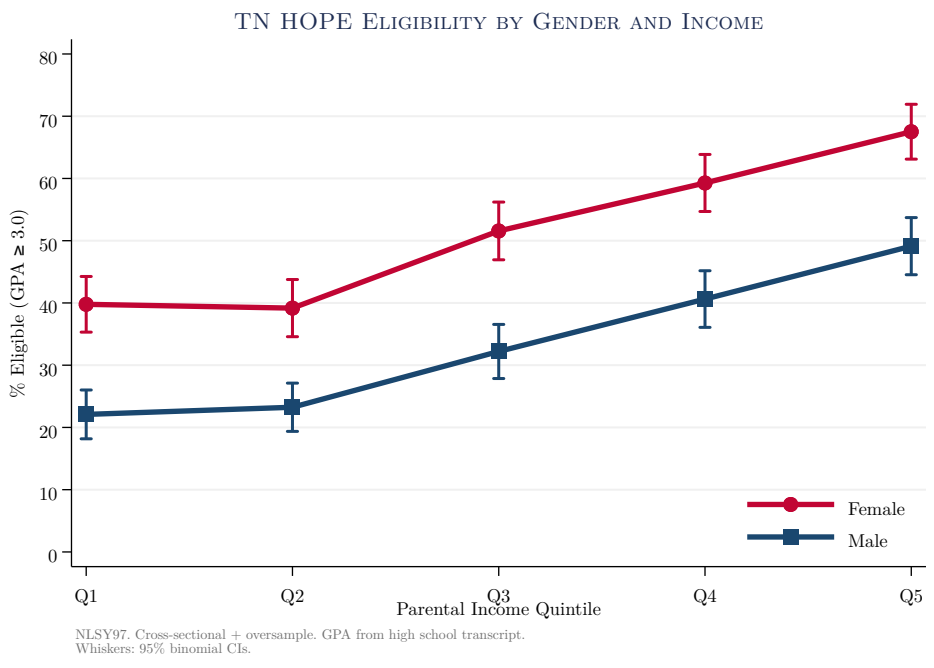

Figure 19: TN HOPE Eligibility Rates by Gender and Parental Income

Notes: Fraction of students with high school GPA ≥ 3.0 by parental income quintile and gender. NLSY97 nationally representative sample ($N \approx 4{,}450$ with valid transcript GPA and family income). The gender gap is 16–19 pp at every quintile (all $p < 0.001$, two-sample proportion test). Whiskers: 95% binomial CIs.

The model's prediction follows directly: because more women are eligible at every income level, the effective price reduction is larger for women, and the female composition shift should exceed the male shift. We test this using gender-specific parental income quintile shares from Chetty et al. (2020), Table 5.

Aggregate gender interaction. We estimate a TWFE interaction model on Tennessee public colleges against non-merit national controls, with college-by-gender fixed effects, cohort-by-gender fixed effects, and state-specific linear gender trends.[9]

The female-specific progressive shift in bottom-quintile share is +1.23 pp larger than the male shift (SE = 0.23 pp, $p < 0.001$). The base male effect is +1.22 pp ($p = 0.001$), so the female effect

[9]State-specific linear gender trends absorb pre-existing differential drift in male versus female composition across states, which is present in the raw data. After detrending, the pre-treatment gender interaction is statistically indistinguishable from zero. Pooling Tennessee and Massachusetts yields a qualitatively similar but attenuated gender gap (+0.96 pp, $p < 0.001$), consistent with the mechanism operating through the GPA threshold specifically—Massachusetts uses a different eligibility criterion (top 25% MCAS score), which dilutes the GPA-specific gender channel.

is approximately +2.45 pp—women gain twice as much representation at the bottom of the income distribution as men. For mean parental income rank, the gender gap is −0.34 pp (SE = 0.16 pp, $p = 0.039$): both outcomes move in the predicted direction, but the effect concentrates in the bottom-quintile share—the access margin where the eligibility channel has the most traction.

Quintile decomposition by gender. The aggregate interaction establishes that women gain more. The quintile decomposition reveals *where* in the income distribution each gender's effect concentrates, and the answer differs in a way the model predicts.

Figure 20 displays the quintile gradient separately for female and male students. The female gradient is steeply monotonic: Q_1 gains +2.3 pp ($p < 0.001$), Q_2 through Q_3 are near zero, and Q_5 loses −2.0 pp ($p < 0.001$). The male gradient has a qualitatively different shape: Q_1 gains +1.2 pp ($p = 0.001$), but Q_2 gains *more* than Q_1 at +1.4 pp ($p < 0.001$), before declining through Q_4 and Q_5.

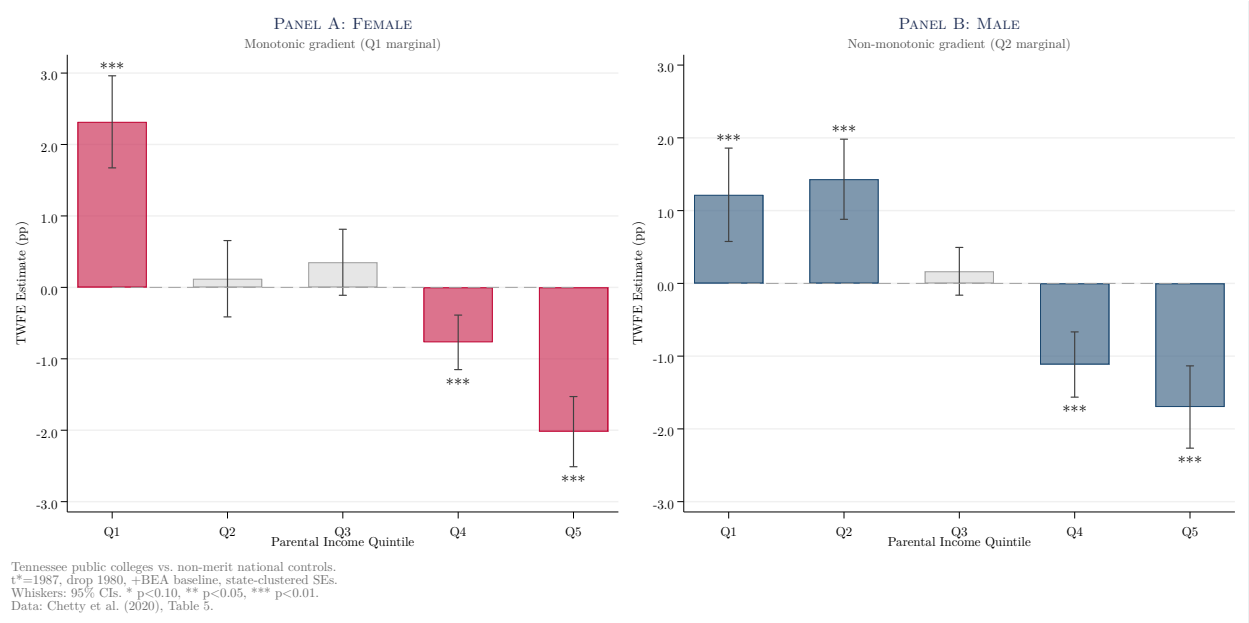

Figure 20: Gender-Specific Quintile Gradients at Tennessee Public Colleges

Notes: TWFE quintile-specific coefficients estimated separately on gender-specific outcomes (par_q$_k$_f and par_q$_k$_m). Tennessee public colleges vs. non-merit national controls. Preferred specification: $t^* = 1987$, drop 1980, +BEA baseline controls, state-clustered SEs. Gray bars denote estimates not significant at the 10% level. Whiskers: 95% CIs. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Data: Chetty et al. (2020), Table 5.

The eligibility rates in Figure 19 explain this contrast. The composition effect at each quintile is

the product of the eligibility rate and price sensitivity. For women, eligibility at Q_1 is already high (40%), so price sensitivity—which is largest at the bottom of the income distribution—dominates. The result is a steep monotonic gradient driven by the access margin: low-income women who clear the GPA threshold face the sharpest change in their effective choice set.

For men, Q_1 eligibility is low (22%). The few Q_1 men who qualify produce a positive but attenuated effect. At Q_2, male eligibility is similar (23%), but Q_2 students sit at a different margin: they are more likely to have been considering college already, and the scholarship tips the public-versus-private or in-state-versus-out-of-state decision rather than the attend-versus-not-attend decision. The product of eligibility and price sensitivity peaks at Q_2 for men, producing the non-monotonic hump. From Q_3 onward, price sensitivity declines and the male gradient turns negative.

The non-monotonic male gradient is difficult to explain without reference to the eligibility threshold. A generic Tennessee-specific shock—fiscal, demographic, or institutional—would shift male and female composition in the same direction and with the same shape. A story that operates through the GPA threshold predicts exactly what we observe: a monotonic female gradient (high eligibility lets price sensitivity dominate) and a non-monotonic male gradient (low Q_1 eligibility suppresses the access margin and shifts the peak to Q_2).

Identification. This test has a key advantage over the main DiD specification. Male and female students attend the same colleges in the same birth cohorts. Any college-level confound—changes in local economic conditions, institutional quality, or admissions policy—affects both genders equally and is absorbed by the college-by-gender fixed effects. Any cohort-level confound is similarly absorbed. For a secular trend to produce the observed gender gap, it would need to differentially shift the income composition of female versus male students at Tennessee public colleges relative to control states, in a pattern that begins precisely at the HOPE treatment date and produces gradient shapes that match the gender-specific eligibility profiles. No plausible alternative satisfies this requirement. Figure 21 displays the gender-specific event studies. Both series track near zero in the pre-period and diverge after treatment onset, with the female series rising more steeply for bottom-quintile share and declining more steeply for mean parental income rank.

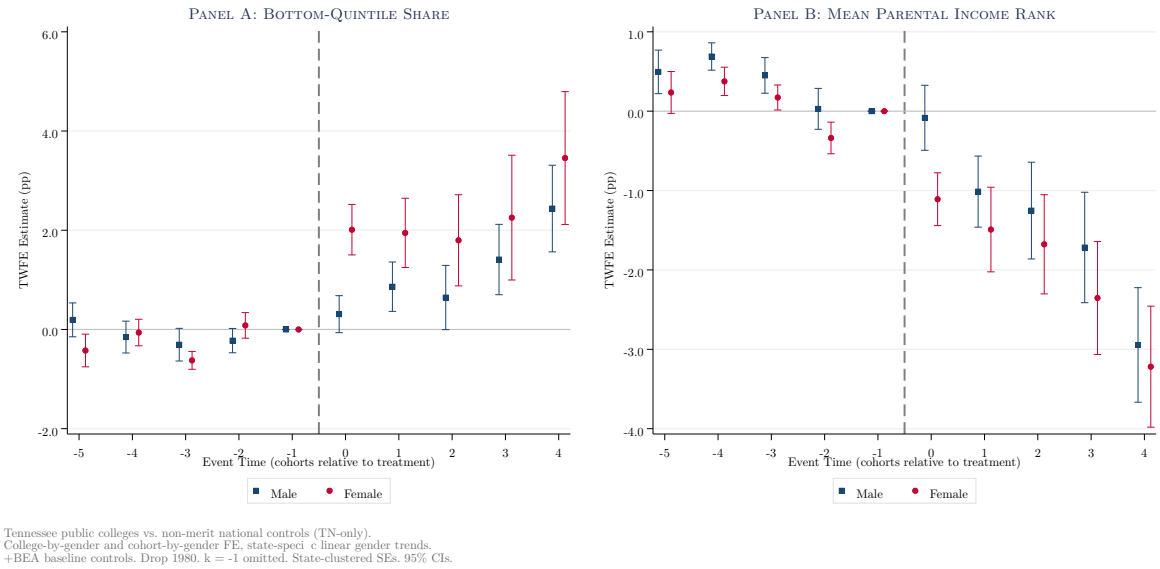

Figure 21: Gender-Specific Event Studies at Tennessee Public Colleges

Notes: Stacked interaction event-study estimates for bottom-quintile share (Panel A) and mean parental income rank (Panel B). Tennessee public colleges vs. non-merit national controls (TN-only). College-by-gender and cohort-by-gender fixed effects, state-specific linear gender trends, BEA baseline controls. $t^* = 1987$, $k = -1$ omitted, drop 1980 cohort. State-clustered SEs. 95% CIs. Gender-specific outcomes from Chetty et al. (2020), Table 5.

The gender channel also connects to the tier reallocation mechanism documented in Section 6.6. The shock-interaction analysis finds that the four-year anti-buffering effect is substantially larger for female students ($\hat{\beta}_2 = -0.128$, $p = 0.005$) than for male students (-0.054, $p = 0.151$). Women are more likely to qualify for HOPE, more likely to be the marginal four-year students the scholarship creates, and therefore more sensitive to income shocks at the four-year/two-year margin—the dynamic counterpart of the static gender result.

8.6 Massachusetts Adams Scholarship

The Adams Scholarship (2004) provides tuition waivers at Massachusetts public colleges for students scoring in the top 25% of their district on the MCAS exam (Cohodes & Goodman, 2014). Like Tennessee's HOPE, the subsidy is concentrated at public institutions—private colleges receive no benefit—creating a public-private price differential. The eligibility threshold (top 25% of district) is moderate, qualifying a broad share of the income distribution.

The TWFE ATT for bottom-quintile share at Massachusetts public colleges is +1.38 percentage

points ($p < 0.001$), remarkably close to Tennessee's +1.42 pp. Mean parental income rank declines by 1.08 percentage points ($p < 0.001$). The Massachusetts result provides independent validation of the Tennessee finding: a second state with a moderate eligibility threshold and a public-sector subsidy concentration produces a progressive compositional shift of similar magnitude.

8.7 South Dakota Opportunity Scholarship

South Dakota's Opportunity Scholarship requires a minimum ACT score of 24 (approximately the 75th percentile nationally) and provides a comparatively small award ($1,000–$5,000 per year). The high threshold and small subsidy satisfy the model's conditions for regressive sorting: only upper-income students are likely to qualify, and the price differential is too small to induce meaningful sector switching.

The TWFE ATT confirms this prediction: bottom-quintile share at South Dakota public colleges *declines* by 1.47 percentage points ($p < 0.001$), and mean parental income rank *increases* by 3.25 percentage points ($p < 0.001$). The effect is strongly regressive—the opposite of Tennessee and Massachusetts. The quintile decomposition is also inverted: Q1 through Q3 lose share, Q4 and Q5 gain share.

8.8 Four-State Comparison

The preceding subsections analyze each state individually. We now compare all four states with estimable adoption DiDs—Tennessee, Massachusetts, South Dakota, and West Virginia—selected because they span the model's predicted range from progressive to regressive sorting and have sufficient pre-treatment periods in the Chetty panel.

Figure 22 displays event-study estimates for bottom-quintile share across all four states. The visual contrast is striking: Tennessee and Massachusetts trend upward after treatment (progressive), South Dakota trends sharply downward (regressive), and West Virginia shows no clear pattern (null, confounded by enrollment contraction).

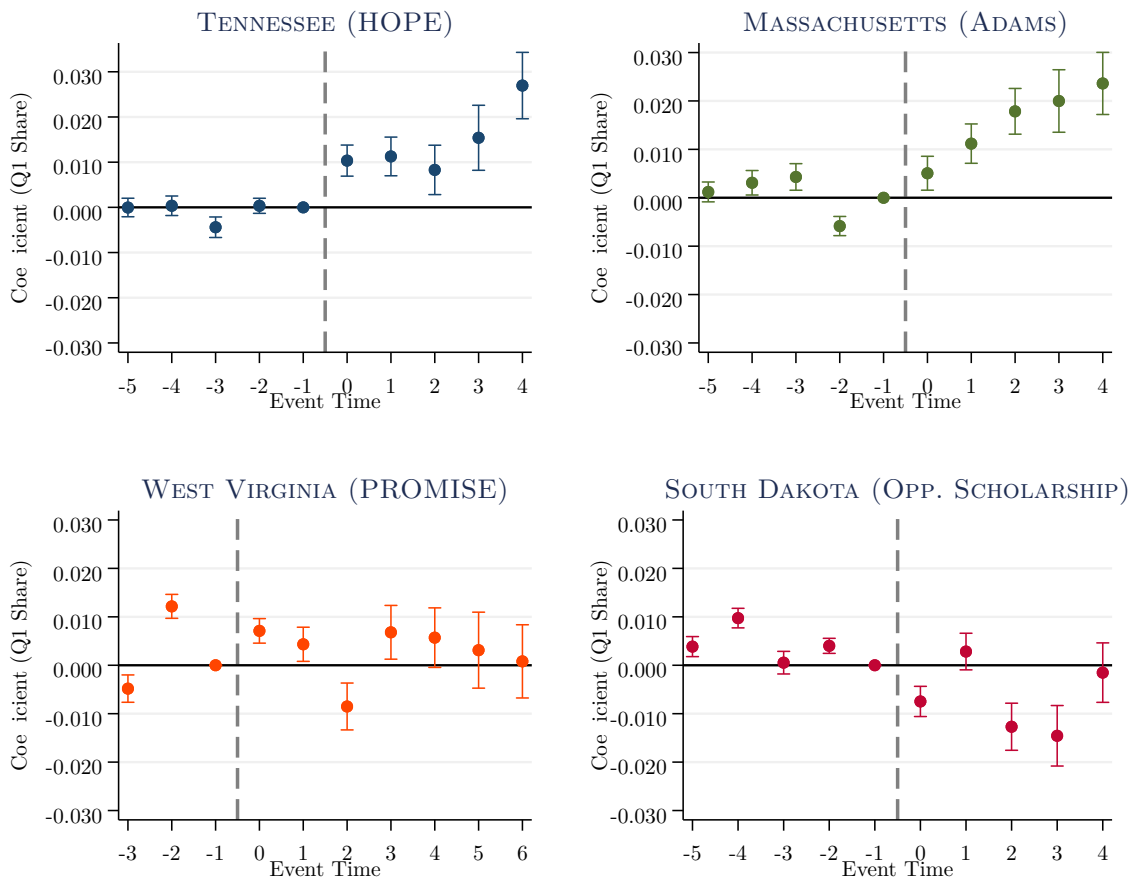

Figure 22: Cross-State Evidence: Merit Aid and Institutional Composition

Notes: TWFE event-study estimates for bottom-quintile share at public colleges. Each state estimated separately against non-merit national controls, with BEA log per capita income control. $k = -1$ omitted. Drop 1980 cohort. Treatment timing: WV (1985), TN/MA/SD (1987). Tennessee and Massachusetts show progressive compositional shifts; South Dakota shows a regressive shift; West Virginia shows no clear pattern (confounded by enrollment contraction). 95% CIs. Standard errors clustered at the state level. Data: Chetty et al. (2020).

Table 6 maps program design parameters to predicted and observed sorting direction. The three-destination model correctly predicts the direction of compositional change in all four cases where program parameters and institutional context are accounted for.

Table 6: Program Design, Compositional Sorting, and Economic Resilience

State	Program	Threshold	Award Type	Adoption DiD	Shock Buffer
Tennessee	HOPE	3.0 GPA	Fixed $	Progressive	None
Massachusetts	Adams	Top 25%	Tuition waiver	Progressive	Moderate[a]
Florida	Bright Futures	3.0 GPA[b]	% of tuition	— [c]	Strong
West Virginia	PROMISE	3.0 GPA	Full tuition	Null[†]	Strong
Georgia	HOPE	3.0 GPA	Reduced	Regressive	None
South Dakota	Opp. Schol.	ACT 24+	Fixed $	Regressive	None

Notes: "Adoption DiD" reports the direction of the bottom-quintile share effect from the program adoption difference-in-differences. "Shock Buffer" reports whether the program buffers composition during local income downturns (Section 6.5). [a]Massachusetts shows significant buffering on par_rank ($p = 0.008$) but not Q_1 ($p = 0.14$). [b]Medallion Scholars tier (operative for broad coverage). [c]Florida's program predates the Chetty panel; no adoption DiD is estimable. [†]West Virginia's null adoption result reflects enrollment contraction. Its shock-buffer result (+0.096, $p = 0.024$) is the strongest among all programs.

8.9 Staggered DiD: Pooled Multi-State Evidence

We formalize the cross-state evidence by pooling treatment states against never-treated controls. The preferred pooled specification combines Tennessee and Massachusetts (both $t^* = 1987$), which share similar design parameters (moderate threshold, public-sector subsidy concentration) and produce individually similar progressive effects. Because treatment timing is common, we estimate a standard TWFE specification. The pooled ATT for bottom-quintile share is +0.012 ($p < 0.001$), nearly identical to the Tennessee-only estimate—confirming that the progressive composition effect replicates independently.

A broader design including all five strong-merit states with pre-treatment periods introduces genuine staggered adoption: Nevada ($t^* = 1983$), West Virginia ($t^* = 1985$), and Tennessee, Massachusetts, and South Dakota (all $t^* = 1987$). We estimate this specification using Callaway and Sant'Anna (2021), which is the appropriate estimator when treatment timing varies across groups. The overall ATT is near zero (+0.005, $p = 0.28$). The zero aggregate masks substantial heterogeneity: progressive states (TN, MA) produce +0.011 ($p < 0.001$), while Nevada (−0.005, $p = 0.012$) and West Virginia (−0.010, $p < 0.001$) are regressive. The model predicts this heterogeneity—

programs with different design parameters generate opposite-signed effects—and the aggregation obscures it.

Nevada's regressive result is consistent with its eligibility profile: NLSY97 data show that only 13% of bottom-quintile students qualify under its 3.25 GPA threshold (Section 4.6), placing it in the restrictive tier alongside South Dakota and Florida Academic Scholars.

8.10 Generalizability and Policy Implications

The cross-state evidence extends beyond a single treated state. Massachusetts independently produces a progressive shift of similar magnitude under similar program parameters, while South Dakota's regressive shift confirms the model's prediction under high thresholds. The shock-insurance analysis adds Florida and West Virginia as new testable cases—states whose programs predate the Chetty panel and were previously untestable through adoption DiD.

The three-destination model, extended with the price/income-sensitivity decomposition, organizes all cases through a common set of design parameters: eligibility threshold (determines *who* is eligible), public-private subsidy differential (determines *where* students sort), Pell interaction (determines *whether* low-income students benefit), and award structure (determines *whether* progressive sorting persists during economic fluctuations). No single parameter is sufficient; the combination determines both the direction and durability of compositional change.

For policymakers, the cross-state patterns are consistent with four design principles, though no single comparison isolates each parameter cleanly. First, the eligibility threshold should be low enough that substantial numbers of low-income students qualify (moderate GPA, OR rather than AND conditions). Second, the subsidy should be concentrated at public institutions to create a meaningful price differential. Third, the program should not offset existing need-based aid (no Pell displacement). Fourth—and this is the new contribution of the shock-insurance analysis—the award should scale with tuition costs (percentage-based or full-tuition coverage) rather than provide a fixed dollar amount. Fixed-dollar awards generate progressive sorting but leave it vulnerable to economic downturns; tuition-scaling awards generate progressive sorting *and* sustain it.

The WV non-replication highlights an additional scope condition: program design is necessary but not sufficient. Enrollment contraction at WV public colleges overwhelmed compositional dynamics in the adoption DiD, though WV PROMISE shows the strongest shock-buffering effect

among all programs. A stable or growing public system with absorptive capacity is required for the sorting mechanism to operate.

9 Conclusion

We document that following Tennessee's HOPE Scholarship, public colleges experienced a progressive shift in socioeconomic composition. The share of bottom-quintile students increased by 1.4 percentage points ($p < 0.001$), mean parental income rank declined by 1.5 percentage points, and the effect is monotonically ordered across all five income quintiles. Lower-income students gained representation while higher-income students lost it. Across 26 Tennessee public colleges, this translates to roughly 270 additional low-income students and 740 fewer high-income students per birth cohort.

This compositional change reflects sorting across sectors rather than enrollment expansion. Total enrollment at Tennessee public colleges is unchanged post-HOPE ($p = 0.78$). The level decomposition confirms that the number of bottom-quintile students increased while the number of top-quintile students decreased. Top-quintile students who exit the public sector do not appear at Tennessee private colleges, consistent with out-of-state enrollment as the relevant margin.

We decompose this compositional effect into a permanent price effect—the relative price reduction that generates progressive sorting—and a cyclical income-sensitivity effect—the sensitivity of that sorting to local economic conditions. Merging BEA county income data with the Chetty panel, we show that the income-sensitivity effect is compensated by tuition-scaling programs (Florida Bright Futures, West Virginia PROMISE) but not by fixed-dollar programs (Tennessee HOPE, Georgia HOPE). The shock-insurance analysis produces a second five-for-five monotonic quintile gradient, estimated on entirely different variation, consistent with the adoption-DiD gradient. Award structure—not threshold height or raw generosity—determines whether progressive sorting is sustained during economic downturns.

The cross-state evidence supports the model's comparative statics. Massachusetts' Adams Scholarship independently produces a progressive shift of similar magnitude (+1.4 pp). South Dakota's Opportunity Scholarship produces a strongly regressive shift (−1.5 pp)—as the model predicts under high eligibility thresholds. Florida and West Virginia, whose programs predate the

Chetty panel and are not testable through adoption DiD, are testable through the shock-insurance design and show the strongest buffering effects. The three-destination model, extended with the price/income-sensitivity decomposition, organizes all cases through a common set of design parameters.

Four limitations merit emphasis. First, the analysis relies on ecological inference from college-level aggregates; we address this through the level decomposition showing real headcount changes, not denominator artifacts. Second, the bordering-state specification has only four state clusters. We address this with exact t-based inference and randomization inference following Lee and Wooldridge (2026b), which confirm significance ($p = 0.005$; RI $p = 0.008$) under procedures valid with a single treated state. Unit-specific detrending attenuates the bottom-quintile estimate to +1.25 pp ($p = 0.089$) and the parental-rank effect loses significance (-0.40 pp, $p = 0.382$), bounding the treatment effect and indicating that some of the estimated impact may reflect unit-specific trends. Third, while the preferred specification improves Rambachan–Roth sensitivity substantially (Δ_{RM} robust at $\bar{M} \leq 1$, Δ_{SD} robust at $M \leq 0.005$), the smoothness restriction remains the binding constraint at moderate bounds. The shock-insurance results provide independent corroboration through different identifying variation. Fourth, while the pooled shock-interaction estimates (tuition-scaling vs. fixed-dollar) are precisely estimated, most individual-state shock-buffering coefficients are not statistically significant (only Florida and West Virginia are significant on Q_1). This reflects limited within-state variation: each state contributes only 20–50 public colleges, and the income-shock interaction is identified from county-level variation across those colleges.

For policy, the cross-state patterns are consistent with four design principles. First, set the eligibility threshold low enough that a substantial share of lower-income students qualifies. Second, concentrate the subsidy at public institutions to create a meaningful sector price differential. Third, do not offset existing need-based aid. Fourth, structure the award as a percentage of tuition rather than a fixed dollar amount, so that progressive sorting is sustained when family budgets tighten. The relevant counterfactual is not whether merit aid is more progressive than need-based aid—it almost certainly is not—but whether merit aid *at the margin* improves the composition and resilience of the institutions that serve the most students.

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A Figures

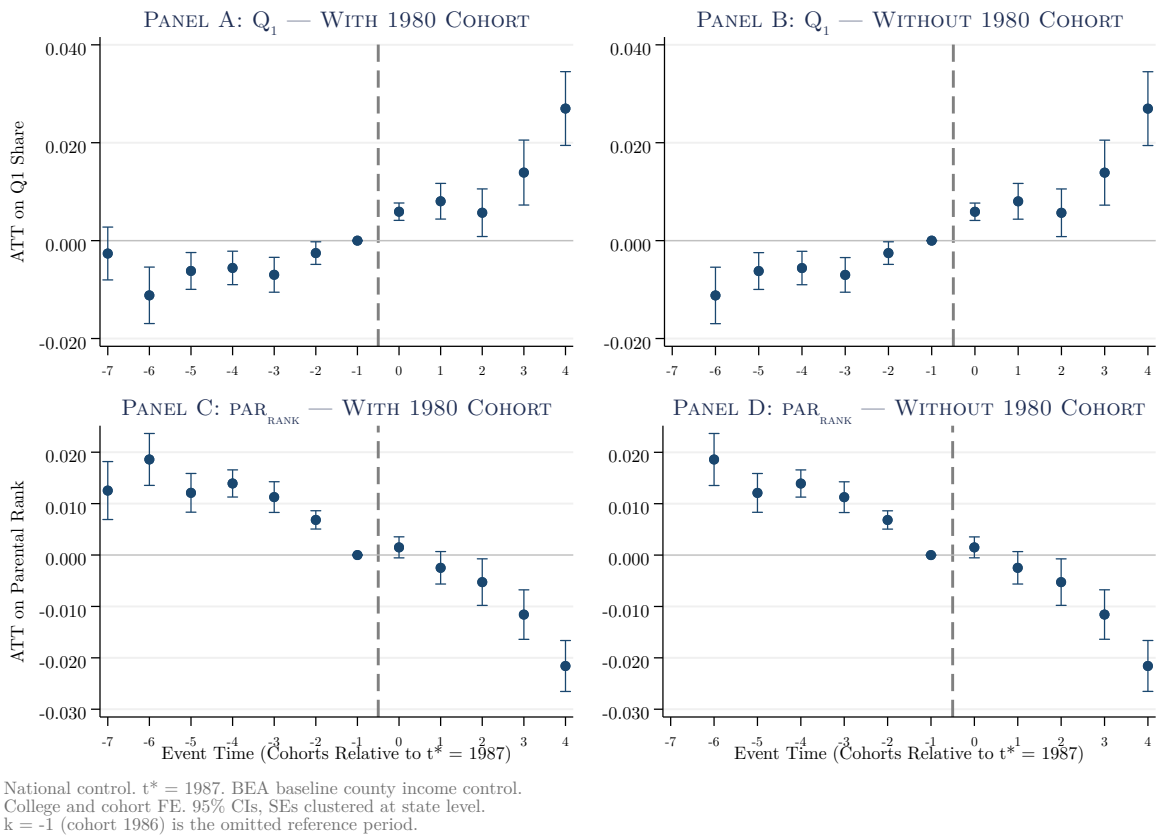

Figure A1: 1980-Cohort Diagnostic: Event Studies With and Without the 1980 Birth Cohort
Notes: TWFE event-study estimates under the preferred specification ($t^* = 1987$, national control, BEA baseline county income control). Left panels include the 1980 birth cohort ($k = -7$); right panels exclude it. Top row: bottom-quintile share (Q_1). Bottom row: mean parental income rank (*par_rank*). Pre-treatment coefficients at $k = -7$ through $k = -2$ trend away from zero when the 1980 cohort is included, indicating a pre-trend that violates parallel trends at the longest horizons. Dropping the 1980 cohort does not materially change the remaining coefficients ($k = -6$ through $k = +4$), confirming that the exclusion removes an anomalous observation without leveraging the post-treatment estimates. $k = -1$ (cohort 1986) is the omitted reference period. College and cohort fixed effects. 95% confidence intervals. Standard errors clustered at the state level. Data: Chetty et al. (2020).

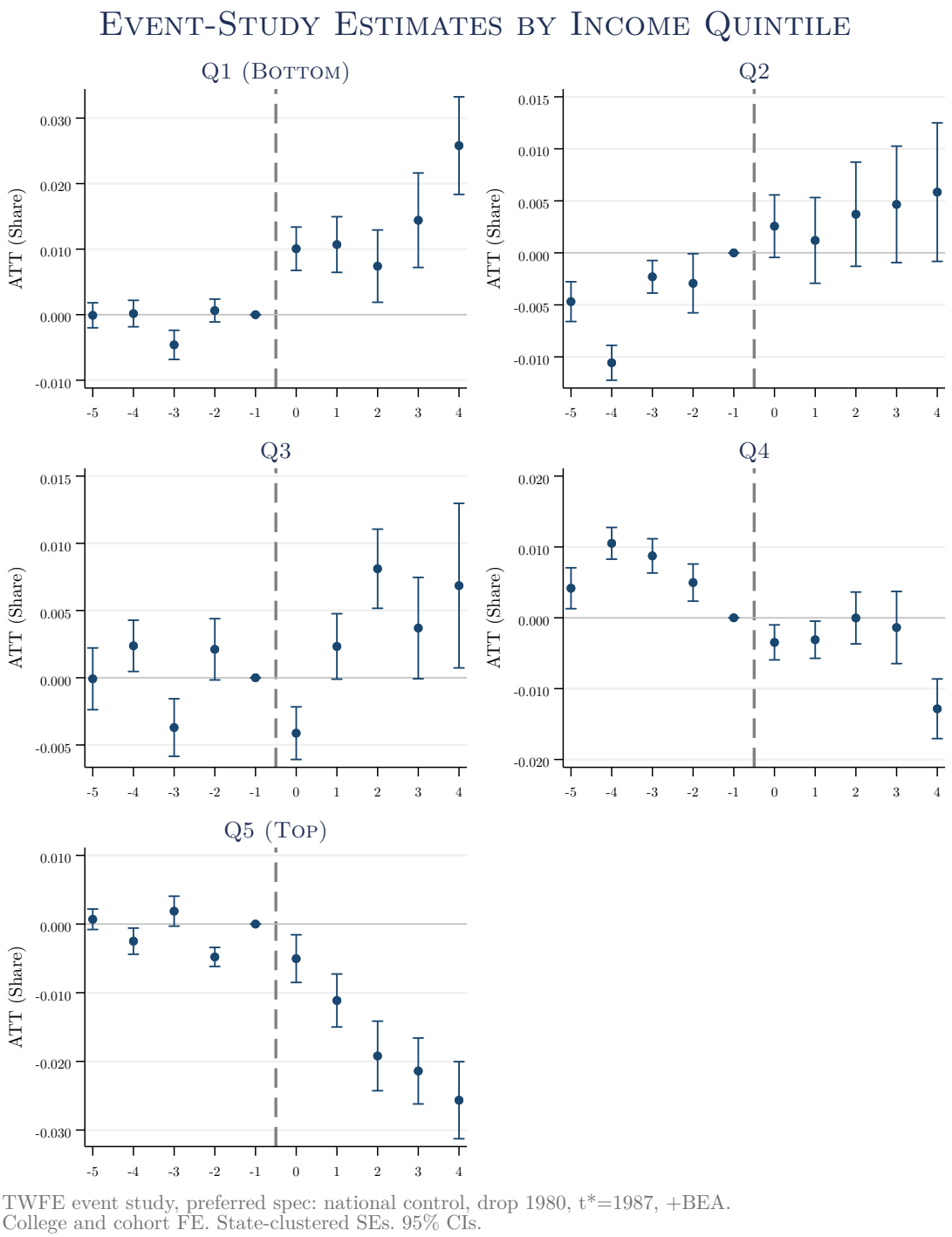

Figure A2: Event-Study Estimates by Income Quintile

Notes: TWFE event-study estimates for each income quintile share under the preferred specification ($t^* = 1987$, drop 1980 cohort, BEA county income control). Bottom quintiles (Q1–Q2) trend upward post-treatment (gaining share); top quintiles (Q4–Q5) trend downward (losing share); Q3 is near zero. The monotonic gradient confirms that HOPE reshuffled composition across the full income distribution. National control group (26 non-merit states). $k = -1$ omitted. Standard errors clustered by state. Data: Chetty et al. (2020).

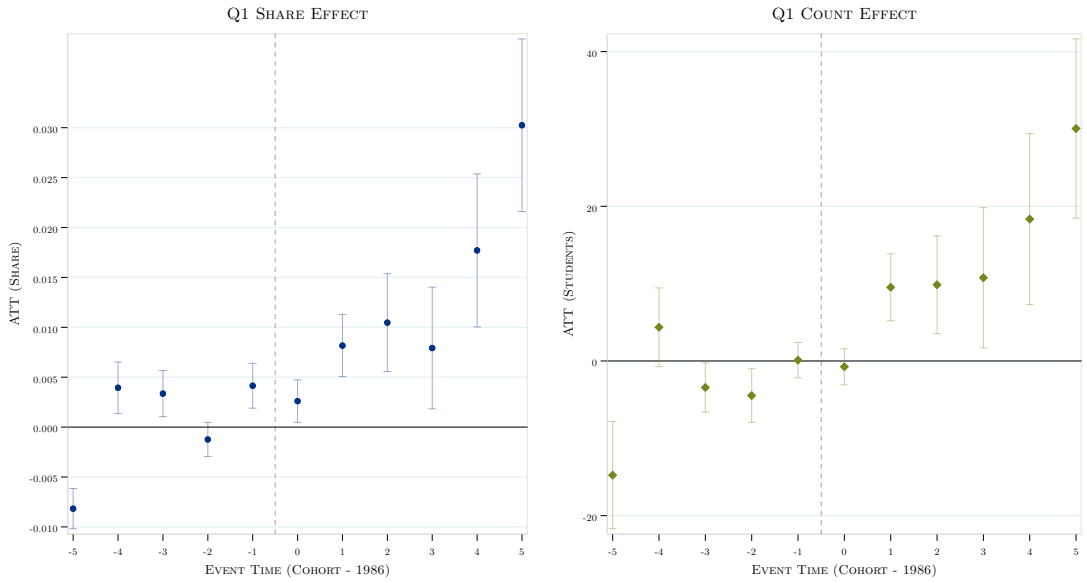

Figure A3: Share Effects vs. Count Effects: Bottom-Quintile Students

Notes: Comparison of share-based (left) and count-based (right) event-study estimates for bottom-quintile students. The share increase is accompanied by a count increase, confirming that the composition shift reflects genuine sorting rather than a denominator artifact. Count outcomes are inherently noisier than share outcomes because they inherit enrollment variation across colleges (coefficient of variation ≈ 1.0). The wider confidence intervals on count estimates reflect this additional noise, not weaker identification. TWFE estimator. Data: Chetty et al. (2020).

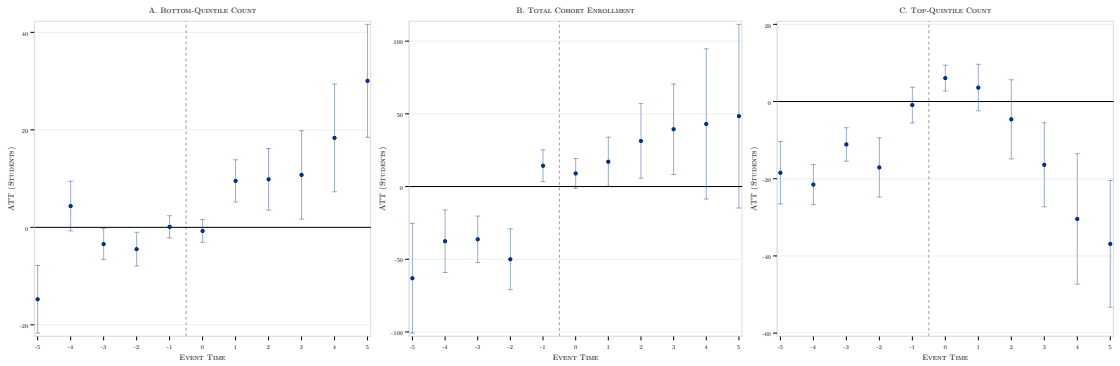

Figure A4: Level Decomposition: Bottom-Quintile Count, Top-Quintile Count, and Total Enrollment

Notes: TWFE event-study estimates for count outcomes (quintile share × cohort enrollment). The critical finding is that total enrollment (Panel B) is unchanged post-treatment, ruling out the denominator-artifact explanation for share-based results. Q1 count increases and Q5 count decreases, consistent with real compositional sorting. Count outcomes exhibit wider confidence intervals than share outcomes because they compound share-level variation with enrollment variation (enrollment coefficient of variation ≈ 1.0 across colleges). The pre-treatment noise in count event studies reflects this inherent volatility rather than a failure of parallel trends for the share-based specifications. Data: Chetty et al. (2020).

Figure A5: Spillover Test: Tennessee Private Colleges

Notes: TWFE event-study estimates comparing Tennessee private colleges (treated) to private colleges in control states. No significant composition change is detected, indicating that top-quintile students who exit Tennessee public colleges do not re-sort to in-state private institutions. $N = 5{,}336$. Data: Chetty et al. (2020).

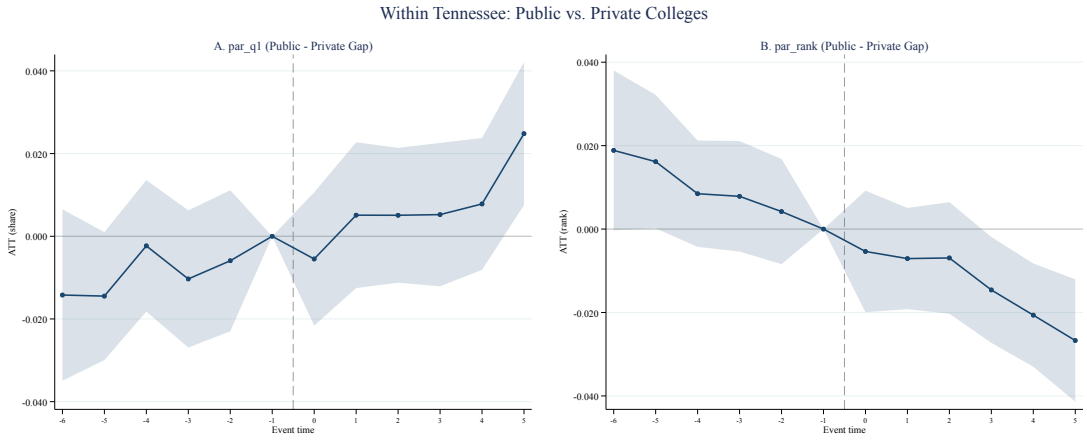

Figure A6: Within-Tennessee Comparison: Public vs. Private Colleges

Notes: Event-study estimates restricting the sample to Tennessee institutions only. Public colleges (treated) are compared to Tennessee private colleges (control). The composition shift is concentrated in the public sector. College and cohort fixed effects included. $N = 571$. Data: Chetty et al. (2020).

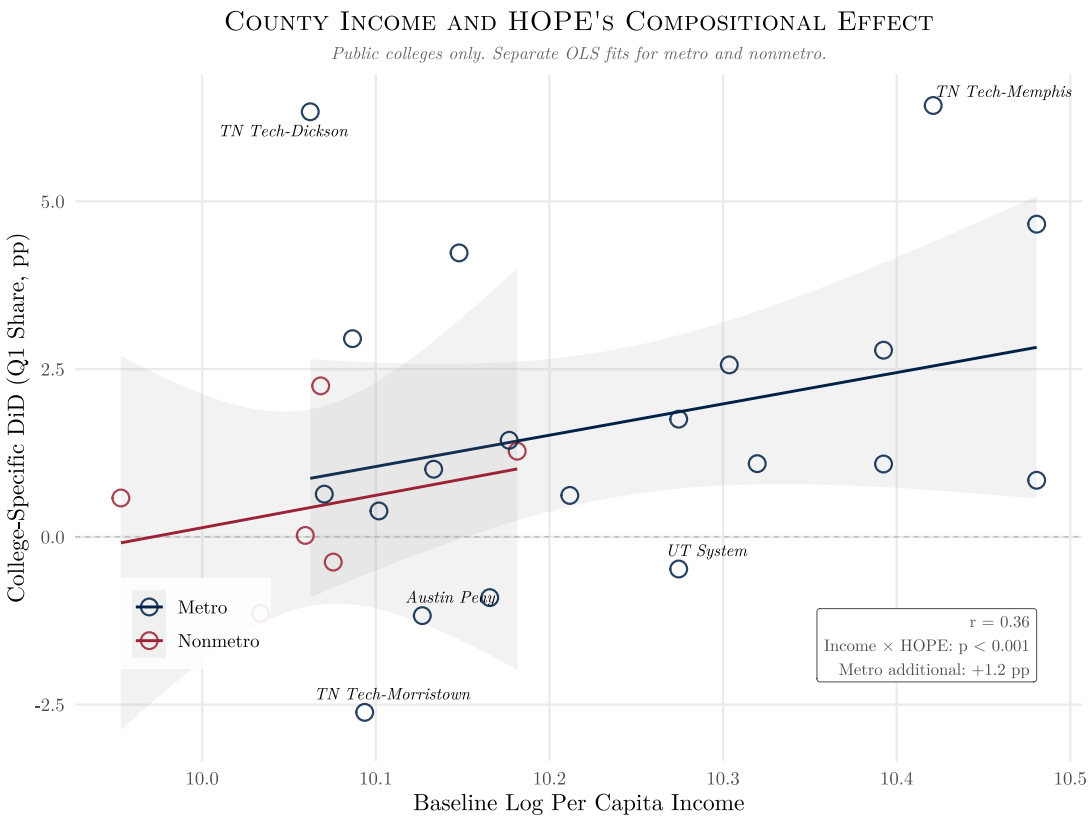

Figure A7: College-Level DiD Effects vs. Baseline County Income

Notes: Each point represents a Tennessee public college. The horizontal axis is baseline log per capita income (BEA county average, 1998–2003); the vertical axis is the college-specific TWFE DiD estimate for bottom-quintile share. Separate OLS fitted lines and 95% confidence bands are shown for metro (navy) and nonmetro (cranberry) colleges. The overall correlation is $r = 0.36$; richer counties host colleges with larger progressive effects. Data: Chetty et al. (2020); BEA CAINC1.

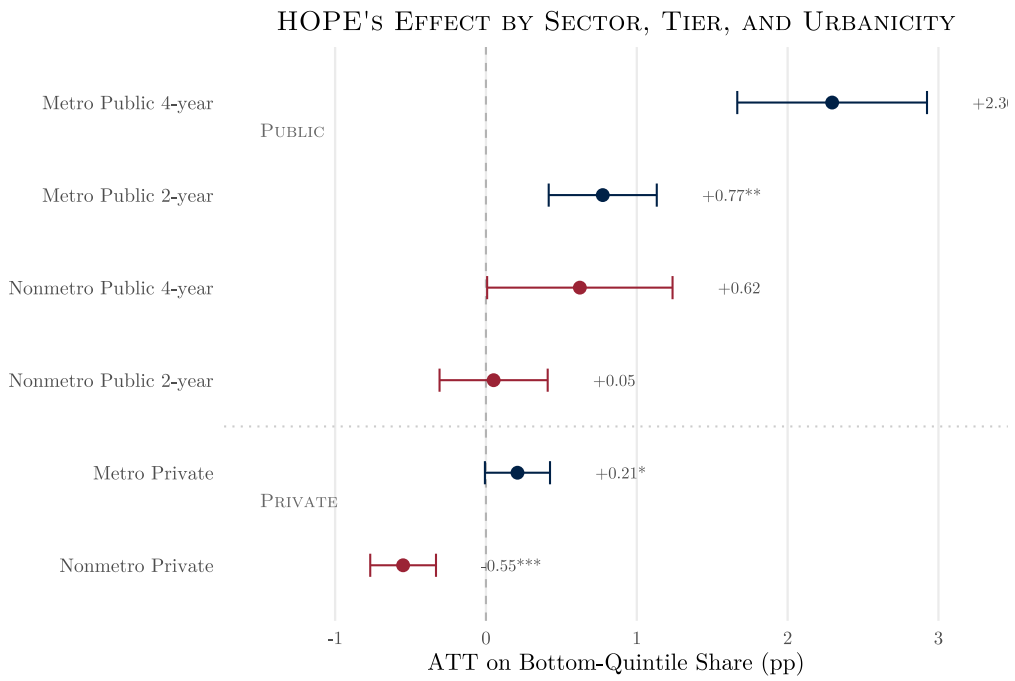

Figure A8: Subgroup ATT Estimates: Metro/Nonmetro × Sector and Tier

Notes: TWFE static DiD estimates for bottom-quintile share, estimated separately by metro status, sector, and tier. Metro public colleges—especially two-year institutions—drive the aggregate effect. Nonmetro public colleges show small, insignificant effects. Private colleges show an attenuated metro pattern consistent with HOPE's smaller private-institution subsidy. Horizontal bars are 95% confidence intervals. Standard errors clustered at the state level. Data: Chetty et al. (2020).

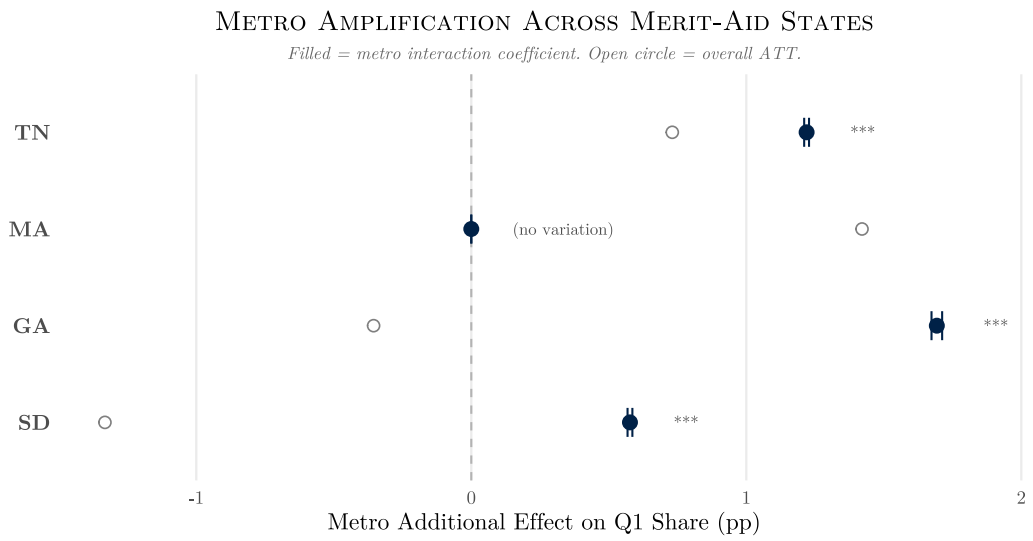

Figure A9: Metro Interaction Coefficients Across States

Notes: Metro × treatment interaction coefficients from TWFE static DiD, estimated separately for each state. Open circles show overall ATT for context. Tennessee, South Dakota, and Georgia all show positive metro interactions, indicating that the metro amplification is not specific to Tennessee. Massachusetts shows no metro variation because virtually all public colleges are in metro areas. Horizontal bars are 95% confidence intervals. Standard errors clustered at the state level. Data: Chetty et al. (2020).

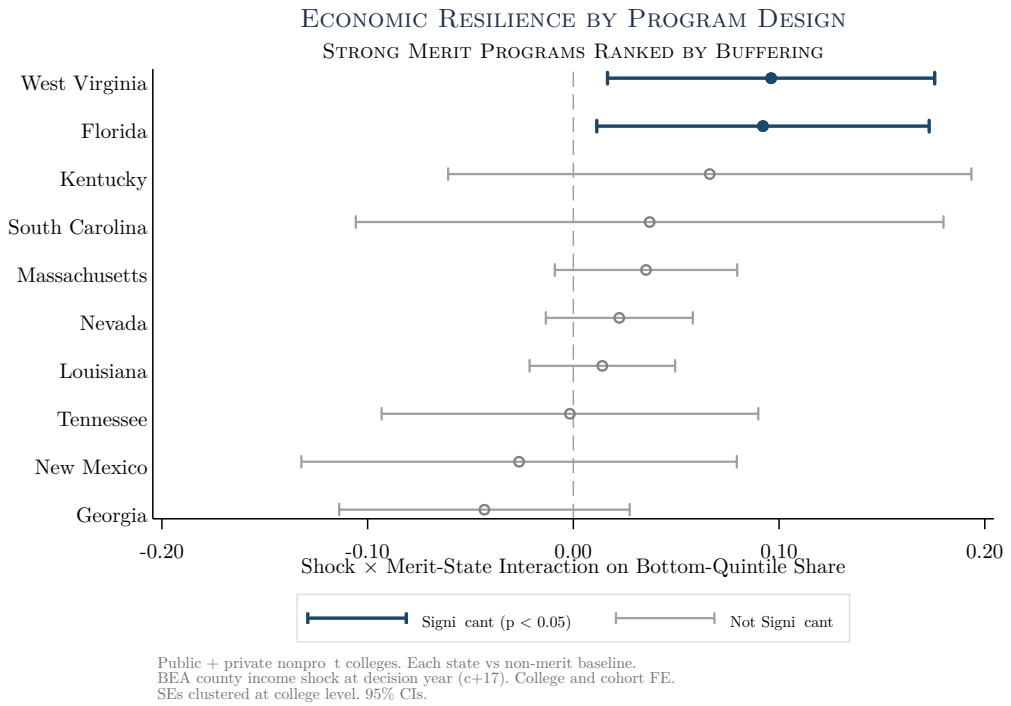

Figure A10: Economic Resilience by Program Design

Notes: Shock × merit-state interaction coefficient on bottom-quintile share for each strong merit program (Sjoquist & Winters, 2015, plus Massachusetts). Each state estimated separately against non-merit baseline. Public and private nonprofit colleges. BEA county income shock at decision year (cohort + 17). College and cohort FE, college-clustered SEs. 95% CIs.

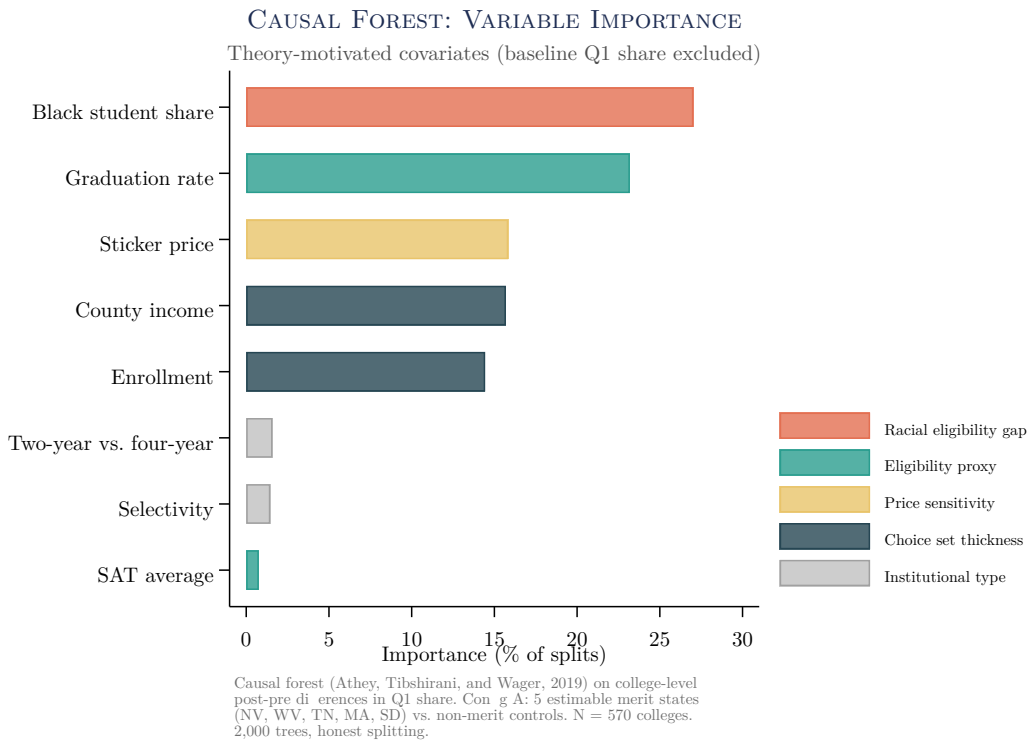

Figure A11: Causal Forest Variable Importance

Notes: Variable importance from a causal forest (Athey et al., 2019) estimated on college-level post-minus-pre differences in bottom-quintile share. Config A: 5 estimable merit states (NV, WV, TN, MA, SD) vs. non-merit controls (N = 570 colleges). Baseline Q_1 share excluded to avoid mechanical correlation with the outcome. Bars show share of splits allocated to each covariate, colored by theoretical channel. 2,000 trees, honest splitting.

B Tables

Table A1: Distribution of HOPE Scholarship Recipients by Sector, Fall 2004–2008

	2004	2005	2006	2007	2008
Panel A: Number of Recipients					
TBR Universities	7,454	7,054	7,637	7,956	8,121
TBR Community Colleges	4,506	3,909	4,720	4,996	4,970
UT Campuses	5,380	5,310	5,576	5,988	6,327
Private Colleges	3,109	3,387	3,492	3,885	3,902
Total	**20,449**	**19,660**	**21,425**	**22,825**	**23,320**
Panel B: Share of Recipients (%)					
TBR Universities	36	36	36	35	35
TBR Community Colleges	22	20	22	22	21
UT Campuses	26	27	26	26	27
Private Colleges	15	17	16	17	17

Notes: First-time freshmen receiving Tennessee HOPE Scholarship awards. TBR = Tennessee Board of Regents; UT = University of Tennessee system. Panel B shares may not sum to 100 due to rounding. Public institutions (TBR + UT) account for 83–85% of recipients in all years. Source: Tennessee Student Assistance Corporation (Tennessee Higher Education Commission, 2025).

Table A2: Difference-in-Differences: Pre- and Post-Treatment Means

	Treated (TN Public)	**Control** (Non-Merit States)
Panel A: Pre-Treatment Means		
Bottom-quintile share	0.138	0.120
Panel B: Post-Treatment Means		
Bottom-quintile share	0.159	0.129
Panel C: DiD Estimate		
Treated × Post	0.0148^{***}	
	(0.0016)	
College FE	✓	
Cohort FE	✓	
Observations	23,395	
R^2	0.948	
Clusters	27	

Notes: Standard errors clustered at the state level in parentheses. Treated = Tennessee public colleges. Control = public colleges in 26 states without broad-based merit programs (Sjoquist & Winters, 2015). Post = birth cohorts 1985 and later. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A3: Callaway–Sant'Anna ATT Estimates (Robustness)

	(1) Bottom-Quintile Share	(2) Parental Income Rank
ATT	0.0126^{***}	-0.0081^{***}
	(0.0023)	(0.0018)
95% CI	[0.0080, 0.0171]	[−0.0117, −0.0046]

Notes: Aggregate ATT from Callaway and Sant'Anna (2021), reported for robustness. In this non-staggered setting, C&S reduces to a non-parametric pre/post comparison (Roth, 2026). Preferred specification: $t^* = 1987$, 1980 cohort dropped, BEA county income control. The comparison group consists of never-treated public colleges in 26 non-merit-program states. Analytical standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A4: Static TWFE Difference-in-Differences Estimates

	(1) Bottom-Quintile Share	(2) Parental Income Rank
TN Public × Post	0.0154^{***}	-0.0185^{***}
	(0.0030)	(0.0025)
Constant	0.138^{***}	0.540^{***}
	(0.0001)	(0.0001)
Observations	6,289	6,289
Adjusted R^2	0.935	0.974
Clusters	27	27

Notes: Two-way fixed effects estimates with college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Treated = Tennessee public colleges. Post = birth cohort $\geq$ 1986. The sample includes public colleges in Tennessee and 26 non-merit-program states. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A5: Quintile Decomposition: TWFE Static DiD by Income Quintile

	(1)	(2)	(3)	(4)	(5)
	Q1 (Bottom)	Q2	Q3	Q4	Q5 (Top)
TN Public × Post	0.0142***	0.0071***	0.0032*	−0.0091***	−0.0154***
	(0.0029)	(0.0023)	(0.0017)	(0.0018)	(0.0019)
Observations	5,630	5,630	5,630	5,630	5,630

Notes: Each column reports the TWFE static difference-in-differences estimate from a separate regression in which the dependent variable is the share of students from the indicated income quintile. Preferred specification: $t^* = 1987$, 1980 cohort dropped, BEA county income control. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Quintile shares sum to one by construction. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A6: Level Decomposition: Share and Count Outcomes

	(1)	(2)	(3)	(4)	(5)
	Q1 Share	Q1 Count	Total Count	Q5 Share	Q5 Count
TN Public × Post	0.0154***	11.23**	−2.11	−0.0179***	−34.38***
	(0.0030)	(4.256)	(25.42)	(0.0025)	(8.635)
Constant	0.138***	167.9***	1,586***	0.200***	449.7***
	(0.0001)	(0.104)	(0.622)	(0.0001)	(0.211)
Observations	6,289	6,289	6,289	6,289	6,289
Adjusted R^2	0.935	0.948	0.985	0.978	0.989
Clusters	27	27	27	27	27

Notes: Two-way fixed effects estimates with college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Count outcomes are constructed as quintile share × cohort size. The Q1 share increase (column 1) is accompanied by a Q1 count increase (column 2), while total enrollment is unchanged (column 3), confirming that share changes reflect compositional sorting rather than a denominator artifact. Post = birth cohort ≥ 1986. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A7: Heterogeneity by Institution Tier: Two-Year vs. Four-Year Colleges

	Two-Year			Four-Year		
Outcome	Coef.	SE	N	Coef.	SE	N
Bottom-quintile share	0.0159^{***}	(0.0034)	3,632	0.0084^{***}	(0.0022)	1,985
Parental income rank	-0.0124^{***}	(0.0025)	3,632	-0.0180^{***}	(0.0017)	1,985
Q1 count	3.11	(7.89)	3,632	22.19^{***}	(6.50)	1,985
Q5 count	-16.80^{**}	(6.84)	3,632	-36.04^{**}	(17.33)	1,985
Total count	-48.80	(45.63)	3,632	146.07^{***}	(39.23)	1,985

Notes: TWFE static difference-in-differences estimates estimated separately for two-year (community colleges) and four-year institutions. Preferred specification: $t^* = 1987$, 1980 cohort dropped, BEA county income control. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Two-year colleges show displacement (composition shifts, enrollment unchanged); four-year colleges show expansion (enrollment grows, composition shifts). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A8: Spillover Test: Tennessee Private Colleges

Outcome	Coef.	SE	p-value	N
Bottom-quintile share	0.0001	(0.0011)	0.914	5,010
Parental income rank	-0.0043^{**}	(0.0018)	0.023	5,010

Notes: TWFE static DiD comparing Tennessee private colleges to private colleges in non-merit-program control states. Preferred specification: $t^* = 1987$, 1980 cohort dropped. The null effect on bottom-quintile share indicates that top-quintile students exiting Tennessee public colleges do not re-sort to in-state private institutions. Standard errors clustered at state level in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A9: Within-Tennessee Comparison: Public vs. Private Colleges

Outcome	Coef.	SE	p-value	N
Bottom-quintile share	0.0156^{***}	(0.0037)	< 0.001	537
Parental income rank	-0.0193^{***}	(0.0033)	< 0.001	537

Notes: TWFE static difference-in-differences restricted to Tennessee institutions only. Treated = Tennessee public colleges; control = Tennessee private colleges. Preferred specification: $t^* = 1987$, 1980 cohort dropped. College and cohort fixed effects included. Robust standard errors in parentheses. The small sample reflects the within-state restriction ($N = 537$). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A10: Robustness Summary Across Estimators, Timing, Control Groups, and Sensitivity

	Bottom-Quintile Share		Parental Income Rank	
Specification	Coef.	SE	Coef.	SE
Panel A: Estimator (preferred spec: $t^* = 1987$, drop 1980, +BEA)				
C&S	0.0126^{***}	(0.0023)	-0.0081^{***}	(0.0018)
TWFE static	0.0142^{***}	(0.0029)	-0.0148^{***}	(0.0022)
Panel B: Treatment Timing (TWFE, national, drop 1980, +BEA)				
$t^* = 1986$	0.0135^{***}	(0.0041)	-0.0159^{***}	(0.0035)
$t^* = 1987$ (preferred)	0.0142^{***}	(0.0029)	-0.0148^{***}	(0.0022)
Panel C: Control Group (TWFE, $t^* = 1987$, drop 1980, +BEA)				
National (preferred)	0.0142^{***}	(0.0029)	-0.0148^{***}	(0.0022)
Bordering (AL, VA, NC)	0.0124^{***}	(0.0023)	-0.0104^{***}	(0.0027)
Panel D: BEA Controls (TWFE, $t^* = 1987$, national, drop 1980)				
Without BEA	0.0158^{***}	(0.0043)	-0.0184^{***}	(0.0038)
With BEA level (preferred)	0.0142^{***}	(0.0029)	-0.0148^{***}	(0.0022)
With BEA level + change	0.0132^{***}	(0.0041)	-0.0155^{***}	(0.0035)
Panel E: Rambachan–Roth Sensitivity (preferred spec)				
Original CI	[0.007, 0.013]		[−0.009, −0.003]	
Δ_{RM}: robust at $\bar{M} \leq$	1.0		0.5	
Δ_{SD}: robust at $M \leq$	0.005		0.005	

Notes: Panels A–D report static difference-in-differences estimates under alternative specifications. All specifications include college and cohort fixed effects. Standard errors clustered at the state level in parentheses. The preferred specification uses the national control group, drops the 1980 birth cohort, sets $t^* = 1987$, and controls for BEA county log per capita income (pre-treatment county average, 1997–2003). Panel E summarizes Rambachan and Roth (2023) sensitivity results for Q_1 under the preferred specification: the original CI excludes zero; Δ_{RM} is robust at $\bar{M} \leq 1.0$ (post-treatment violations as large as the maximum pre-treatment violation); Δ_{SD} is robust at $M \leq 0.005$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A11: Spatial Heterogeneity: Metro Status and County Income Interactions

	(1) Metro Only	(2) Income Only	(3) Joint
Dependent variable: Bottom-quintile share (Q_1)			
TN Public × Post	0.0073**	0.0142***	0.0068**
	(0.0031)	(0.0040)	(0.0032)
× Metro	0.0122†		0.0098***
	†		(0.0015)
× Baseline Log Income		0.0036***	0.0024***
		(0.0008)	(0.0009)
Observations	5,855	5,855	5,855
College & Cohort FE	Yes	Yes	Yes
BEA Controls	Yes	Yes	Yes
Implied effects:			
Metro ATT	0.0196		0.0166
Nonmetro ATT	0.0073		0.0068

Notes: TWFE static difference-in-differences with spatial interactions. Column (1) interacts treatment with metro status (USDA RUCC 1–3 = metro); column (2) interacts treatment with baseline county log per capita income (BEA, 1998–2003 average); column (3) includes both interactions jointly. Metro status is defined at the county level using 2003-vintage Rural-Urban Continuum Codes. All specifications include college and cohort fixed effects and control for BEA county log per capita income (pre-treatment county average). Standard errors clustered at the state level in parentheses. † Fewer than five effective clusters for this interaction; cluster-robust SE is anticonservative. Significance assessed via subgroup decomposition (Figure A8). $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020); BEA CAINC1.

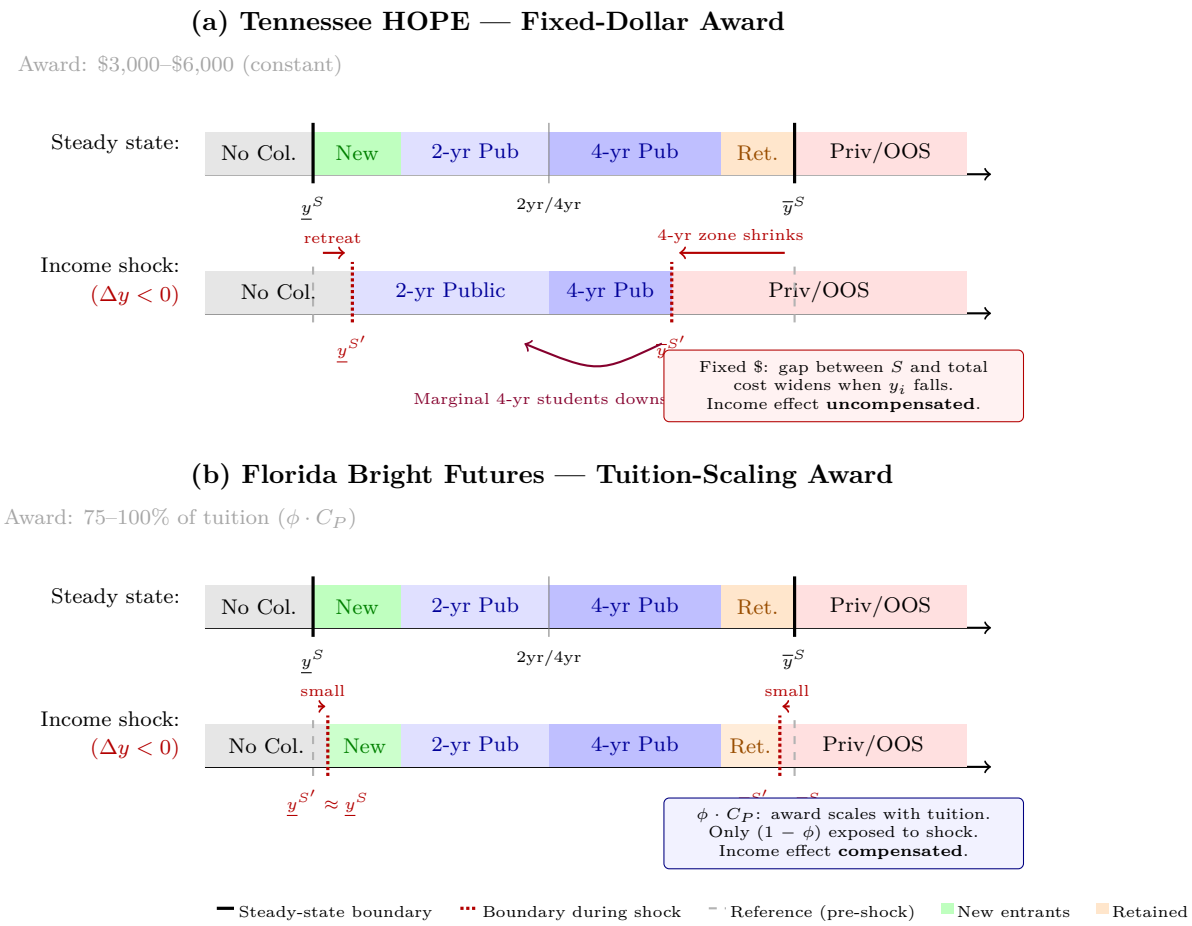

Figure 2: Slutsky Decomposition: How Award Structure Determines Resilience to Income Shocks

Notes: Each panel shows the steady-state sorting (top bar) and the response to a negative income shock (bottom bar). Panel (a): Tennessee HOPE pays a fixed dollar amount ($3,000–$6,000). When income falls, the gap between the award and total cost of attendance widens. Both boundaries retreat substantially: some new entrants at the access margin drop out, and marginal four-year students (created by HOPE) downshift to two-year colleges. The income effect is uncompensated. Panel (b): Florida Bright Futures covers a fixed percentage of tuition (75–100%). When income falls, the award scales with tuition costs, so the gap between scholarship and cost barely changes. Boundaries shift minimally. The income effect is compensated. Both programs produce progressive sorting in steady state (the substitution effect); only the percentage-based program sustains it during downturns.

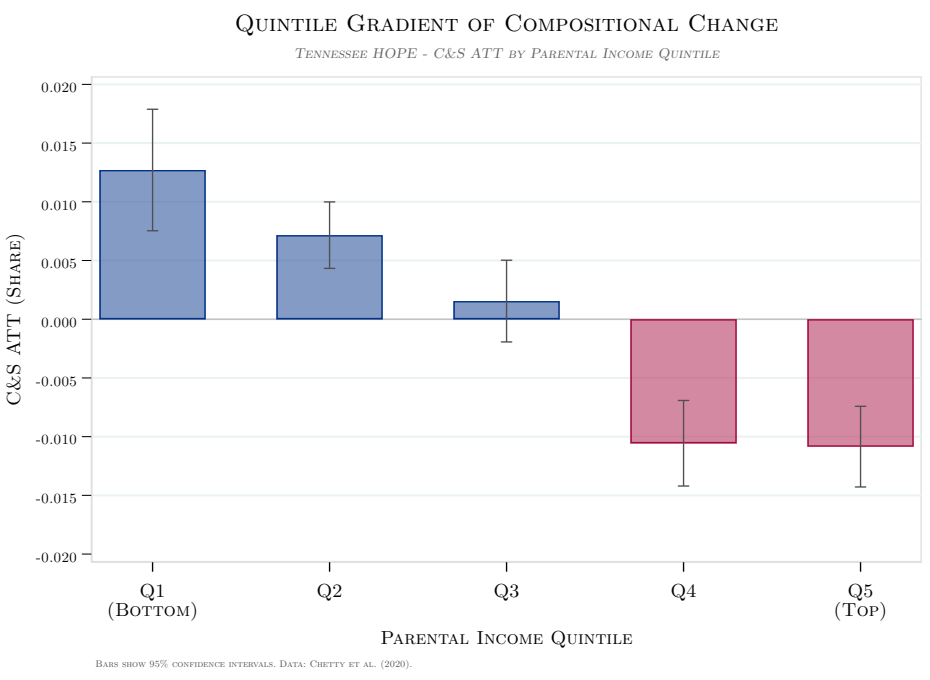

Figure 7: Quintile Gradient of Compositional Change Following Tennessee HOPE

Notes: TWFE static ATT estimates for each parental income quintile share. The effect is monotonically ordered: bottom-quintile students gain the most representation and top-quintile students lose the most. Significance is strongest at the tails (Q1, Q5) where both the theoretical effect and cross-college variance are largest. Bars denote 95% confidence intervals. Data: Chetty et al. (2020).

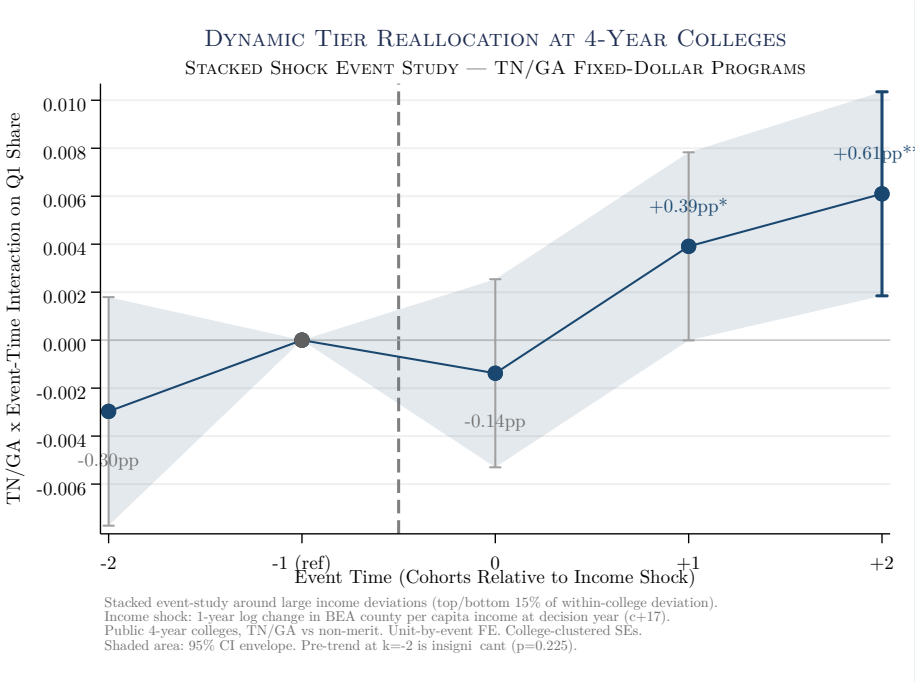

Figure 10: Dynamic Tier Reallocation at Four-Year Colleges

Notes: Stacked shock-event-study around large income deviations (top/bottom 15% of within-college deviation from mean). Shows the TN/GA × event-time interaction coefficient on Q_1 share at four-year public colleges. The pre-trend at $k = -2$ is insignificant ($p = 0.225$). Composition displaces on impact then rebounds by $k = +2$ (+0.61 pp, $p = 0.005$). Unit-by-event FE, college-clustered SEs. Shaded area: 95% CI envelope.